HUMAN COMPUTER INTERACTION

EXPLORING THE GAP: EXPLORING THE SPACE BETWEEN DIGITAL AND ANALOGUE LEARNING ENVIRONMENTS FOR NEURODIVERGENT STUDENTS

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Project rapport

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Abstract

This thesis investigates how neurodivergent university students navigate hybrid learning environments that span digital, analogue, and augmented paper modalities. While hybrid systems are often designed around assumptions of seamlessness and efficiency, this project argues that such framings overlook the embodied, affective, and cognitive variability of neurodivergent learners. Drawing on human-computer interaction, cognitive load theory, and neurodiversity studies, the study employs a qualitative mixed-methods approach, including a literature informed diary study and semi-structured interviews. Through grounded coding and cognitive load analysis, it identifies key breakdowns, adaptation strategies, and design tensions in students' personal tool ecologies. Findings reveal that flexibility, sensory predictability, and user-governed augmentation are more critical than automation or interactivity. The thesis culminates in a set of empirically grounded design principles for hybrid reading systems that centre upon the neurodivergent experience—not as an exception, but as a generative lens for inclusive design. By shifting the design gaze toward cognitive sovereignty and epistemic safety, this work contributes a novel framework for reimagining hybrid learning through user variability rather than universal optimisation.

Danish Abstract

Dette Speciale undersøger, hvordan neurodivergente studerende navigerer i hybride læringsmiljøer, der kombinerer digitale, analoge og augmented paper-baserede modaliteter. Gennem en kvalitativ mixed-methods tilgang, bestående af dagbogsstudier og semistrukturerede interviews, identificeres centrale strategier, sammenbrud og designspændinger i de studerendes artifaktøkologier. Resultaterne viser, at fleksibilitet, sensorisk forudsigelighed og brugerstyret augmentering er vigtigere end sømløshed og automatisering. Afhandlingen præsenterer et antal designprincipper, der tager udgangspunkt i neurodivergente erfaringer og bidrager med en ny ramme for inkluderende design, hvor kognitiv variation ses som en styrke frem for en afvigelse.

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1 INTRODUCTION

Over the last two decades, hybrid learning environments have evolved from niche innovations to mainstream pedagogical frameworks. Blending digital and physical modalities, these environments promise flexibility, personalisation, and multimodal access. Yet, for neurodivergent learners—students whose cognitive and sensory profiles diverge from dominant norms—these same environments often reproduce forms of exclusion, friction, or overload. While hybrid designs aim to optimise learning through seamless transitions and rich interactivity, they may inadvertently privilege cognitive styles aligned with consistency, predictability, and low sensory sensitivity.

This thesis investigates how neurodivergent university students navigate hybrid learning ecologies across analogue, digital, and augmented paper interfaces. Through a qualitative mixed-methods design—comprising a literature review, a diary study, semi-structured interviews, and grounded thematic coding—this project asks:

1. What are the lived strategies, breakdowns, and design tensions experienced by neurodivergent students in hybrid learning environments? And further:
2. How might current design ideas for hybrid learning be reimagined through a neurodivergent-centered lens?

Much of the existing literature on hybrid learning emphasises universal access and efficiency. Design frameworks centre personalisation, collaboration, and technological fluency as means to optimise outcomes. However, these frameworks often lack nuance in addressing cognitive volatility, sensory processing variability, and the embodied tactics neurodivergent students employ to manage attention, memory, and emotional regulation. This thesis engages critically with these gaps, situating design not just as an interface challenge but as an epistemic and affective concern.

Rather than treating neurodivergent users as a special case or add-on to general usability, this study foregrounds their experiences as central to rethinking hybrid learning.

Starting with existing literature, this thesis first reviews works pertaining to the digital versus the physical reading experience, augmented paper, artifact ecologies, embodied learning, and neurodivergent learning processes, concluding the literature review by presenting a set of design principles grounded in the existing literature. Afterward, drawing from human-computer interaction (HCI), cognitive load theory, neurodiversity studies, and embodied learning, the research constructs a set of revised design principles—each informed by empirical accounts of friction, workaround, and adaptation.

1.1 RESEARCH AIMS AND QUESTIONS

The project is guided by two overarching aims:

1. To understand how neurodivergent students engage with and configure their own hybrid learning ecologies, including paper, digital, and augmented systems.
2. To develop design principles that centre neurodivergent learning by prioritising variability, autonomy, and emotional coherence—building on and extending existing literature.

These aims are explored through the following research questions:

- What strategies do neurodivergent students use to navigate hybrid learning environments?
- What breakdowns or tensions emerge when digital or augmented systems assume uniform user behaviour?
- How can hybrid learning design principles be adapted to both reflect established literature and accommodate the lived realities of neurodivergent learners?
- Is it possible to construct design principles that guide future hybrid learning design without disregarding prior theoretical contributions?

1.2 STRUCTURE OF THE THESIS

This thesis is structured as follows:

- Chapter 2 reviews foundational literature on digital and print reading, augmented paper, artifact ecologies, embodied learning, and neurodivergent learning processes, and concludes by presenting design principles for hybrid learning design that adhere to the literature.
- Chapter 3 details the qualitative mixed-methods approach, including data collection, participant recruitment, and analytical strategy as well as ethical considerations.
- Chapter 4 presents empirical findings from diary studies and interviews, structured through grounded coding and Cognitive Load Theory, and concludes by presenting revised design principles that consider the lived experiences of neurodivergent students.
- Chapter 5 synthesises these findings and revised design principles and explores broader theoretical and design implications through a thematic discussion.
- Chapter 6 concludes with a neurodivergent-centered framework for hybrid learning, reflecting on its commitments and potential for future research.

1.3 CONTRIBUTION

The thesis contributes to Human-Computer Interaction by proposing a reframing of hybrid learning design through the lens of neurodivergent experience. Rather than seeking universally seamless systems, it advocates for epistemic safety, opt-in augmentation, and affective legibility as core design values. It also contributes methodologically by integrating diary-based self-observation with grounded theory and Cognitive Load Theory in the study of hybrid academic practices.

Ultimately, this project argues that starting from neurodivergent learners does not limit design—it expands it.

2 RELATED WORK

2.1 THE DIGITAL VS. PRINT READING EXPERIENCE

As digital reading becomes increasingly prevalent, it is essential to consider not only its usability but also its cognitive impact. While augmented paper seeks to combine the advantages of both print and digital mediums, research indicates that digital reading introduces new cognitive challenges that can affect comprehension and engagement.

The shift from print to digital reading has profoundly influenced reading habits, cognitive processing, and information retention. Extensive research has compared the two formats, with a particular focus on comprehension, user engagement, and cognitive load. This section examines key studies on the **differences between digital and print reading**, highlighting their impact on **reading comprehension, text navigation, and cognitive engagement**.

2.1.1 *Reading Comprehension and Medium Effects*

One of the most extensively studied aspects of digital vs. print reading is comprehension performance. Several meta-analyses and empirical studies have consistently demonstrated a print superiority effect in comprehension tasks.

A meta-analysis by Altamura, Vargas, and Salmerón [3] examined 40 studies with a sample of 469,564 participants, finding that while digital reading is increasingly prevalent, it does not support comprehension as effectively as print reading. Their findings revealed a small but significant negative effect ($r = .055$) of digital reading on comprehension, particularly in younger readers. The correlation coefficient (denoted r) quantifies the strength and direction of the relationship between digital reading and comprehension. In this case, a correlation coefficient of .055 indicates a weak but statistically significant negative correlation, meaning that as digital reading increases, comprehension slightly decreases. While this effect is not large, it is consistent across multiple studies, suggesting a trend rather than an isolated finding. This suggests that while digital reading offers

convenience, it may not provide the same level of cognitive engagement and retention as print reading.

This trend aligns with findings from Delgado et al. [11], which analyzed studies between 2000 and 2017 and found a consistent disadvantage for digital reading. The effect size was reported using Hedges' g = -0.21, which quantifies the standardised mean difference between digital and print reading comprehension scores. In this case, a negative Hedges' g value indicates that participants who read in digital formats performed worse in comprehension tasks compared to those who read in print. While this effect size is considered small to moderate, it is statistically meaningful across the studies included in the meta-analysis, reaffirming the superiority of print reading in retaining complex information.

Similarly, Peras et al. [29] conducted a systematic review of 23 studies and identified that prior comprehension skills and reading habits play a crucial role in determining performance across digital and paper formats. They noted that screen inferiority effects are more pronounced in younger students, who may struggle to process and retain information as effectively when reading digitally. However, more advanced readers (high school and university level) exhibit better adaptation to digital reading, likely due to greater familiarity with digital tools and self-regulated learning strategies. These findings highlight the role of experience and reading habits in moderating the effectiveness of digital reading.
Additionally, Singer and Alexander [32] reviewed decades of empirical research and concluded that digital reading often promotes skimming behaviours rather than deep comprehension, particularly in cases where students assume digital texts require less cognitive effort. This highlights that the medium itself influences reading strategies and comprehension outcomes.

2.1.2 *Spatial Navigation and Cognitive Mapping*

Reading on digital devices often lacks the tactile and kinesthetic feedback inherent to print books, which affects the reader's ability to navigate and organise information spatially. This limitation is particularly evident in longer texts, where readers rely on physical markers such as page numbers, thickness, and the ability to flip through pages to locate information.

Mangen, Olivier, and Velay [25] conducted an experimental

study comparing the comprehension of a long narrative text read on a Kindle versus a print book. Their results showed no significant differences in overall comprehension but highlighted that print readers performed significantly better in locating events in a story and reconstructing its timeline. This supports the hypothesis that physical interaction with a book aids in forming a cognitive map of the text, thereby facilitating better event tracking and temporal sequencing. The reduced spatial awareness in digital reading may contribute to difficulties in organising and recalling narrative structures, which can impact comprehension and engagement, particularly for readers who rely on physical cues to track their progress through a text.

Further supporting this, Jeong and Gweon [15] examined visual reading patterns and found that print readers tend to maintain more stable eye movements, reducing cognitive load, while digital readers exhibit more erratic eye-tracking behaviours. This instability in eye movement during digital reading suggests increased cognitive effort, potentially leading to quicker fatigue and reduced comprehension over extended reading periods. Additionally, erratic eye-tracking behaviours in digital formats are often associated with difficulties in maintaining a coherent reading flow, making it harder for readers to mentally map text structures and retain information effectively.

2.1.3 *Digital Reading and Student Preferences*

Despite the cognitive advantages of print reading, digital reading has gained popularity due to its accessibility, portability, and interactive features. Many readers, particularly students, choose digital reading formats for their convenience in storing, searching, and annotating text. However, this preference does not always translate into improved comprehension.

A study by Cahyani and Wahyunengsih [7] among journalism students at UIN Jakarta found that students prefer print books over digital due to superior reading quality, reduced eye strain, and improved focus. Their findings also revealed that students find digital books less engaging for deep reading but useful for quick information retrieval. Their findings align with a broader study by Mizrachi et al. [26], which surveyed over 10,000 university students across 21 countries and found that the vast majority still prefer print for academic reading. The study indicated that students believe they learn better from print materi-

als, particularly for deep reading tasks, despite their frequent use of digital texts for quick reference. This suggests that students' reading preferences are influenced by both functional needs and cognitive engagement, with digital formats being favored for quick reference and print being preferred for immersive learning.

Additionally, Peras et al. [29] conducted a systematic review and found that factors such as socioeconomic status, gender, and prior reading habits significantly influence digital reading preferences. Higher-income students with greater access to books at home tend to perform better in both digital and print reading but still show a preference for print when reading for comprehension. The review also highlighted that students from lower socioeconomic backgrounds are more likely to rely on digital reading due to affordability and accessibility, yet they may face challenges such as lower digital literacy and increased exposure to distractions when reading on screens. Moreover, gender differences were noted, with female students generally exhibiting stronger preferences for print reading, particularly in academic settings, whereas male students were more likely to engage in digital reading for recreational purposes. These findings suggest that the choice of reading medium is shaped by a complex interaction of personal, social, and economic factors, which influence both reading performance and engagement.

2.1.4 *Neurodiversity and Cognitive Load in Digital Reading*

The impact of digital reading varies across different cognitive profiles, particularly for neurodivergent individuals who may experience increased cognitive load when engaging with online texts. Research by Le Cunff, Smith, and Taylor [21] highlights that while both neurotypical and neurodivergent students face challenges in online learning, neurodivergent students report higher levels of extraneous cognitive load—the mental effort required to process and organise information that is unrelated to the inherent complexity of the content itself. In digital reading contexts, this can manifest as difficulties in navigating online platforms, dealing with information overload, and processing dense or poorly structured digital texts.

One key challenge faced by neurodivergent students is reading comprehension in digital environments. Unlike print materials, digital texts often introduce distractions such as hyperlinks, scrolling interruptions, and inconsistent formatting, which can

disrupt focus and increase mental strain. This issue is particularly pronounced in students with ADHD, dyslexia, or ASD, who may struggle with maintaining attention, decoding text efficiently, and organising information within an online reading interface. Furthermore, inaccurate automated transcripts and inaccessible content presentation create additional barriers, making it harder for students to engage meaningfully with digital texts [21].

Neurodivergent readers also report greater difficulties in multitasking between different modes of engagement, such as simultaneously listening to a lecture while reading on-screen materials. This increased cognitive load can lead to quicker fatigue, reduced retention, and a preference for structured, visually supportive materials. As a result, many neurodivergent students find multimedia learning tools, flexible formatting, and alternative content delivery methods (e.g., video explanations, audio-enhanced text, or interactive note-taking tools) more effective than traditional digital reading.

Le Cunff, Smith, and Taylor [21] further highlight that neurodivergent students experience increased extraneous cognitive load due to difficulties with text-heavy formats, inefficient navigation, and distracting digital interfaces. Addressing these challenges requires implementing features such as structured reading layouts, interactive guidance, and adaptable text displays that align with different cognitive needs.

Furthermore, digital environments should incorporate customisable reading aids, such as dyslexia-friendly fonts, adjustable text spacing, text-to-speech functionality, and accurate captioning systems. These features not only reduce cognitive strain but also support self-paced learning, which is particularly beneficial for neurodivergent readers. Additionally, the integration of flexible learning formats—including audio-enhanced text, structured video content, and interactive note-taking tools—can improve engagement and comprehension [21].

2.1.5 *Factors Influencing Digital vs. Print Reading Outcomes*

Age and Educational Stage

Age is a critical factor in determining reading comprehension differences between digital and print formats. Studies suggest that younger students struggle more with digital reading due

to their developing cognitive skills and reliance on spatial markers to process text. In contrast, older students and adults may benefit from digital formats, particularly when they develop strategies to mitigate its limitations.

Altamura, Vargas, and Salmerón [3] found that digital reading has a negative effect in primary and middle school students but becomes neutral or positive in university students. This shift may be attributed to the development of advanced reading strategies, greater familiarity with digital tools, and increased ability to manage distractions. As readers mature, they become more adept at compensating for the cognitive and spatial challenges associated with digital reading, making it a more viable alternative to print. However, these findings are contradicted by Mizrachi et al. [26], who found that even though university students frequently use digital devices, they still prefer print when engaging in deep learning. The preference for print among university students suggests that digital reading, despite greater exposure and potential adaptation, does not necessarily lead to better engagement with complex texts. This contradicts the assumption that older students automatically develop strategies to mitigate the disadvantages of digital reading. Instead, the continued preference for print may indicate that deep comprehension and long-term retention are still better supported by physical texts, regardless of the increasing presence of digital reading in higher education.

Reading Purpose and Content Type

The type of content being read also influences comprehension differences. Delgado et al. [11] conducted a meta-analysis of 54 studies and found that narrative texts show minimal differences in comprehension across digital and print formats, likely because readers engage with narratives in a continuous and immersive manner regardless of the medium. However, informational and expository texts favor print reading due to the need for deep cognitive engagement, reference checking, and structuring of complex ideas.

Singer and Alexander [32] also found that when reading for deep comprehension, print is the preferred medium. They concluded that print supports better comprehension and retention, particularly for texts that require critical analysis, while digital reading is often associated with skimming behaviours.

Studies by Jeong and Gweon [15] and Mizrachi et al. [26] in-

dicate that print reading provides an advantage for subjects requiring detailed analysis, such as scientific or philosophical texts, where comprehension benefits from the ability to cross-reference, annotate, and revisit key points. This suggests that while digital reading may be effective for casual and surface-level reading, print remains superior for tasks that demand higher-order thinking and retention.

Multitasking and Distractions

A major challenge in digital reading is increased exposure to distractions. Unlike print reading, which is a single-focus activity, digital reading often involves multitasking, notifications, and hyperlinks, leading to shallow processing and fragmented reading habits [25]. The presence of interactive elements, such as embedded videos and hyperlinks, can further contribute to cognitive overload, reducing a reader's ability to concentrate and deeply engage with the text.

While some studies suggest that experienced digital readers can develop strategies to minimise distractions, the overall impact of digital reading on sustained attention remains a concern. To mitigate these effects, Mizrachi et al. [26] suggest that digital reading environments should incorporate features that reduce cognitive load and support deeper engagement, such as minimal interface design, distraction-free modes, and enhanced annotation tools.

2.1.6 *Summary of The Digital vs. Print Reading Experience*

In summary, research on digital versus print reading consistently highlights the cognitive advantages of print, particularly in comprehension, spatial navigation, and deep engagement. Print reading supports better information retention, fosters stable reading patterns, and allows for stronger cognitive mapping due to its tactile nature. While digital reading is convenient and widely used, especially among students, it is often associated with reduced comprehension, increased distractions, and a preference for skimming rather than deep reading. Although some studies suggest that older students adapt better to digital reading, findings indicate that even university students frequently prefer print for complex academic tasks. Socioeconomic and gender factors further influence reading preferences, with higher-income students and female readers more likely to favor print.

Additionally, neurodivergent individuals experience unique challenges in digital reading, facing heightened cognitive load due to distractions, poor interface design, and multitasking demands. Research has shown that students with ADHD, dyslexia, and ASD may struggle with digital text processing, leading to increased fatigue and difficulty in organising information. Accessible digital reading environments with structured layouts, multimodal integration, and assistive features can help mitigate these barriers. Ultimately, while digital reading continues to grow in popularity, print remains the superior medium for deep, focused, and comprehensive reading experiences, particularly for individuals requiring high cognitive engagement.

While digital reading continues to shape modern workflows, research suggests that print-based interactions retain cognitive advantages, particularly for comprehension, focus, and spatial navigation. However, rather than viewing digital and analogue media as opposing forces, many researchers have explored hybrid solutions that integrate the strengths of both. This has led to the development of augmented paper systems, which aim to preserve the tactile and cognitive benefits of paper while incorporating digital enhancements. The following section examines how augmented paper technologies have been designed to bridge the gap between physical and digital reading, providing new affordances for knowledge work, education, and collaborative environments.

2.2 AUGMENTED PAPER

2.2.1 *Augmented Paper Systems*

Augmented paper systems integrate digital functionality into traditional paper-based workflows, enabling users to interact with printed materials in novel ways. Unlike fully digital solutions, which attempt to replace paper, augmented paper systems preserve paper's affordances—its portability, flexibility, and ease of annotation—while embedding computational enhancements. The core challenge in this field has been designing systems that allow users to benefit from digital augmentation without disrupting established paper-based practices.

One of the most prominent examples of augmented paper in professional workflows is the *Ariel* system, which was developed as part of the EuroCODE initiative to support engineers working on large infrastructure projects [23]. Engineers on projects such as the construction of the Great Belt Bridge in Denmark relied on paper blueprints for their flexibility and portability, but paper-based documentation often resulted in inconsistencies and information loss due to its static nature. Ariel sought to bridge the gap between traditional engineering documentation and digital systems by introducing projection-based augmentation that allowed engineers to see digital overlays directly on their paper blueprints. The system also implemented barcode-based synchronisation, linking physical drawings to digital records, thereby ensuring that any changes made on paper were reflected in the digital database. Additionally, Ariel facilitated remote collaboration by integrating video conferencing tools that allowed engineers to discuss blueprint modifications in real-time while working with physical documents. Studies of Ariel's use in engineering workflows demonstrated that augmented paper could effectively enhance traditional practices without requiring professionals to abandon their preferred work methods.

In scientific research, augmented paper has also been explored as a means of improving documentation and data management. Mackay et al.'s *The Missing Link* project, examined how augmented laboratory notebooks could be used to integrate handwritten notes with digital systems [24]. Biologists rely on handwritten laboratory notebooks not only for personal record-keeping but also for legal documentation of experimental procedures and intellectual property. However, traditional notebooks present

challenges in terms of searchability, data retrieval, and collaboration. The *a-book*, developed as part of this research, combined a graphics tablet with a PDA-based Interaction Lens that allowed users to browse, annotate, and retrieve information both physically and digitally. By capturing handwriting in real time and linking physical notes to electronic databases, the system provided an effective hybrid model that retained the advantages of paper while incorporating digital search and indexing functionalities.

2.2.2 *Augmented Paper in Education*

Educational applications of augmented paper have been a major focus of research, as traditional paper-based learning methods continue to coexist with digital tools. Unlike fully digital learning environments, which often require fundamental shifts in teaching methods, augmented paper allows teachers and students to maintain familiar workflows while incorporating digital enhancements. A comprehensive review of augmented paper systems in educational settings was conducted by Prieto et al. [30], who examined the challenges of integrating digital-paper hybrids within classroom settings from an orchestration perspective.

One of the key insights from this research was that the success of augmented paper systems in education depends heavily on their ability to empower teachers while minimising additional workload. Teachers must be able to easily manage augmented materials, monitor student engagement, and seamlessly transition between traditional and augmented content without technical barriers. Many systems have struggled with this balance, as they either introduce too much complexity or fail to integrate smoothly with existing educational practices. Another crucial factor is flexibility—augmented paper should allow students to transition between individual work, group activities, and teacher-led instruction without rigid constraints. Systems that impose predefined structures on paper-based activities are often met with resistance from educators who rely on adaptability in their lesson plans. Finally, minimalism is a significant consideration; augmented paper interfaces should be simple and non-intrusive to encourage adoption. Technologies that require extensive setup or dramatically alter the appearance of printed materials tend to face challenges in real-world classroom environments.

A practical implementation of augmented paper in education is seen in the *Paper++* system, which was designed to create dynamic links between printed and digital content [22]. This system allowed students to interact with educational materials by pointing to printed images or text, which then triggered related multimedia content on a digital device. By maintaining the affordances of traditional textbooks while enabling seamless access to supplementary digital resources, Paper++ exemplified how augmented paper could support learning without disrupting the natural interaction between students and printed materials.

The role of augmented paper in educational contexts is further explored in Vincent [36] study on students' use of paper and pen versus digital media in university environments. This cross-cultural research examines how students from various countries engage with paper-based and digital materials in academic settings, highlighting their preferences and perceptions of each medium.
Vincent [36] finds that, despite the increasing availability of digital tools, students continue to value paper for its cognitive benefits, such as improved comprehension, focus, and memory retention. The study reveals that students frequently use a hybrid approach, integrating both digital and paper-based methods in their learning practices. For instance, many students prefer handwriting notes for better recall, even if they later digitise them for organisation and searchability. Furthermore, the study identifies a strong emotional and aesthetic attachment to paper, as students associate it with deeper engagement and reduced distractions compared to digital screens. These findings align with prior research on augmented paper, reinforcing the notion that hybrid learning environments that balance digital convenience with paper's cognitive advantages can optimise educational outcomes.

Another important advancement in augmented paper for education, was made by Jung et al. in their research on *Multiple Page Recognition and Tracking for Augmented Books* [17]. Traditional augmented books often rely on fiducial markers such as barcodes or visual tags for tracking, which can be visually intrusive and limit their usability. Jung and his colleagues developed a markerless tracking method that allowed for the simultaneous recognition of multiple pages in augmented books. This significantly improved the usability of augmented educational materials by enabling natural interactions with printed

content while maintaining high tracking accuracy.

Recent research has also demonstrated the potential of augmented paper systems in therapy and educational interventions for children with Autism Spectrum Disorder (ASD). The *Audio-Augmented Paper* system [2] allows children and therapists to record and interact with audio using standard sheets of paper enhanced with tangible tools. This method has been found to improve engagement, attention control, and cooperative behaviours in therapy sessions, providing a tangible and interactive medium that bridges conventional paper worksheets and multi-sensory digital learning tools.

2.2.3 *Interaction Models and Challenges*

A recurring theme in augmented paper research is the *preservation of paper's inherent affordances* while extending its capabilities. Unlike purely digital solutions, *augmented paper emphasises cognitive offloading, spatial organisation, and tangible manipulation*, which are critical for effective knowledge work.

The concept of discontinuity plays a critical role in augmented paper interactions. Research from *The Mysterious Whiteboard* [19] emphasises that augmented surfaces should *embrace discontinuities rather than remove them*. Whiteboard-based workspaces thrive on their ability to transition between different contexts, allowing content to evolve dynamically rather than being constrained to a rigid digital format. This principle has implications for augmented paper, reinforcing the need to design systems that enhance transitions between physical and digital spaces without disrupting natural workflows.

Research has identified several *key challenges*:

- **Preserving Whiteboard Affordances:** Research from *The Mysterious Whiteboard* [19] emphasises that augmented surfaces should *embrace discontinuities rather than remove them*. The ability to transition content between different contexts in physical space plays a key role in creativity and problem-solving.
- **Hybrid Interfaces:** Augmented paper systems should consider the balance between analogue and digital interactions [23]. The challenge lies in ensuring that digital annotations and interactions do not disrupt the cognitive benefits of tangible paper-based work.

- **Interaction Complexity:** Systems that integrate augmented paper often require additional hardware (e.g., tracking cameras, styluses, or sensors). Studies such as [2] highlight the technical complexity of achieving seamless interactions without introducing usability barriers.

2.2.4 *Augmented Paper and Augmented Reality*

Augmented Reality (AR) has played a crucial role in shaping augmented paper research, as it provides a means of overlaying digital content onto physical objects. Mackay et al.'s research on *Augmenting Reality, Adding Computational Dimensions to Paper* explored a range of AR-based techniques for enhancing paper-based workflows [22]. These included interactive digital annotations, embedded sensor tracking, and projection-based interactions. The core principle of this research was that augmented paper should function as a natural extension of traditional paper use, allowing users to transition seamlessly between physical and digital tasks.

Similar principles were applied in the *Ariel* system, which used AR to enhance engineering drawings. By allowing engineers to interact with projected overlays while maintaining their physical blueprints, the system demonstrated how augmented paper could be used in highly technical fields without requiring a shift to fully digital workflows.

2.2.5 *Recognition and Tracking in Augmented Paper*

One of the major technical challenges in augmented paper research has been the development of reliable recognition and tracking mechanisms. Early systems relied on fiducial markers such as barcodes and RFID tags, but these often disrupted the natural use of paper. Recent advances, such as Jung et al.'s GRF-based multi-page recognition method, have made significant improvements in enabling markerless tracking [17].

Similarly, *The Missing Link* project explored methods for tracking laboratory notebook content, employing a combination of barcode recognition and graphical interfaces to link handwritten notes with digital records. These advancements highlight the ongoing effort to create seamless augmented paper systems that maintain the flexibility and usability of traditional paper while integrating digital functionality.

2.2.6 *Cognitive Load and Usability in Augmented Paper Systems*

A central challenge in augmented paper research is ensuring that digital enhancements do not introduce unnecessary *extraneous cognitive load*, which can hinder usability and efficiency. While augmented paper aims to preserve the natural affordances of print, the integration of digital overlays, interactive elements, and tracking mechanisms requires users to cognitively switch between analogue and digital modes of interaction. This transition can impose additional mental effort, particularly when digital elements disrupt established reading or annotation habits.

Studies on human-computer interaction suggest that cognitive offloading—The process of offloading cognitive tasks onto external representations is a fundamental mechanism in knowledge work, as it reduces cognitive load, enhances problem-solving, and enables the manipulation of complex information structures [18]. Traditional paper enables efficient cognitive offloading by allowing users to spatially organise information, annotate freely, and use tactile markers for navigation. However, augmented paper systems may unintentionally increase *intrinsic cognitive load* if users must frequently adjust their interaction strategies to accommodate digital augmentation.

For example, research on engineering workflows using the *Ariel* system [23] found that while digital overlays improved access to real-time data, users experienced occasional disruptions in task flow due to system latency and inconsistencies between digital and paper-based updates. Similarly, in educational settings, *Paper++* [22] introduced dynamic links between printed text and multimedia content, but studies revealed that students faced challenges when managing transitions between physical textbooks and digital screens. This aligns with findings in digital reading research, which indicate that frequent shifts between print and screen-based modalities can increase *germane cognitive load*, affecting comprehension and engagement [21].

In an extensive review of writing tools, Wollscheid, Sjaastad, and Tømte [37] highlight how digital devices and pen-and-paper media impose different cognitive demands on learners, especially for younger or developing writers. Their findings suggest that while digital interfaces can improve efficiency and provide quick editing options, they may simultaneously hamper deeper processing and long-term retention by diluting the sensorimotor feedback inherent in handwriting. This aligns with the no-

tion that rapid toggling between physical and digital spaces can disrupt cognitive rhythms and undermine the benefits of traditional paper-based workflows.

To mitigate cognitive overload in augmented paper systems, designers must prioritise seamless interaction models that reduce the need for excessive attentional shifts. Features such as *passive augmentation*, where digital enhancements remain unobtrusive until actively triggered, and *context-aware interfaces* that adapt to user behaviour, have been proposed as solutions [19]. Additionally, providing customisation options—such as adjustable overlay intensity, annotation syncing preferences, and simplified interface controls—can help users tailor the system to their cognitive needs.

By integrating principles from cognitive load theory, augmented paper systems can be designed to enhance traditional workflows without introducing unnecessary mental strain. This approach ensures that hybrid paper-digital interactions remain intuitive, allowing users to benefit from digital functionality while maintaining the cognitive efficiency of print-based work.

2.2.7 *Summary of Augmented Paper*

In summary, augmented paper systems bridge the gap between traditional paper-based workflows and digital functionalities, preserving the inherent benefits of physical paper while enhancing interactivity. Research in this field has demonstrated its value in professional, scientific, and educational contexts, providing tools that facilitate collaboration, knowledge retention, and accessibility without disrupting existing practices. While augmented paper enables seamless transitions between analogue and digital spaces, challenges such as interface complexity, usability, and tracking mechanisms remain central to its development. Advances in Augmented Reality (AR) and recognition technologies continue to refine these systems, emphasising non-intrusive, intuitive interactions that support natural workflows. As research progresses, augmented paper holds significant promise for enhancing productivity, learning, and accessibility in a variety of domains.

Research on augmented paper highlights the increasing importance of hybrid workflows, where users dynamically transition between physical and digital tools to optimise their tasks. However, augmented paper is rarely used in isolation. Instead, it

becomes part of a broader artifact ecology—a network of interrelated tools and technologies that individuals configure based on their needs. Understanding how users navigate and adapt their personal tool ecosystems is crucial for designing seamless, context-aware hybrid interactions. The next section explores artifact ecologies, examining how individuals and groups orchestrate multiple tools, particularly in knowledge work and educational settings.

2.3 ARTIFACT ECOLOGIES

Digital and physical tools rarely exist in isolation; rather, they form interconnected systems that evolve to meet users' needs. The concept of artifact ecologies describes the dynamic configurations of tools, both digital and analogue, that individuals and communities rely on to perform tasks, manage information, and support cognitive processes [6]. Unlike traditional studies of single devices or interfaces, research on artifact ecologies focuses on how people orchestrate multiple tools within their workflows, transitioning fluidly between physical and digital spaces.

2.3.1 *Artifact Ecologies in Knowledge Work*

In professional and academic settings, individuals construct hybrid ecologies of tools that allow them to balance the affordances of analogue and digital media. The emergence of augmented paper systems (discussed in Section 2.2) reflects a broader trend in knowledge work: instead of replacing traditional media, professionals and students integrate paper-based and digital tools to leverage their respective advantages. Studies of engineers, for example, show that hand-drawn sketches and physical documents remain essential even in highly digitalised workflows [23]. Similarly, research on academic note-taking has highlighted that students frequently combine handwritten and digital notes, using handwriting for conceptual engagement and digital tools for organisation and searchability [36].

Artifact ecologies offer a lens through which to examine these transitional workflows, where physical and digital artifacts are continuously reconfigured based on task demands. Bødker and Klokmose [5] introduced the concept of non-deterministic ecologies, highlighting how users shape their toolsets through an ongoing process of adaptation rather than adhering to predefined structures. This adaptability is particularly crucial in collaborative environments, where shared artifact ecologies emerge through collective practices, such as co-annotation of documents or hybrid meeting setups that involve both physical whiteboards and digital screens.

2.3.2 *Artifact Ecologies and Neurodivergent Users*

In my recent research, I extended the concept of artifact ecologies to neurodivergent populations, particularly in education and workplace settings. Neurodivergent students often develop personalised tool ecologies that support their unique cognitive strategies, incorporating a mix of physical, digital, and assistive technologies [27]. For example, students with ADHD may use physical planners for externalising memory, while simultaneously employing digital reminders and note-taking applications for long-term organisation. Similarly, students with autism may rely on structured paper-based materials to reduce sensory overload, while using selective digital tools for information retrieval and flexibility.

Artifact ecology research in neurodivergent contexts underscores the importance of flexibility, customisation, and redundancy in tool use. Unlike standard digital workflows, which often assume a uniform cognitive style, neurodivergent users tend to develop layered and adaptive artifact configurations, switching between paper and digital tools based on task demands and cognitive load. This has implications for the design of augmented paper systems, suggesting that rather than enforcing rigid paper-digital transitions, such systems should prioritise user-controlled augmentation, allowing for gradual adaptation rather than full replacement of existing workflows.

2.3.3 *Challenges in Hybrid Artifact Ecologies*

Despite their advantages, hybrid artifact ecologies introduce several design challenges:

- **Fragmentation and Synchronisation:** One of the primary difficulties in maintaining hybrid paper-digital workflows is the synchronisation of analogue and digital artifacts. While augmented paper solutions aim to bridge this gap, many current systems require manual integration between handwritten notes and digital records, creating friction in the workflow.
- **Cognitive Load and Task Switching:** Research on cognitive offloading suggests that hybrid workflows can either reduce or increase cognitive load, depending on how well transitions between tools are managed [18]. Frequent switching between physical and digital modalities may in-

troduce extraneous cognitive load, particularly when tools require different interaction paradigms.

- **Sociotechnical Barriers:** Collaborative artifact ecologies often involve heterogeneous tool preferences, where different users rely on different combinations of paper and digital tools. This can lead to incompatibility issues, particularly in institutional settings where standardised digital systems (e.g., Learning Management Systems) are expected to replace traditional paper workflows.

2.3.4 *Designing for Adaptive Artifact Ecologies*

Given the complexity of modern artifact ecologies, future research in augmented paper systems should emphasise flexibility, user control, and seamless integration. Some key design principles include:

- **Cross-Device Continuity:** Ensuring that physical and digital artifacts remain linked, even as users transition between different contexts (e.g., classroom, workplace, home).
- **User-Driven Augmentation:** Allowing users to control when and how digital enhancements appear, rather than imposing automatic transitions that disrupt workflows.
- **Support for Neurodivergent Workflows:** Recognising the diversity of cognitive styles and ensuring that augmented paper solutions provide customisable interfaces that accommodate different sensory and organisational needs.

By addressing these challenges, artifact ecology research can inform the development of more inclusive and adaptable augmented paper systems, supporting fluid transitions between paper and digital tools in diverse user populations.

While artifact ecologies provide insight into how users configure and transition between different tools, they do not fully capture how these interactions unfold at a cognitive and physical level. Many knowledge practices—such as writing, sketching, or manipulating objects—are inherently embodied, relying on spatial, motor, and sensory engagement. Emerging research on embodied learning and interaction suggests that digital augmentation should not only integrate with existing workflows but also align with the way humans naturally think, move, and engage with physical artifacts. The following section examines how embodied cognition theories inform the design of hybrid

digital-physical systems, emphasising gesture, spatial reasoning, and tangible interactions in augmented environments.

2.4 EMBODIED LEARNING AND AUGMENTED INTERACTIONS

Embodied approaches to learning emphasise the idea that cognition emerges not solely in the mind, but through continuous, reciprocal interactions between body, mind, and environment. In the context of augmented paper systems and broader paper-digital workflows, embodiment offers a lens to understand how physical actions, gestures, and sensory experiences shape the learning process.

2.4.1 *Embodied Design in Augmented Learning*

Abrahamson et al. [1] explores the concept of *embodied design*, illustrating how movement and sensorimotor interactions can ground complex ideas, particularly in STEM education. Rather than viewing learning as an internal process of storing and retrieving abstract symbols, Abrahamson stresses that meaningful engagement arises from opportunities to *enact* underlying concepts physically. In mathematics, for instance, students might perform carefully designed motions that reflect mathematical relationships (e.g., proportional reasoning or geometry). By integrating these bodily experiences with more conventional representations—like symbolic notation—students can build richer, more transferable knowledge.

Applied to augmented interactions, *embodied design* suggests that digital overlays and sensor-based feedback can amplify learners' awareness of critical concepts. For example, a system that tracks hand movements as users manipulate paper-based diagrams could provide real-time visual or auditory feedback, enabling learners to "see" the correspondence between their embodied actions and the conceptual content. Such systems blend the tangibility of paper with digital enhancements that highlight key mathematical or scientific properties. This approach helps maintain the cognitive and sensory advantages of physical interaction while leveraging dynamic affordances of digital media—aligning well with the objective of *augmented paper* solutions.

A key insight from Abrahamson et al. [1] is the potential for *deliberate movement forms* to trigger conceptual insights. Students who physically manipulate objects or move their bodies in choreographed sequences are effectively exploring the conceptual terrain at a bodily level. Through reflection and facilitation, they gradually connect these movement patterns to formalised dis-

ciplinary knowledge. When applied to augmented paper, such movement-based tasks might involve highlighting critical dimensions on printed diagrams or dynamically altering sketches on a tablet overlay, ensuring that motor actions and conceptual reflection evolve in tandem.

2.4.2 *Embodied Cognition and Paper-Based Interactions*

Expanding on embodiment in education, Flood [12] offers a broader interdisciplinary synthesis, drawing from psychology, anthropology, and learning sciences to illustrate how *embodied cognition* shapes academic contexts. Within paper-digital ecosystems, Flood argues that a purely digital approach can neglect the sensory richness and motor engagement inherent to paper. Embodied cognition posits that learners do not simply receive information; they enact and co-construct knowledge through bodily, social, and material interactions.

In paper-based workflows—augmented or otherwise—*touch, gesture, and spatial arrangement* of notes serve as crucial external resources, grounding learners' mental activities. Flood [12] underscores that when students manually annotate, highlight, or rearrange paper materials, they anchor cognitive processes in embodied acts. These physical marks act as external scaffolds, supporting memory retention, comprehension, and problem-solving.

When merging these paper affordances with digital augmentation, educators and system designers must attend to how technology modulates learners' embodied experiences. For instance, a digital overlay that disrupts the natural tangibility or haptic feedback of paper may inadvertently diminish the advantages associated with tactile manipulation. Conversely, subtle augmentations—such as layering context-sensitive hints or interactive hotspots—can reinforce embodied comprehension by directing student movement to relevant areas of a text or diagram without overshadowing the inherent strengths of paper.

Flood [12] also emphasises the social dimensions of embodiment. Classroom environments often rely on face-to-face dialogue, collaborative material manipulation, and gestures to negotiate meaning. Augmented paper can mediate these interactions by capturing or enhancing these gestures, thus enriching group-based cognition rather than replacing it with purely individual, screen-based tasks.

2.4.3 *Synthesis: Bridging Embodiment and Augmented Paper*

Together, Abrahamson et al. [1] and Flood [12] reveal how an embodied perspective can inform the design of hybrid paper-digital environments. *Embodied design* encourages carefully orchestrated movement-based activities that naturally integrate with conceptual tasks, while *embodied cognition* highlights the fundamental role of sensory and bodily processes in all learning. By merging these perspectives, augmented paper solutions can:

1. **Preserve Tangibility:** Retain the haptic and spatial affordances of paper-based tasks that support cognitive offloading.
2. **Encourage Deliberate Movements:** Incorporate directed or guided physical interactions that align with conceptual objectives, drawing on choreographed gestures or manipulation.
3. **Enable Social Collaboration:** Support group-based sense-making, capturing and amplifying gestural communication in real time.
4. **Integrate Seamless Feedback:** Use unobtrusive digital cues to guide attention without overwhelming the embodied, paper-centric experience.

By foregrounding the body's role in knowledge construction, these embodied theories emphasise that effective augmented paper systems should not simply overlay digital functions onto static pages, but rather cultivate meaningful physical interactions that heighten learners' conceptual understanding. In doing so, educators can leverage both the cognitive benefits of print media and the flexibility of digital enhancements, forging a more holistic environment for interactive, embodied learning.

While embodied learning highlights the importance of physical interactions, gestures, and spatial engagement in cognition, learning experiences are further shaped by individual cognitive differences. Neurodivergent individuals, including those with autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD), often process information in unique ways, requiring tailored educational approaches. The interplay between sensory perception, attention regulation, and cognitive organisation influences how these learners interact with digital and print-based materials, as well as embodied and augmented

learning environments. The following section explores neurodivergent learning processes, examining how systemising tendencies, hyperfocus, sensory modulation, and kinesthetic learning preferences shape engagement with educational content and technology. By understanding these cognitive profiles, we can design more inclusive learning environments that support both traditional and digital learning modalities.

2.5 NEURODIVERGENT LEARNING PROCESSES

Neurodivergent learners experience unique cognitive and sensory processing differences that shape how they engage with reading materials, digital tools, and collaborative learning environments. Variations in sensory perception, attention regulation, and cognitive organisation contribute to diverse learning styles and needs. Research in Autism Spectrum Disorders (ASD) and attention deficit hyperactivity disorder (ADHD) highlights distinctions in systemising tendencies, sensory modulation, hyperfocus, and kinesthetic learning preferences. This chapter synthesises findings from relevant literature to provide a foundation for understanding how neurodivergent individuals interact with educational material and how pedagogical approaches can be adapted to support their learning effectively.

2.5.1 *Hyper-Systemising and Cognitive Processing in ASD*

Structured Learning and Knowledge Representation

Research on Autism Spectrum Disorder (ASD) indicates a pronounced tendency for hyper-systemising, a cognitive style characterised by a strong drive to analyze and construct systems based on identifying rules and patterns [4]. Systemising involves recognising predictable relationships within structured domains, such as mathematics, programming, and taxonomy, by applying "if p, then q" reasoning to understand and predict how systems behave. This cognitive approach enables individuals with ASD to excel in highly structured environments where rules and patterns are consistent, but it can also present challenges in tasks requiring flexible, high-level categorisation. Unlike neurotypical learners, who often rely on prototype-based categorisation, hyper-systemisers focus on exhaustive detail. For example, rather than summarising that "a beetle is an insect," a hyper-systemiser might attempt to list thousands of beetle species, providing precise taxonomic classifications [4].

Traditional assessment methodologies often fail to accommodate this cognitive processing style, as standard evaluations, such as timed tests and essay-based assignments, tend to emphasise summarisation rather than detailed systemisation. Hyper-systemisers may struggle with broad, open-ended questions but perform exceptionally well in structured problem-solving tasks that require identifying precise rules or systematic relationships. To support these learners, educators should consider

alternative assessment strategies, such as structured problem-solving tasks, applied demonstrations, or scaffolded frameworks that allow for systematic exploration of concepts rather than requiring broad generalisations. These approaches align better with the cognitive strengths of hyper-systemising individuals while ensuring fair and accurate assessments of their knowledge.

Sensory Processing and Learning Environments

Sensory hypersensitivity is another defining characteristic of ASD, affecting attention regulation and engagement in educational settings [35]. Research has documented enhanced perceptual acuity, with some individuals exhibiting visual discrimination at levels comparable to birds of prey (Snellen score of 20:7) [4]. However, this heightened sensory perception often results in sensory overload, leading to difficulties in conventional classroom environments.

Accommodations for sensory differences include modifying environmental factors such as lighting, noise levels, and tactile inputs. Sensory-friendly classrooms that incorporate noise-canceling devices, flexible seating, and structured visual layouts can mitigate overstimulation and enhance information retention [35].

2.5.2 *Hyperfocus in ADHD and Its Educational Implications*

Mechanisms of Hyperfocus

Contrary to traditional conceptions of ADHD as a deficit in attention, recent research suggests that it represents a dysfunction in attentional regulation rather than a lack of focus [14]. Hyperfocus (HF) is characterised by prolonged engagement with specific tasks, often at the expense of peripheral awareness and task-switching ability. While this cognitive state can enhance productivity and knowledge acquisition, it presents challenges when task transitions are required.

Pedagogical Approaches to Harnessing Hyperfocus

Educational strategies designed to align with HF tendencies include:

- **Interest-Based Learning**: Structuring curricula around subjects that naturally engage the student's focus.

- **Flexible Pacing**: Allowing extended time for deep work while providing structured transition cues.
- **Gamified Learning Environments**: Utilising interactive software to sustain engagement and mitigate off-task behaviour [28].

Studies indicate that game-based learning frameworks significantly improve engagement and reduce distractibility in students with ADHD, making them a viable supplement to traditional instruction [28].

2.5.3 *Kinesthetic Learning and its Impact on Cognitive Retention*

The Role of Movement in Learning

Kinesthetic learning approaches have been found to be particularly beneficial for neurodivergent students, as they integrate movement into the cognitive process. Research suggests that physical activity enhances executive function, memory retention, and task engagement, particularly in individuals with ADHD [13].

Implementation of Kinesthetic Learning Techniques

Common kinesthetic learning strategies include:

- **Flexible Seating and Movement Integration**: Incorporating wobble stools, resistance bands, and standing desks.
- **Embodied Learning**: Using role-play, hands-on activities, and interactive movement-based tasks to reinforce conceptual understanding [16].

Studies indicate that movement-based pedagogical approaches lead to improved concentration, reduced anxiety, and higher retention rates, making them particularly relevant for students with attention regulation difficulties [13].

2.5.4 *Collaborative Learning and Digital Adaptations*

Group Work Challenges and Strategies

Collaborative learning presents unique challenges for neurodivergent students, particularly in STEM fields. Difficulties in communication, group conflict, and workload distribution can create barriers to effective participation [31]. Additionally, fast-paced discussions may exclude neurodivergent students from

key decision-making processes.

Strategies for facilitating neurodivergent-friendly group work include:

- **Structured Role Assignments**: Clearly defined roles help mitigate confusion and ensure equitable participation.
- **Flexible Interaction Formats**: Providing multiple modes of communication, such as written input or asynchronous discussions, allows for diverse engagement.
- **Instructor-Led Mediation**: Faculty support in setting clear expectations and resolving conflicts enhances inclusivity [31].

Digital Learning and Assistive Technologies

Advancements in digital learning platforms offer both opportunities and challenges for neurodivergent students. While digital interactions may increase cognitive load, adaptive technologies can improve accessibility.

Key digital accommodations include:

- **Customisable Text Formats**: Adjustable fonts, text spacing, and colour overlays improve readability for students with dyslexia [10].
- **Assistive Tools**: Speech-to-text, text-to-speech, and AI-based predictive typing applications reduce cognitive barriers in writing-intensive tasks [9].
- **Multisensory Digital Learning**: Interactive simulations, haptic feedback, and gamified content enhance engagement and retention [16, 28].

Emerging technologies such as virtual reality (VR) and augmented reality (AR) show promise in fostering immersive, structured learning experiences tailored to neurodivergent cognitive styles.

2.5.5 *Neurodivergent Reading Challenges in Digital Contexts.*

Research on digital versus print reading has shown that digital reading environments often introduce increased cognitive load, decreased spatial navigation cues, and fragmented reading experiences due to distractions and scrolling interruptions. These

challenges can be particularly pronounced for neurodivergent individuals, whose cognitive profiles influence how they engage with text.

Autism Spectrum Disorder (ASD) and Digital Reading

Autistic individuals often exhibit hyper-systemising cognitive styles, meaning they process information through structured, rule-based frameworks rather than flexible, high-level inference [4]. This preference for detail-oriented processing can lead to difficulties with inferential comprehension—the ability to integrate information across paragraphs and make abstract connections. Digital reading, which often promotes skimming behaviours and fragmented text structures (e.g., hyperlinked documents, scrolling interfaces), can exacerbate comprehension difficulties by reducing access to spatial markers that support text organisation [25].

Furthermore, sensory sensitivities commonly experienced by autistic readers—such as aversion to bright screens, inconsistent text formatting, or eye strain from excessive scrolling—can increase extraneous cognitive load, reducing their ability to maintain attention on digital texts. Many autistic students report a preference for structured, visually clear print materials over fluid digital text, as physical books provide a stable reading experience that aligns with their cognitive processing strengths [21].

To support autistic readers in digital environments, customisable formatting options such as adjustable text size, dyslexia-friendly fonts, and controlled scrolling can help mitigate sensory distractions and enhance comprehension. Additionally, providing structured, linear reading interfaces—such as paginated e-books rather than infinite-scroll web pages—can improve retention and reduce cognitive fatigue.

ADHD and Digital Reading

In contrast to ASD, individuals with attention-deficit/hyperactivity disorder (ADHD) face different but equally significant challenges when engaging with digital reading materials. ADHD is characterised by difficulties in sustained attention, working memory, and impulse control, all of which impact reading comprehension [14].

Digital reading environments, particularly those that encour-

age multitasking or contain embedded distractions (e.g., hyperlinks, notifications, interactive elements), can worsen attentional drift and make it harder for students with ADHD to maintain focus on the text. Studies indicate that students with ADHD exhibit more erratic eye-tracking behaviours when reading on screens, suggesting increased cognitive effort to regulate focus compared to print reading [15]. This results in fragmented comprehension, where readers struggle to maintain a coherent understanding of the text due to frequent lapses in attention.

However, digital reading also presents unique affordances that can be beneficial for ADHD learners. Features such as text-to-speech tools, interactive annotation functions, and structured note-taking integrations can help compensate for working memory deficits and enhance engagement [9]. Additionally, chunking text into smaller sections, incorporating gamified reading interventions, and utilising immersive reading modes (e.g., focus view tools that remove distractions) can help ADHD readers maintain attention and improve retention.

Comparing Digital Reading Challenges in ASD and ADHD

While both autistic and ADHD readers experience difficulties with digital reading, the underlying causes differ:

- Autistic readers struggle with spatial disorientation, sensory overload, and inferential comprehension, making digital reading environments with excessive scrolling, hyperlinks, and unstructured layouts challenging.
- ADHD readers struggle with sustained attention, impulsivity, and working memory, making digital environments with distractions, multitasking demands, and cognitive overload difficult to navigate.

To optimise digital reading environments for neurodivergent students, flexible assistive tools, structured layouts, and sensory-friendly design principles should be incorporated. For autistic readers, reducing visual clutter and stabilising text presentation enhances comprehension, whereas for ADHD readers, incorporating focus-enhancing features and interactive engagement tools improves retention.

2.5.6 *Summary of Neurodivergent Learning Processes*

Neurodivergent learning is influenced by distinct cognitive processing styles, attentional regulation mechanisms, and sensory profiles. Autistic individuals demonstrate strengths in systemising and pattern recognition but may require structured learning environments to mitigate sensory overload. ADHD learners exhibit hyperfocus and benefit from interactive, interest-driven educational approaches, yet often struggle with task transitions. Sensory-friendly and kinesthetic learning strategies, including adaptive seating and multimodal instruction, enhance engagement for both groups. Collaborative learning presents additional challenges, requiring structured role assignments and inclusive interaction formats to support neurodivergent students. Digital tools, including assistive technologies and multisensory learning platforms, further enhance accessibility and engagement. By integrating structured, flexible, and strengths-based educational methods, learning environments can be optimised to support neurodivergent students effectively.

2.6 SUMMARY OF RELATED WORK

The growing prevalence of digital reading has prompted extensive research comparing it to traditional print reading, focusing on comprehension, engagement, and cognitive load. Studies consistently highlight the cognitive advantages of print reading, particularly in fostering deeper comprehension, stable reading patterns, and stronger spatial navigation. Print reading supports better information retention due to its tactile nature, while digital reading, though more accessible and convenient, is often associated with increased distractions and fragmented reading behaviours. Despite a growing adaptation to digital reading among students, particularly at the university level, research indicates a continued preference for print for deep learning tasks.

A key concern in digital reading is the increased cognitive load, particularly for neurodivergent individuals, who may struggle with navigating online texts, dealing with information overload, and processing dense digital content. For individuals with Autism Spectrum Disorder (ASD), digital reading often lacks the structured, stable layout needed to support inferential comprehension and spatial organisation. Those with ADHD, on the other hand, face challenges with sustaining attention in digital environments due to distractions, scrolling interruptions, and multitasking demands. Implementing structured and customisable digital reading environments can help mitigate these challenges, enhancing accessibility for neurodivergent learners.

In response to these findings, researchers have explored hybrid solutions, particularly through augmented paper systems, which aim to preserve the affordances of print while integrating digital enhancements. Augmented paper technologies facilitate seamless transitions between physical and digital interactions, offering enhanced accessibility, annotation capabilities, and multimodal integration. Applications of augmented paper have been explored across professional, educational, and scientific domains, demonstrating potential in improving knowledge retention, collaboration, and accessibility. However, challenges remain in ensuring that digital augmentation does not introduce excessive cognitive load or disrupt established paper-based workflows.

The broader perspective of artifact ecologies provides insight into how individuals dynamically configure and transition between different tools in their workflows. Rather than relying

solely on digital or print media, knowledge workers and students orchestrate hybrid systems that optimise efficiency and engagement. This adaptive approach is particularly evident among neurodivergent individuals, who construct personalised artifact ecologies to accommodate their unique cognitive and sensory needs. Research highlights the importance of designing flexible, user-driven augmented paper systems that align with real-world hybrid workflows.

Additionally, embodied cognition theories provide a valuable framework for understanding how physical interaction, spatial organisation, and movement influence learning and engagement. Augmented paper solutions that incorporate embodied interaction—such as gesture recognition, tangible manipulation, and spatial tracking—can enhance cognitive engagement and retention. By integrating these principles into augmented learning environments, researchers aim to bridge the gap between digital flexibility and the cognitive benefits of physical interaction.

Ultimately, research across digital reading, augmented paper, artifact ecologies, and embodied cognition converges on the importance of designing systems that align with human cognitive processes. Effective hybrid solutions should leverage the strengths of both digital and print mediums while minimising cognitive barriers and enhancing usability. This literature review lays the groundwork for the methodological approach in the next chapter, where I evaluate augmented paper and hybrid interaction designs aimed at fostering inclusivity. Through a diary study followed by semi-structured interviews, I seek to formulate and empirically validate design principles that inform future innovations in hybrid reading and learning environments.

2.7 DESIGN PRINCIPLES FOR HYBRID LEARNING ENVIRONMENTS

Building on the literature reviewed across digital reading, augmented paper, artifact ecologies, embodied interaction, and neurodivergent learning, this chapter concludes by outlining a consolidated set of design principles that synthesise insights from these domains. These principles serve a dual role: they offer a normative framework for inclusive hybrid reading environments and function analytically as sensitising concepts in the empirical phases that follow.

Digital Reading Design Principles

1. **Support diverse reading goals and attentional states** — Enable transitions between deep focus and overview via distraction-free modes and structurally adaptive views.
   Rationale: Digital reading contexts are marked by multitasking and fragmented attention. Readers, particularly those with ADHD, benefit from interfaces that accommodate hyperfocus as well as periods of low engagement. Literature on attentional regulation and cognitive pacing suggests that interfaces should support toggling between immersive deep-reading modes and skimmable overview states [21, 26].

2. **Centre accessibility and personalisation** — Ensure user control over sensory input, reading style, and assistive tools; keep systems simple and compatible with cognitive diversity.
   Rationale: Neurodivergent users vary widely in sensory profiles and cognitive styles. Research on ASD and dyslexia highlights the importance of customisable text formatting, adjustable visual density, and assistive technologies (e.g., text-to-speech) to reduce extraneous cognitive load [11, 29]. Personalisation ensures the system adapts to the reader—not vice versa.

3. **Restore spatial sense-making** — Use stable layouts, visual markers, and navigational cues to support spatial mapping and reading flow.
   Rationale: Print reading affords a spatially stable reading environment which enhances memory and cognitive mapping. Studies show that digital scrolling reduces spatial orientation, especially for autistic readers or those with executive function differences [15, 25]. Reintroducing stable spatial markers helps mitigate disorientation and supports linearity when needed.

4. **Enable active processing and feedback** — Provide tools for marking, summarising, and tracking reading comprehension through notes, self-tests, or feedback loops.
   Rationale: Active annotation and metacognitive monitoring foster deeper understanding and retention. Especially for learners with working memory challenges (e.g., ADHD), tools that scaffold summarisation and visual organisation (like highlighting or comprehension prompts) are essential [32, 37].

5. **Activate the body and senses** — Integrate haptics, stylus-based interaction, and touch-based gestures to bridge analogue-digital boundaries.
   Rationale: Embodied cognition literature emphasises that reading and learning are sensorimotor acts. Stylus use, touch gestures, and subtle haptic feedback enhance memory and engagement by grounding digital interactions in bodily motion [1, 12].

6. **Facilitate co-located collaborative reading** — Support synchronous shared reading using AR, whiteboards, or multi-user tablets.
   Rationale: Collaboration is key in educational settings, yet neurodivergent students often require alternative interaction formats. AR-enhanced or shared screen reading systems can enable inclusive group learning by combining shared attention with flexible participation modes [30, 36].

Augmented Paper Design Principles

1. **Preserve and amplify the role of paper** — Augment rather than replace; preserve handwriting, tactile feedback, and physical navigation.
   Rationale: Paper affords spatial, tactile, and embodied learning benefits that support cognitive offloading, particularly for neurodivergent learners. Research consistently finds that learners value paper for its sensory grounding and annotation ease [12, 24, 36].

2. **Let the user control digital augmentation** — Use passive overlays activated by embodied interaction; avoid overstimulation or automatic triggers.
   Rationale: Studies show that overactive or automatic digital augmentations disrupt embodied learning and increase cognitive load, especially for users with sensory sensitivities or executive function challenges [19, 21]. Systems should instead respond to user-initiated cues.

3. **Adapt to user needs and rhythms** — Offer varied input/output modes; adapt to energy levels, learning styles, and contexts.
   Rationale: Neurodivergent users develop adaptive tool ecologies tailored to fluctuating sensory thresholds and cognitive rhythms. Systems should support handwriting, audio feedback, and visual augmentation flexibly [6, 31].

4. **Integrate seamlessly without disruption** — Augmentation should feel like a natural extension of existing work-

flows.
Rationale: Both professional and educational users resist tools that alter their routine workflows. Seamless augmentation reduces the learning curve and cognitive overhead by allowing digital affordances to emerge only when useful [22, 30].

5. **Ensure simplicity and robustness** — Prioritise low-tech, modular, and reliable components that align with users' existing tools.
Rationale: Complex technical setups often deter use. Artifact ecology studies show that users prefer stable, interoperable tools that work across time and settings without requiring high expertise or hardware dependency [6, 17].

6. **Harness bodily engagement in learning** — Leverage movement, gesture, and tangible interaction as core learning modalities.
Rationale: Kinesthetic and embodied learning strategies support retention, executive functioning, and concept formation, particularly for ADHD learners. Augmented paper can scaffold such interactions using tracked gestures or manipulable overlays [1, 12, 13].

7. **Design with simplicity and coherence across tools** — Design for predictable interfaces and cross-platform compatibility to support users' organisational strategies.
Rationale: Users form personal ecologies of tools that evolve over time. Neurodivergent learners often layer paper, digital, and assistive tools, requiring coherence and modularity rather than all-in-one solutions [6, 21].

Design Dilemmas and Tensions

These principles are not without conflict. Several cross-domain dilemmas emerged in the review that underscore the complexity of hybrid design:

- **Activation vs. Passivity:** Digital reading encourages real-time feedback (e.g., comprehension tracking), while augmented paper emphasises user-triggered augmentation. Solutions must balance proactive support with embodied control.

- **Shared Screens vs. Paper Anchoring:** Digital environments support shared views, while paper supports shared space. Collaborative tools must bridge these modalities without undermining embodied grounding.

- **Virtual Navigation vs. Physical Manipulation:** Digital layouts offer virtual maps and scrolling, whereas paper requires tactile layouting. Hybrid systems must balance spatial metaphors and physical handling.
- **Technological Richness vs. Simplicity:** While digital systems encourage feature-rich stylus and AR interactions, augmented paper calls for minimal complexity. Designing scalable systems that flex across contexts becomes crucial.

These tensions highlight that hybrid learning environments are not merely technical systems but cognitive and embodied ecologies that must negotiate user needs, sensory capacities, and contextual expectations.

2.7.1 *Transition to Empirical Inquiry*

The design principles and dilemmas outlined here serve as the analytical scaffolding for the empirical investigation presented in the following chapters. Through a diary study and a series of semi-structured interviews with neurodivergent university students, I explore how these principles manifest—or fail to manifest—in lived experiences of hybrid reading. The next chapter introduces the methodological approach, research instruments, and participant sample, laying the foundation for a grounded, cognitive-load-informed analysis of hybrid learning ecologies.

3 METHODOLOGY

This project employs a qualitative mixed-methods design, integrating a diary study, semi-structured interviews, cognitive load analysis, and an in-depth literature review. The methodological framework was crafted to explore how neurodivergent university students engage with, adapt to, and make sense of hybrid reading environments that span physical print, digital platforms, and augmented paper systems. The aim of the study is to generate design principles for inclusive, cognitively sensitive reading tools grounded in the lived experiences of neurodivergent learners.

The study draws on theoretical perspectives from Human-Computer Interaction (HCI), cognitive science, educational theory, and neurodiversity studies. The methodology is both iterative and exploratory, allowing empirical insights to feed back into the evolving analytical framework.

3.1 RESEARCH DESIGN

The study follows a *qualitative mixed-methods* approach with three interconnected phases: (1) a literature-driven theoretical grounding, (2) a longitudinal diary study, and (3) semi-structured interviews that build on diary data. This structure enables a layered analysis that captures both macro-level patterns (e.g., artifact use and design tensions) and micro-level, embodied experiences of reading across modalities.

The design is rooted in the recognition that reading is not a neutral cognitive act but an embodied, context-dependent, and materially mediated activity. Therefore, the methods were selected to uncover both the cognitive strategies and sensorimotor practices neurodivergent students employ to manage hybrid learning environments.

3.2 LITERATURE REVIEW AND DESIGN PRINCIPLES

The study began with a comprehensive literature review structured around five thematic domains:

1. Digital vs. Print Reading Experience
2. Augmented Paper Systems
3. Artifact Ecologies
4. Embodied Learning and Augmented Interaction
5. Neurodivergent Learning Processes

Over 50 academic publications were initially reviewed, from which 33 key sources were selected to form the foundation of the RELATED WORK chapter. These texts were read critically, and recurring concepts, contradictions, and recommendations were extracted and mapped.

From this mapping process, a dual set of design principles was systematically constructed: 22 principles for digital reading and 26 for augmented paper. Through iterative synthesis and thematic refinement, these were distilled into six core design principles for digital reading and seven for augmented paper systems. Additionally, the review identified several design conflicts—situations where different principles contradicted or competed with one another—highlighting the complexity of designing inclusive tools for cognitively diverse users.

These principles and tensions were used to evaluate participant tool use in the empirical phases and formed the backbone of the interview guide and coding framework.

3.3 PARTICIPANTS

Seven neurodivergent university students participated in the study. All were enrolled in STEM programs at Aarhus University and had formal diagnoses of ADHD, ADD, and/or Autism Spectrum Disorder. Four of the participants identified as having ADHD, three as autistic, and one as having ADD. One participant had dual diagnoses (ADHD and autism).

Participants were recruited through a combination of longitudinal engagement (some had taken part in my prior research project) and peer referral. The group included students from computer science, physics, and mathematics. Six of the seven participants identified as women, and one as nonbinary. Ages ranged from 20 to 25, and academic progression varied: two were in their 4th semester, one in their 6th, two in their 8th, and two in their 10th semester.

Due to academic stress or life changes, four participants were unable to complete the full study. Three participants completed all phases and contributed diary data and interviews.

3.4 DIARY STUDY

The diary study was designed to capture the situated, reflective, and embodied study experiences of participants over time. Diaries focused on participants' use of reading tools and workflows across analogue, digital, and augmented modalities. Participants were provided with either written or verbal onboarding explaining the project goals, their role, and ethical considerations. Each participant gave informed consent in accordance with GDPR protocols and received a document outlining data storage, anonymity, and usage.

Participants were encouraged to document their study activities in a semi-structured format, including reflections on:

- Cognitive and sensory experiences
- Which tool have you used?
- How did you use it, and what did you use it for?
- In what context did you use the tool?
- Where did you use the tool?

The first item on the list was verbally added 3 days into the diary study after further refining the study's goal. A copy of the written guide can be seen in APPENDIX A: DIARY GUIDE.

3.4.1 *Digitalisation and Data Preparation*

Each diary was transcribed and digitised using LaTeX (Overleaf) to facilitate qualitative analysis. The digital versions were annotated with a structured artifact summary at the beginning of each file, listing the tools and technologies mentioned and their primary uses.

From this annotated corpus, an *artifact ecology* was generated for each participant—mapping their constellation of tools and the transitions between them. The written artifact ecologies were returned to participants to be checked and fine-tuned to the specific individual, ensuring accuracy and participant validation.

3.4.2 *Qualitative Analysis*

Diaries were analyzed using a three-phase coding process based on the grounded theory research by Corbin and Strauss [8].

Open Coding

The first step in the analysis involved open coding, a process through which the data was analytically broken down into discrete incidents, actions, and interactions. These fragments were examined comparatively and assigned conceptual labels that reflected their core meanings. Over time, similar instances were grouped into broader categories and subcategories, each of which was further specified through the identification of properties and dimensions. For example, repeated references to stress-regulation strategies might be grouped under a category such as *cognitive offloading*, with subdimensions such as duration, tool-dependency, or setting. Open coding served to challenge default assumptions and cultivate theoretical sensitivity by prompting generative questions about the conditions, forms, and implications of participant practices. This stage of analysis was iterative and exploratory, laying the foundation for subsequent relational modeling of the data.

Axial Coding

Following the open coding phase, the analysis proceeded through axial coding, in which categories were systematically related to their subcategories. This stage focused on articulating the internal structure of each category by linking conditions, contextual factors, strategies (actions/interactions), and consequences. For instance, a category such as *comfort work* (as explored in the grounded theory tradition) would be further unpacked to explore the specific circumstances that give rise to such practices, how they are enacted, and what outcomes they produce. Axial coding enabled a deeper conceptual integration by testing preliminary hypotheses derived from the open codes against the full dataset. These hypotheses were considered provisional and were continuously revised based on emerging variation and complexity. The goal of this phase was to identify underlying mechanisms and patterned relationships that could explain the data beyond descriptive levels.

Selective Coding

The final phase, selective coding, aimed to integrate the analysis by identifying a *core category*—a central concept that cap-

tured the overarching theme of the study and organised the remaining categories around it. This integrative process involved reviewing all categories and their interrelations in order to construct a cohesive theoretical narrative. Categories were positioned in relation to the core as either conditions, strategies, contextual influences, or outcomes. During this phase, poorly developed categories were identified and either elaborated through re-engagement with the data or excluded from the final theory. Theoretical memos and diagramming were used to refine the model and ensure conceptual density. Selective coding marked the transition from analytic saturation to theory generation, culminating in an abstract yet grounded representation of how neurodivergent students navigate and structure hybrid reading environments.

3.4.3 *Cognitive Load Theory as Analytical Framework*

To extend and deepen the interpretive scope of the grounded theory process, Cognitive Load Theory (CLT) was incorporated as a secondary analytical framework. Rather than functioning solely as theoretical background, CLT served as a dialogical lens during the axial and selective coding phases—providing a structured typology through which participant strategies, tool selections, and study behaviours could be interpreted in relation to their cognitive demands. Originally developed by John Sweller [33], CLT offers a model of how working memory is constrained during learning and distinguishes between three interrelated types of cognitive load: *intrinsic*, *extraneous*, and *germane*. These categories are particularly useful for understanding hybrid learning environments where sensory sensitivity, multitool navigation, and emotional regulation intersect with complex academic content.

Before detailing how CLT was analytically applied in this study, the following sections introduce each load type in turn. This step is necessary to contextualise how the theory informs the interpretation of cognitive demands in neurodivergent learners' study ecologies—laying the groundwork for the subsequent coding layer that classifies participant strategies according to these load types.

Intrinsic Cognitive Load

Intrinsic cognitive load refers to the inherent complexity of the material being learned and is closely tied to the element interac-

tivity within a task. Some concepts are more demanding than others due to their internal structure, and tasks that require deeper levels of processing—such as application, synthesis, or evaluation—naturally impose a higher intrinsic load. According to Sweller [34], intrinsic cognitive load is determined by both the nature of the content and the learner's prior knowledge. In the context of this study, intrinsic load was observed in participant reflections on challenging texts, abstract mathematical reasoning, and the cognitive demands of switching between digital and analogue reading modalities. Understanding intrinsic cognitive load is essential for evaluating how task complexity interacts with individual cognitive resources, especially for neurodivergent learners who may experience increased difficulty with certain types of conceptual processing.

Extraneous Cognitive Load

Extraneous cognitive load is generated by the way information is presented rather than by the content itself. It includes all elements of the learning environment that do not directly support the construction of knowledge—such as poorly organised materials, distracting interfaces, or irrelevant visual and textual elements. Sweller emphasises that extraneous load arises from instructional design choices that fail to align with learners' cognitive capabilities, thereby overburdening working memory [34]. In the diary study, participants frequently described moments of disorientation or distraction when interacting with cluttered interfaces, redundant text-image pairings, or multitasking between tools—each contributing to unnecessary cognitive strain. Minimising extraneous cognitive load is thus a critical concern in the evaluation of tools used in hybrid reading workflows, especially for learners with heightened sensory or attentional sensitivities.

Germane Cognitive Load

Germane cognitive load refers to the cognitive effort devoted to the construction and automation of schemas—mental structures that support long-term learning. Unlike intrinsic or extraneous load, germane load represents productive cognitive investment: it includes processes such as reflection, summarisation, elaboration, and the use of mnemonic strategies that facilitate deep learning. Although not an independent type of load in the same sense as the others, germane load captures the extent to which working memory resources are intentionally directed toward schema-building activities [34]. In this study, ger-

mane load was inferred from participant strategies that aimed at meaning-making and integration—such as the use of colour-coded annotations, handwritten summaries, or the deliberate combination of print and digital materials to reinforce comprehension. Supporting germane cognitive load through effective tool design and layout is crucial for fostering engagement and deeper understanding in neurodivergent learners.

3.4.4 *Cognitive Load Theory as a Secondary Analytical Framework*

Having established the conceptual distinctions between intrinsic, extraneous, and germane cognitive load, these categories were subsequently operationalised as analytical filters during the interpretation of coded data. Rather than functioning as static classifications, the load types served as heuristics for re-engaging with grounded theory memos and diary segments — allowing me to revisit previously identified categories through the lens of cognitive resource distribution. In this sense, CLT became a dialogical framework: its typology helped evaluate the cognitive implications of different study strategies and tool configurations, while also enabling cross-participant comparisons in terms of how neurodivergent learners adapt to complex academic demands. The following analytic layer summarises how these load categories were practically applied to the coded material.

- Assessing diary segments for signs of **intrinsic load**, such as abstract conceptual tasks, multi-step problem solving, or cumulative workload spikes;
- Identifying sources of **extraneous load**, including interface fragmentation, platform switching, sensory distraction, or spatial-context mismatches;
- Inferring **germane load** from behaviours indicating intentional schema-building, such as analogue-to-digital sequencing, summarisation strategies, or environmental optimisation for deep work.

This CLT-guided lens enabled triangulation across participants' practices and made it possible to distinguish between productive and obstructive forms of effort. It also allowed for the theorisation of individual differences in coping mechanisms, as well as the identification of overlapping affordances across artifact ecologies.

To summarise, CLT was not applied prescriptively but dialogically: its principles were used to refine grounded categories,

test thematic coherence, and structure the comparative cross-case analysis that followed. This integration contributed to a nuanced understanding of how neurodivergent students dynamically manage cognitive resources across tools, spaces, and time.

3.5 SEMI-STRUCTURED INTERVIEWS

Following the diary study, each participant engaged in an individual semi-structured interview designed not only to triangulate but to deepen and problematise the empirical insights developed during the initial data collection phase. Interviews were conducted in person and lasted between 40 and 50 minutes. All interviews were audio-recorded with participant consent and transcribed verbatim. In total, the interviews generated approximately 21,000 words of raw transcript material.

The interviews were not structured as confirmatory follow-ups but as dialogic elaborations—spaces where participants were invited to recount specific learning episodes, react to tentative interpretations from their diary data, and critically reflect on the relevance (or irrelevance) of emerging design principles. This situated, reflective mode of interviewing allowed not only for validation of prior patterns (e.g., modality switching, affective thresholds, or spatial scaffolding) but also surfaced conceptual contradictions, emotional undercurrents, and strategy rationales that had remained implicit in the diaries.

3.5.1 *Structure and Protocol*

The interview protocol was semi-structured and flexible, organised around five overarching domains derived from the diary coding schema and literature-based design principles:

1. Reading preferences and attentional rhythms across digital and analogue formats
2. Analogue-digital transitions in note-taking, ideation, and academic writing
3. Embodied routines, sensory regulation, and spatial-scaffolding strategies
4. Breakdown episodes, trust boundaries, and interface rejection thresholds

5. Reactions to preliminary design principles for inclusive hybrid learning

While these domains provided thematic scaffolding, the actual interview structure was conversational and example-driven. Participants were explicitly asked to walk through specific academic events—such as preparing for an oral exam, solving a mathematics problem set, or navigating dense PDF readings — and to comment on how tools, environments, and bodily actions shaped their cognitive engagement. Many questions took the form of situated prompts: "Can you describe the last time you switched from paper to digital in a study session?" or "What happens when a digital tool gives you feedback you didn't ask for?"

Importantly, interviews were not conducted in isolation from the diary material. Each participant's interview guide was customised based on themes that had emerged from their own entries. For instance, if a participant had described "colour-coded page layouts" or "working with a peer on a blackboard," they were invited to expand on these moments with concrete elaboration and evaluative reflection. In some cases, participants were directly read excerpts from their diary entries or shown simplified summaries of their artifact ecologies and asked to comment, revise, or complicate them. This dialogical integration created a recursive data layer in which prior findings could be affirmed, critiqued, or recontextualised.

3.5.2 *Theoretical Framing and Analytic Role*

The interview methodology followed the spirit—if not the strict procedural order—of Kvale and Brinkmann's seven-stage model [20], adapted to support an exploratory, constructivist approach grounded in neurodivergent experience. Specifically, the model was operationalised as follows:

1. **Thematisation:** The interview guide was thematically structured around key constructs drawn from the literature review and diary coding: cognitive load, modality transition, embodied pacing, user control, and epistemic friction.

2. **Design:** Each guide was tailored to the participant's prior diary data, enabling situated reflection and maximising relevance and comfort.

3. **Interviewing:** Conversations were kept conversational, with the researcher providing minimal scaffolding and frequent

invitations to "think aloud" or "explain what it feels like" in relation to specific study contexts.

4. **Transcription:** Verbatim transcripts were produced for each session. Attention was paid to affective phrasing, metaphorical language, and implicit contrasts (e.g., "It's like for kindergarten" or "It's just me writing—that's why it works.").

5. **Analysis:** Transcripts were coded inductively using the same open/axial/selective framework as the diary data. Codes were cross-referenced with diary themes to assess convergence, divergence, and reframing. Memos were written for each participant interview to track emerging tensions between their self-reports and prior interpretations.

6. **Validation:** Participants were offered opportunities during the interview to reflect on earlier diary observations and correct or nuance the researcher's interpretations. This member-checking function was informal but valuable, ensuring both empirical robustness and epistemic respect.

7. **Reporting:** The interview data is reported in Chapter 4 through in-depth participant narratives, followed by a cross-case synthesis. Each interview functions not only as a personal case but as a theoretical site—where latent assumptions from the literature are surfaced, contested, or expanded.

3.5.3 *Role in Research Design*

The interviews played three intersecting roles within the overall research design. First, they served as an **elaboration tool** — enabling participants to articulate the motivations, breakdowns, and adaptations behind their hybrid study ecologies. Where the diaries documented what happened, the interviews illuminated why and how. Second, they operated as a **validation mechanism** — allowing the researcher to test the preliminary design principles for inclusive hybrid learning developed in Chapter 2. These principles were not imposed but introduced conversationally, with participants invited to affirm, reject, or modify them in light of their own needs. Third, the interviews became a site of **empirical-theoretical friction**. In many cases, participants voiced strong resistance to principles that were theoretically well-supported (e.g., interactive feedback or sensory augmentation), thereby exposing crucial gaps in existing HCI frameworks around consent, pacing, and emotional predictability.

Rather than treating participants as passive respondents, the interview phase positioned them as co-analysts of their own learning ecologies. Their contributions not only clarified diary-based patterns but shaped the formulation of revised design principles. For example, Participant 1's rejection of unsolicited digital corrections directly led to a revision of Digital Reading Principle 4, while Participant 3's tactical use of gesture in response to aphantasia prompted an expansion of Augmented Paper Principle 6. Similarly, Participant 2's critique of scanned PDFs and their preference for structured analogue review materials refined Principle 3 around spatial sense-making.

In sum, the semi-structured interviews were essential in translating diary-based observation into actionable, participant-grounded design insight. They acted not merely as a secondary data source, but as a generative space for interpretive refinement—bridging empirical narrative with design theory and surfacing the tensions, workarounds, and affective thresholds that must inform the design of hybrid learning systems for neurodivergent learners.

3.6 ETHICAL CONSIDERATIONS

All participants received a written informed consent form outlining the scope, purpose, and ethical parameters of the study. This included a clear explanation of the types of data collected, how the data would be used, and how long it would be stored. Participation was entirely voluntary, and individuals retained the right to withdraw from the study at any point without justification or consequence.

To ensure compliance with the General Data Protection Regulation (GDPR), all personal data were anonymised and stored in encrypted, access-restricted environments. Special attention was given to ensuring clarity and accessibility of communication, including the language used in consent materials and study guides. These considerations were especially important in light of the diverse cognitive and sensory needs among the neurodivergent participant group.

3.7 SUMMARY OF METHODOLOGY

This thesis employed a qualitative mixed-methods design to investigate how neurodivergent students navigate hybrid reading

environments across analogue, digital, and augmented modalities. The methodological framework combined a literature-grounded development of design principles, a longitudinal diary study, and semi-structured interviews, all analyzed through iterative grounded theory coding enriched by Cognitive Load Theory (CLT). This approach enabled both descriptive and conceptual modeling of how learners manage cognitive demands and sensory complexity in their academic workflows.

The study began with a comprehensive literature review from which a dual set of preliminary design principles was derived. These principles guided the construction of diary prompts and interview questions and later served as points of comparison for empirical data. Diary entries were collected and digitised to document situated reading practices, multimodal tool transitions, and embodied study routines. These data were subjected to a three-stage grounded theory analysis—open, axial, and selective coding—aimed at surfacing categories of strategy, breakdown, and cognitive adaptation.

To deepen this analysis, semi-structured interviews were conducted with each participant, using personalised guides informed by their diary material. Participants were asked to reflect on concrete academic episodes and to evaluate the relevance, accuracy, or limitations of emerging theoretical interpretations and design principles. These interviews served not only to elaborate empirical patterns but also to identify conceptual tensions, such as discomfort with unsolicited augmentation or the emotional thresholds that trigger tool abandonment.

Throughout both data sources, CLT was used as a dialogical framework, allowing grounded categories to be reinterpreted in terms of intrinsic, extraneous, and germane cognitive load. This provided a nuanced lens for comparing how different participants regulate their learning environments and prioritise certain tools or modalities under varying affective and cognitive conditions.

The resulting analysis forms the foundation for a revised set of design principles grounded in lived experience, cognitive diversity, and the affective realities of studying in hybrid environments. The following chapter presents the findings derived from this analytical process, beginning with the coding of diary data and moving through interview elaborations and cross-participant synthesis.

4 FINDINGS AND ANALYSIS

4.1 DIARY STUDY

This section presents a cross-case thematic analysis of three neurodivergent university students' engagement with hybrid learning environments, drawing on diary data, artifact ecology mapping, and semi-structured interviews. The analysis follows the methodological scaffolding described in Chapter 3, structured around three coding phases from grounded theory (open, axial, and selective coding) and afterward interpreted through the lens of Cognitive Load Theory (CLT). For each participant, I present a structured narrative composed of four analytic dimensions: technological diversity, contextual challenges, proposed design solutions, and reflections on individual preferences. The section concludes with a cross-participant synthesis and cognitive load typology.

4.1.1 *Participant 1: Rhythmic Self-Regulation in Hybrid Learning*

Technological Diversity

Participant 1 navigated a richly layered artifact ecology that combined physical textbooks, digital stylus-based notetaking, LaTeX, and planning tools. Analogue annotation was central to their mathematics workflow, using highlighters coded semantically (e.g., green for translations, red for supplementary notes). Chemistry tasks involved dynamic sketching in OneNote using a stylus and foldable tablet. This analogue-digital split mirrored the participant's division of content types: tactile structuring for abstract math and visual spatialisation for molecular diagrams. Planning was supported via weekly calendars and structured activities (e.g., poledance), serving both academic and cognitive pacing functions.

Their modality preference aligns with findings in Vincent [36] and Mizrachi et al. [26], where students often preserve paper-based strategies for deeper engagement and use digital tools for flexibility and indexing. This hybrid strategy supports cognitive offloading through spatial arrangement and bodily anchoring [12].

Contextual Challenges

Participant 1 reported high intrinsic load during abstract tasks such as topology exercises and dense lectures. This was compounded by the temporal structure of the academic week—long consecutive teaching days often left little energy for task recovery. Extraneous cognitive load emerged during transitions between learning modalities or when faced with unpredictable instructional changes (e.g., late release of exercise content). Environmental sensitivity was acute: silence led to cognitive drift, while unfamiliar auditory input disrupted focus. Strategic sensory regulation (e.g., sitcoms or audiobooks) became essential during late-day study sessions, especially when ADHD medication wore off.

These findings are consistent with Le Cunff, Smith, and Taylor [21], who note that neurodivergent students often struggle with digital modality switching and benefit from sensory-stabilising environments. As Jeong and Gweon [15] show, erratic digital eye-tracking increases fatigue, reinforcing the importance of tactile and embodied supports.

Proposed Design Solutions

Participant 1's workflow highlights the need for tools that support temporal and sensory self-regulation. Design opportunities include:

- **Rhythm-based planning tools:** Interfaces that allow users to scaffold study tasks around embodied routines (e.g., physical exercise, calendar-linked motivational cues), echoing embodied learning strategies [1].
- **Hybrid annotation ecosystems:** Seamless integration of analogue and digital markup, supporting user-driven augmentation in line with Mackay [22] and Klokmose and Bertelsen [19].
- **Structured environmental modulation:** Systems that embed ambient audio control or personalised white noise to reduce auditory overstimulation, aligned with neurodivergent-friendly design as outlined in Le Cunff, Smith, and Taylor [21].

Reflections on Preferences

Participant 1 emphasised a desire for structured flexibility—environments where they could alternate between control and surrender, effort and rest. Their use of embodied pacing (e.g., poledance)

echoes the embodied cognition perspective in Flood [12], which highlights physical movement as a support for executive function. Collaboration with a study partner served as a cognitive and emotional anchor, supporting not only task execution but metacognitive reassurance.

Cognitive Load Integration

- **Intrinsic Load:** Managed via collaborative ideation and tool-specific breakdowns of complex material.
- **Extraneous Load:** Minimised through rhythm, environmental curation, and predictable routines.
- **Germane Load:** Actively supported via structured annotation systems, embodied time anchors, and co-regulatory learning.

4.1.2 *Participant 2: Affective Modulation and Fragmented Ecologies*

Technological Diversity

Participant 2's artifact ecology was characterised by fragmentation, multitasking, and mood-contingent tool use. Their core tools included VS Code for programming, Brightspace and GitHub for course management, and analogue sketching for pseudocode generation. Academic work was interwoven with podcasts, YouTube, Spotify playlists, and digital games—each serving distinct affective and regulatory functions.

This multitool layering reflects the hybrid ecologies described in Bødker and Klokmose [6] and mirrors affect-driven tool-switching found in ADHD profiles [14]. Digital media served both as stimulation and escape, reflecting affective regulation strategies discussed in Le Cunff, Smith, and Taylor [21].

Contextual Challenges

The participant encountered frequent cognitive strain from tool fragmentation and content-switching. Programming tasks introduced high intrinsic load, particularly when overlapping with dramaturgical assignments. Extraneous load stemmed from simultaneous platform use, emotional fluctuations, and unstructured planning. Social tools (Messenger, Instagram) had ambivalent roles—sometimes providing co-regulation, sometimes derailing focus.

These observations align with research on multitasking fatigue in digital reading [25] and the difficulties ADHD readers

face in sustaining coherent cognitive flows in digital spaces [15]. Participant 2's reliance on passive media in low-energy states exemplifies feedback loops described in Le Cunff, Smith, and Taylor [21].

Proposed Design Solutions

Participant 2's workflow suggests a need for environments that adapt to energy states and centralise fragmented tools. Design concepts include:

- **Mood-responsive dashboards:** Interfaces that suggest tools/tasks based on real-time affective state—consistent with adaptive scaffolding proposals in Ota and DuPaul [28].
- **Multimodal recovery spaces:** Embedded low-stakes interactivity (e.g., ambient games, audio narratives) to aid transitions.
- **Cross-platform integration:** Systems that consolidate digital artifacts to reduce fragmentation and extraneous load [6].

Reflections on Preferences

Participant 2 exemplified a responsive rather than proactive study ecology, governed by moment-to-moment mood navigation. Their strategy reflects strong emotional intelligence and flexible self-regulation, though susceptible to inefficiency without structural supports. This matches Flood [12] on the need for contextual scaffolding that aligns with cognitive rhythms.

Cognitive Load Integration

- **Intrinsic Load:** Driven by recursion-heavy code tasks and dramatic analysis; buffered by sketching and audio layering.
- **Extraneous Load:** Elevated due to multitool fragmentation and affective shifts; reduced via ambient tools.
- **Germane Load:** Supported indirectly by analogue-first coding, familiar environments, and affective feedback loops.

4.1.3 *Participant 3: Structured Modularity in High-Abstraction Domains*

Technological Diversity

Participant 3 maintained a modular artifact ecology centered around an analogue-to-digital pipeline. Mathematical reasoning began with chalk or paper and culminated in structured digital output via LaTeX. Obsidian, Google Calendar, and Brightspace supported knowledge management. They used gloves and headphones to block out sensory inputs—an example of environmental modulation consistent with neurodivergent sensory strategies [21].

Their reliance on spatial segmentation echoes principles from Vincent [36] and Klokmose and Bertelsen [19], where physical separation of thinking modalities enhances cognitive coherence.

Contextual Challenges

Participant 3 reported high intrinsic load from abstract content (e.g., Morse Theory), requiring strong internal scaffolding. Extraneous load was low due to their strict space-bound workflow—academic tasks were kept out of the home. Friction emerged during analogue-to-digital transitions, echoing challenges from augmented paper literature regarding hybrid synchronisation [24].

Proposed Design Solutions

Participant 3's workflow suggests design priorities around modality transitions and semantic organisation:

- **Analogue capture tools:** Tools that digitise paper or blackboard work without syntax reentry [24].
- **Modular planners:** Systems that reflect discrete task phases (ideation → formalisation).
- **Conceptual tag systems:** Semantic organisers that match structured thinking patterns of hyper-systemisers [4].

Reflections on Preferences

Participant 3 preferred predictability and segmentation, minimising task-switching and social distractions. Their learning mirrored hyper-systemising styles [4], where rule-based structure supports comprehension. Unlike the others, their design logic was solitary, quiet, and rigid by choice—not by constraint.

Cognitive Load Integration

- **Intrinsic Load:** High, especially in mathematical formalism; managed via analogue scaffolding and paced formalisation.
- **Extraneous Load:** Low due to modular routine and spatial separation.
- **Germane Load:** Maximised through reflective journaling, Obsidian-based indexing, and iterative concept mapping.

4.1.4 *Comparative Analysis Across Participants*

Commonalities

Despite variation in tool use and structure, all participants showed:

- **Multimodal engagement:** Aligning analogue and digital modalities with distinct task types.
- **Auditory regulation:** Podcasts and music used to sustain attention and modulate affect.
- **Cognitive load sensitivity:** Strategic ecological modulation based on workload and energy.
- **Embodied regulation:** Use of physical activity, sensory barriers, and gesture-rich interfaces, supporting embodied cognition [1, 12].

Variations in Ecological Structure

- **Participant 1:** Sought rhythm and co-regulation; workflow centered on embodied time anchors.
- **Participant 2:** Responsive and affect-driven; flexible but fragmented tool use.
- **Participant 3:** Structured and modular; prioritised stability and solitary control.

4.1.5 *Cognitive Load Across Cases*

Theoretical Synthesis

The findings support broader research that neurodivergent learners do not optimise solely for tool efficiency but for *cognitive continuity*, *sensory modulation*, and *adaptive flexibility* [21, 27]. These

Load Type	Participant 1	Participant 2	Participant 3
Intrinsic	Topology, back-to-back lectures	Programming recursion, dramaturgy	Morse Theory, Theoretical Exercises
Extraneous	Tool layering, sensory disruption	Platform switching, energy fluctuation	Manual digital transcription, reference navigation
Germane	Co-writing, colour-coded notes, body anchors	Pseudocode sketches, media reuse	Chalkboard derivation, Obsidian notes, print review

Table 1: Cognitive Load Dimensions Across Participants

personalised ecologies reflect the logic of artifact ecologies [6], where each participant configures hybrid systems according to embodied, cognitive, and emotional rhythms. Augmented paper and hybrid environments should thus prioritise user-driven control, low-friction transitions, and affordances that support embodied and distributed cognition.

4.1.6 *Linking Findings to Design Principles*

The hybrid workflows described by the participants closely echo the concerns raised in the literature review, particularly those summarised in the concluding design principles for digital reading and augmented paper. This section draws together key connections between participants' empirical practices and the design directions proposed in Chapter 2.

Digital Reading

Several patterns across the cases support the principle of **designing for accessibility and personalisation** (Principle 2). All participants demonstrated a high degree of sensitivity to cognitive context—fluctuating energy levels (P2), executive function pacing (P1), or sensory control (P3). Their reliance on self-curated auditory environments, adaptive tool switching, and personalised spatial routines highlight the importance of systems that are configurable to sensory, emotional, and cognitive needs.

Participant 1 and 3's use of colour-coded annotations, spatial segmentation, and consistent layouting strategies also under-

score the relevance of **restoring spatial orientation in digital text** (Principle 3). Their practices illustrate how visual anchors and physical arrangement support comprehension in abstract and high-load tasks.

Moreover, Participant 2's layered media use and preference for responsive switching between modalities speaks to the value of **supporting multiple reading modes and attentional states** (Principle 1). Rather than following a single linear path, their ecology demonstrates the need for fluid, mood-adaptive navigation that can accommodate affectively charged states and cognitive fragmentation.

Augmented Paper

All three participants relied on paper not just as a fallback, but as a core cognitive medium—particularly for ideation, decomposition, and reflection. This strongly validates the principle to **preserve and strengthen the role of paper** (Principle 1). Whether through chalkboards, physical annotation, or pen-based sketching, participants used paper as a thinking substrate that afforded stability, tactile feedback, and freedom of expression.

Participant 1's dynamic balance between control and surrender, as well as Participant 2's frequent task modulation, aligns with the principle of **letting users control the digital layer** (Principle 2). Their practices suggest that augmentation should remain in the background until deliberately activated—ensuring cognitive focus remains anchored in embodied activity.

In addition, Participant 3's modular workflows, characterised by predictable transitions and minimal disruptions, support the call to **integrate without disturbing existing routines** (Principle 4) and to **design with stability and coherence** (Principle 7). Their ecological stability appears not as a restriction but as a scaffold for deep cognitive work.

Finally, across all cases, embodied regulation—through movement (P1), gesture (P3), or tactile media switching (P2)—illustrates the practical value of **engaging the body and senses in meaning-making** (Principle 6). Participants did not separate cognition from the body, but instead drew on proprioception, movement, and affective state to navigate complex material.

In sum, the findings do not merely reflect the theoretical de-

sign principles—they enact them. Each participant's ecology demonstrates the lived reality behind the literature: that hybrid study is not a balancing act between tools, but a fluid orchestration of space, body, and cognition. The design implications are clear: support adaptability, preserve tactile and embodied engagement, and treat digital augmentation as a responsive companion—not a controlling layer.

4.2 SEMI-STRUCTURED INTERVIEWS

Having examined the rich empirical material generated through the diary study—including multimodal workflows, cognitive load profiles, and personalised artifact ecologies—this chapter now turns to the second strand of data collection: semi-structured interviews. The purpose of these interviews was twofold. First, they served to extend and elaborate the diary-based findings by prompting participants to reflect more deeply on their own tool use, transitions between analogue and digital media, and the embodied processes of externalising, translating, and archiving academic knowledge. In particular, participants were asked to walk through concrete episodes—such as exam preparation routines or modality shifts during note-taking—to unpack how analogue materials were integrated, transformed, or retained across different media contexts.

Second, the interviews provided a space to test and challenge the applicability of the design principles identified during the literature review and diary analysis. Participants were invited not only to affirm or refine design directions that resonated with their practices, but also to critically engage with principles that did not appear immediately relevant to their workflows. In doing so, the interviews served as both a validation tool and a source of theoretical friction—allowing tensions between idealised design frameworks and lived cognitive ecologies to surface. The following section presents key interview findings, organised by participant, with a focus on those moments where tool evaluation, medium transition, and embodied learning converged in ways that either reinforced or complicated the initial coding schema.

4.2.1 *Participant 1: Semi-Structured Interview Analysis*

The semi-structured interview with Participant 1 enriched the earlier diary-based account of analogue-digital transitions and embodied learning by clarifying the specific material flows, cog-

nitive pacing strategies, and augmentation preferences shaping their hybrid academic practices. The conversation also illuminated their stance on interface trust, technological friction, and the neurodivergent need for user-controlled digital environments. A particularly salient theme was the rhythmical cycling between media types—especially in mathematics—which enabled iterative refinement and supported both dyslexia accommodations and collaboration with a trusted peer.

Task-Based Media Distribution and Tool Flexibility

Participant 1 maintained a strict separation between analogue and digital tools, determined by task type, cognitive load, and modality. Chemistry notes were created exclusively in OneNote on a foldable tablet, primarily due to stylus affordances that enabled quick colour switching, shape correction, and content rearrangement:

> "I can change colours fast, erase quickly, and move stuff around if the lecturer jumps around on the board... sometimes I just press a button and it erases—super smart."

They highlighted that chemistry lectures often compressed large volumes of visual information into short time spans—requiring a tool capable of high-speed sketching, colour-based semantic grouping, and post-hoc layout manipulation. Conversely, theoretical chemistry exercises (*TØ*) and mathematics problem-solving were performed strictly on paper. Here, the affordance of spatial overview, tactile feedback, and seamless cross-reference between materials outweighed any digital benefit:

> "When I'm in tablet mode, it's harder to jump between pages. But on paper, I can just flick through and write down things fast—way easier."

This tool use pattern reflects a deliberate strategy for minimising extraneous load and preserving task fluency across cognitive domains.

Iterative Transitions as Conceptual Refinement

In mathematics, the participant described a multi-stage process spanning paper, blackboard, and finally digital notation in LaTeX. Initial idea generation occurred on analogue worksheets, supported by margin notes, colour-coded segments, and intentionally preserved failed attempts ("notation hell") that were later revisited. Study sessions with a peer were conducted

around a blackboard, allowing for collaborative ideation and verbal rehearsal before returning to paper and eventually entering LaTeX:

> "We go from paper to board and back again—like five times. And when I type it up, I usually change the wording, clean up the arguments... it's not just copying, it's tidying up my thinking."

The final digital transcription marked a stage of notational reorganisation, where semantic coherence was prioritised. This workflow corresponds with principles of germane load regulation: conceptual clarity emerged not from initial articulation, but from repeated transformation across modalities.

Notation, Layout, and Dyslexic Accommodations

Participant 1 employed deliberate spatial structuring to compensate for dyslexia-related disorientation. During oral exam preparation, they always used analogue one-page dispositions written in column format to reduce line-tracking difficulty:

> "I always write in columns... I can't find the next line otherwise. Shorter lines just make it easier."

This format was augmented with visual markers—e.g., red or green ink to denote key concepts or links—allowing for dynamic annotation during rehearsal. These visual routines were paired with self-directed reading aloud and gestural engagement, highlighting the interdependence between layout, memory consolidation, and embodied pacing.

Technological Aversion and Friction Management

While adept with select tools like OneNote and LaTeX, Participant 1 was notably resistant to unfamiliar digital systems, describing them as mentally exhausting unless accompanied by clear, overly simplified guides:

> "If I don't get a proper walkthrough, I just won't use it... it needs to be like for kindergarten."

They viewed Word's autocorrect features as invasive and personified it as a "pretentious little snob," contrasting this with LaTeX, which they appreciated for its quieter, less judgmental interface. New software was generally met with skepticism unless demonstrably simple or recommended by trusted peers. This aversion underscores the need to design tools that are cognitively light, predictably structured, and fail gracefully.

Augmentation Control and Interface Negotiation

Although appreciative of stylus-based augmentation features — such as shape correction, colour layering, and retroactive formatting — Participant 1 stressed that these must be fully user-controlled. They tolerated unsolicited suggestions only because they could be quickly undone:

> "Sometimes it thinks I'm drawing a triangle when it's just an 'e'... but I can just undo it, so it's fine."

Their commentary affirms the importance of **opt-in augmentation** and **low-friction reversibility**, both essential for maintaining cognitive sovereignty in neurodivergent learners. OneNote's passivity was praised as a reason it remained usable: "It's just me writing—if I want more, I can add it."

Digital Reading and Spatial Anchoring

Participant 1 reported mild-to-moderate spatial disorientation when reading dense digital texts, particularly longer PDFs. They developed improvised solutions such as using a piece of paper to track progress:

> "I keep reading the same line three times. I'd love a little arrow telling me where I am."

These comments validate both the principle of **restoring spatial sense-making** (Digital Reading Principle 3) and the pragmatic appeal of tactile anchoring strategies even within digital environments. The participant's desire for minimal but helpful indicators further echoes the broader theme of *non-intrusive augmentation*.

Cognitive Overload and Self-Limiting Ecologies

Perhaps most striking was the participant's sharp threshold for technological frustration. If OneNote malfunctioned or became unresponsive, they would simply stop taking notes rather than attempt a workaround:

> "The more annoyed I get, the less technology I use. If OneNote starts acting up—then there are no notes that day."

This behavioural cut-off highlights a protective mechanism against overload and affirms the importance of stability, simplicity, and emotional safety in hybrid learning tools. Even curiosity-driven exploration was inhibited by perceived complexity—features like PDF embedding in OneNote were viewed as "magical" but ultimately inaccessible due to poor onboarding.

Summary: Rhythmic Transitions and Neurodivergent Agency

The interview with Participant 1 reinforced and extended earlier diary findings in six central areas:

- **Domain-specific analogue-digital workflows** that optimise for content type and pacing needs.
- **Iterative transitions across modalities** supporting gradual conceptual clarification.
- **Layout strategies and colour coding** tailored to mitigate dyslexia-related disorientation.
- **Collaborative externalisation routines** grounded in embodied co-regulation.
- **Passive augmentation** that supports but never presumes user intention.
- **Strict friction thresholds** defining when technology remains viable or is abandoned.

Together, these findings affirm nearly all principles discussed in Chapter 2, especially those concerning user control (Principles 2, 4), embodied engagement (Principle 6), and cognitive load minimisation (Principle 7). Participant 1's hybrid ecology was not driven by novelty or efficiency but by rhythmic, embodied, and affectively stable engagement—where analogue and digital resources were orchestrated in response to domain demands, emotional state, and sensory regulation.

Summary: Affirmations, Reinterpretations, and Emerging Tensions

While the interview with Participant 1 largely reinforced prior diary-based insights—particularly the rhythmic analogue-digital transitions and embodied pacing strategies—it also surfaced several meaningful tensions that challenge or qualify the original design principles proposed in Chapter 2.

First, although Participant 1 affirmed the need for **user-controlled digital augmentation** (Augmented Paper Principle 2), the interview revealed a sharper boundary than previously assumed. Their disdain for unsolicited digital corrections (e.g., autocorrect in Word) went beyond simple preference: it reflected a deeper resistance to interfaces that "talk back" or impose unrequested feedback. This directly contradicts aspects of Digital Reading Principle 4, which advocates interactive feedback on

reading behaviour. For Participant 1, such feedback is not supportive but cognitively and emotionally intrusive: "I know I can't spell. You don't need to tell me that."

Second, the participant strongly rejected **haptic feedback** and other forms of digital sensory augmentation (Augmented Paper Principle 4), which had been tentatively supported in the diary analysis via embodied regulation routines. In the interview, however, Participant 1 dismissed tactile cues as "weird" and unnecessary—adding sensory noise rather than meaningful engagement: "I know where my finger is. I don't need it to vibrate." This calls for a critical revision of Principle 6, acknowledging that embodiment may be beneficial in some domains (e.g., movement-based pacing) but disruptive when digitally simulated.

Third, while previously characterised as a confident navigator of hybrid systems, the interview exposed the participant's deep reliance on others for technological onboarding and their avoidance of new systems without childlike instruction. They repeatedly invoked a "kindergarten" threshold for interface clarity and expressed strong distrust in emerging tools unless demonstrated by a peer. This complicates assumptions behind Augmented Paper Principle 5 (robust simplicity) and reveals that simplicity alone is insufficient; onboarding pathways must be explicitly scaffolded to avoid total abandonment of unfamiliar systems.

Finally, the interview revealed a conditional openness to collaboration. Whereas diary data emphasised solitary control over tools, the participant expressed interest in co-authoring shared OneNote environments—provided that access and navigation were made transparent. This nuanced stance suggests that Principle 7 (supporting modular collaboration) must accommodate users who value shared artifacts but resist shared screens or synchronous pace-setting due to reading-related anxiety.

In short, the interview functioned not just as a triangulation tool but as a site of conceptual friction. It highlighted where design principles need refinement, where neurodivergent trust breaks down, and where embodiment is welcomed in physical but not digital form. These tensions sharpen the direction of this thesis: hybrid tools must be not only user-controlled but emotionally considerate, offering choice without surveillance and augmentation without assumption.

4.2.2 *Participant 2: Semi-Structured Interview Analysis*

The semi-structured interview with Participant 2 offered further insights into the affect-driven, fragmented ecological patterns observed during the diary study. It also deepened the understanding of how digital reading environments and analogue-digital transitions impact cognitive load, tool preferences, and affective regulation.

Reflections on Digital Reading and Attentional States

Participant 2 elaborated on the pronounced difficulty they experience when reading digital texts, particularly scanned PDFs. They emphasised a recurrent loss of place within the digital document, which severely increased the cognitive load and made sustained concentration challenging. Unlike reading on paper, where tactile page-turning supports spatial memory and reading flow, digital formats disrupted their sense of progression and made navigation effortful. This aligns closely with research findings by Mangen, Olivier, and Velay [25] and Jeong and Gweon [15] regarding impaired spatial orientation and erratic eye movements during digital reading.

> "The biggest problem is that I lose my place in the text. I forget where I've gotten to—and that makes reading digitally so much harder and so much more energy-intensive."

Participant 2 expressed a strong preference for paper-based reading, stating that the physical act of holding a text anchors their focus. However, due to the volume of required readings and logistical constraints (e.g., printing costs and transport burden), they often had no choice but to read digitally. The interview validated the importance of **supporting diverse reading goals and attentional states** (Digital Reading Principle 1) and **restoring spatial sense-making** (Digital Reading Principle 3). The participant speculated that more controlled reading environments—such as Kindle-style progress bars or viewport restrictions—could alleviate these challenges.

Tool Fragmentation and Affective Modulation

Extending findings from the diary study, Participant 2 described a highly fragmented artifact ecology shaped by mood and task energy. Platforms like VS Code, GitHub, Brightspace, YouTube, and various music services were used interchangeably depending on cognitive load and emotional needs. This fragmenta-

tion led to frequent extraneous load, especially when emotional states fluctuated. Social media tools played ambiguous roles, oscillating between co-regulation and distraction.

> "My ecology changes a lot depending on how I'm feeling—sometimes music helps me get started, sometimes it's just noise. It depends on how much energy I have."

The interview reinforced the need for **mood-responsive dashboards** and **cross-platform integration** to streamline fragmented ecologies (linking to proposed design solutions in Section 4). Participant 2 clearly indicated that current tool fragmentation made it harder to achieve cognitive coherence during study sessions.

Augmented Paper and Analogue Preferences

Participant 2 strongly endorsed the principle of **preserving and strengthening the role of paper** (Augmented Paper Principle 1). Despite their engagement with digital tools, handwritten notes remained central to their cognitive workflow, especially for ideation and preliminary problem-solving in coding assignments.

> "I always write notes by hand—during lectures and when solving problems. A lot of them never get transferred to digital format because it just takes too much energy."

They expressed interest in emerging technologies like reMarkable tablets but also concern that multifunctional devices such as iPads would function more as "just another computer" than as genuine hybrid tools. These preferences validate the principles **letting users control the digital layer** (Augmented Paper Principle 2) and **designing with simplicity and coherence across tools** (Augmented Paper Principle 7).

Transitioning analogue Work into Digital Spaces

Participant 2 highlighted that transferring handwritten notes to digital format offered limited cognitive benefit unless the transformation involved active reworking. They explicitly contrasted scanning with rewriting:

> "When I just scan my notes, I don't remember them better. But when I rewrite them into my bachelor thesis, I understand them more and find small mistakes I didn't notice before."

These reflections strongly support findings on germane load and active restructuring [21]. Participant 2 described past efforts to digitise paper archives from high school and noted that passive archiving did little for recall or reuse.

Embodied Learning and Spatial Anchoring

Embodied learning emerged as a crucial but often implicit strategy. During oral exam preparation, Participant 2 reported standing at large blackboards to rehearse explanations—highlighting how spatial and physical enactments scaffolded memory:

> "I stand at the blackboard and go through the whole thing—writing and moving helps me remember. Even if I'm not memorising it, the body remembers the gestures."

They emphasised the importance of gesture and physical layout in organising and recalling mathematical procedures. These practices reaffirm **activate the body and senses** (Augmented Paper Principle 6) and align closely with embodied cognition theories [1].

Summary: Key Tensions and Design Implications

The interview with Participant 2 illuminated key tensions already surfaced in the diary study:

- **Activation vs. Passivity:** Digital systems must support active user engagement without enforcing constant interaction.
- **Shared Screens vs. Paper Grounding:** The participant's preference for tangible, spatially anchored materials suggests that hybrid tools should preserve embodied spatiality rather than fully digitise the reading space.
- **Technological Richness vs. Simplicity:** While multimodal systems are necessary, feature creep must be avoided; simpler, stability-oriented designs like specialised e-ink tablets may better serve users like Participant 2.

Participant 2's preferences and challenges strongly substantiate the design principles articulated in Chapter 2, and their lived practices offer nuanced empirical grounding for future hybrid learning system designs that aim to scaffold embodied cognition, affective modulation, and flexible attention regulation.

Summary: Affirmations, Reinterpretations, and Emerging Tensions

While the interview with Participant 2 largely reinforced prior diary-based insights—particularly regarding affect-driven tool-switching, embodied rehearsal, and analogue sketching for ideation — it also revealed key contradictions and reinterpretations that challenge assumptions embedded in several proposed design principles.

First, although Participant 2 affirmed the value of **affective modulation and multimodal scaffolding** (Digital Reading Principle 1), the interview complicated the assumption that all multimodal input aids regulation. For example, while music and audio-based tools were sometimes helpful, their benefit was deeply state-dependent: "Sometimes music helps me get started, sometimes it's just noise." This ambivalence indicates that affective tools must be not only selectable but dynamically tuned to shifting energy states. The principle remains valid but demands more flexible implementation.

Second, Participant 2 reaffirmed the centrality of **analogue tools** — especially for ideation and memory encoding—thus reinforcing **Augmented Paper Principle 1 (Preserve and Strengthen Paper)**. However, they also noted that much of their handwritten work was either discarded or forgotten unless it was actively rewritten into digital formats. Simple digitisation via scanning was dismissed as cognitively ineffective: "When I just scan my notes, I don't remember them better." This insight suggests that **active transformation**—not preservation alone—must be emphasised in Principle 1 and expanded within Principle 3 (Tailor Interaction).

Third, while Participant 2 was skeptical of multi-functional devices like iPads, they did not reject digital augmentation altogether. Instead, they articulated a preference for **dedicated, low-friction hybrid tools** such as digital notebooks or e-ink devices—if these offered improved structure and did not replicate computer-like distractions. Their concern that a tablet might become "just another computer" points to a subtle reinterpretation of **Augmented Paper Principle 7 (Design with simplicity and coherence across tools)**: simplicity must not only mean technical restraint but also *emotional legibility*—tools should feel qualitatively different from laptops.

Fourth, despite early impressions of a reactive, fragmented study

ecology, the interview revealed deliberate embodied learning strategies. Participant 2 described rehearsing oral exam content at a blackboard and using gesture-based memory prompts to recall procedures: "Even if I'm not memorising it, the body remembers the gestures." This confirms the relevance of **Augmented Paper Principle 6 (Activate the Body and Senses)** but also highlights that such strategies were rarely acknowledged as "designed." Designers should therefore treat embodied practice as latent expertise rather than an intervention to be introduced.

Finally, the interview surfaced an unexpected contradiction around **digital reading environments**. While the diary suggested adaptation to PDF-based readings, the participant expressed acute frustration during the interview: "I lose my place in the text... it makes reading digitally so much harder." Poorly scanned PDFs were particularly harmful, with participants describing increased eye strain and disorientation. These observations intensify the urgency of **Digital Reading Principle 3 (Restore Spatial Sense-Making)**, but also cast doubt on Principle 4 (Offer Interactive Feedback). The participant did not want help "finding their place" via eye-tracking or alerts. Instead, they favored **low-interference spatial aids** like Kindle-style progress bars that supported navigation without monitoring.

In summary, the interview deepened rather than simply confirmed diary findings. It brought forward tensions around passive digitisation, emotionally intrusive augmentation, and the fine line between integration and overwhelm. Participant 2 demonstrated clear design preferences—but also underscored that good design must adapt to affective variability, avoid technocentric assumptions, and respect the emotional contours of cognitive load. As with Participant 1, their feedback sharpens the role of this thesis: hybrid tools should not only scaffold learning but do so with humility—offering support without excess and structure without control.

4.2.3 *Participant 3: Semi-Structured Interview Analysis*

The interview with Participant 3 provided deeper insight into their modular and segmented learning practices, revealing how their tool choices, analogue-to-digital transitions, and embodied cognitive strategies are shaped by both neurodivergence (autism and aphantasia) and personal preferences for control, spatial consistency, and cognitive rigor.

Analogue Primacy and Controlled Transitions

Participant 3 confirmed a strong preference for initiating academic tasks in analogue form. They emphasised the value of handwriting not only for memory retention but also for cognitive pacing:

> "I prefer to process it with my hands... writing slowly makes it easier to remember."

While some materials—such as theoretical exercises—remained entirely analogue, the majority of their academic workflow involved a staged transition from paper to Obsidian (for atomic concept mapping) or to LaTeX (for structured formal outputs). This transition was not merely a transcriptional step but a deliberate phase of conceptual refinement:

> "The transfer makes me think through which ideas actually matter."

This aligns closely with the principle of **supporting active processing and feedback** (Digital Reading Principle 4), as well as **preserving and strengthening the role of paper** (Augmented Paper Principle 1). Participant 3's comments resonate with research on germane cognitive load: active reorganisation during analogue-to-digital transition enhances understanding more than passive digitisation would [21].

Digital Augmentation: Deliberate, Not Intrusive

Participant 3 was particularly sensitive to interface behaviours that reduced their sense of cognitive sovereignty. They expressed clear skepticism toward any augmentation that occurred automatically:

> "Unless I've asked for it, it should stay out of the way."

While open to the idea of augmented paper tools such as reMarkable—particularly if they faithfully mimicked paper and permitted delayed activation of digital features—they stressed the need for user-controlled timing. This preference validates two key principles: **let the user control digital augmentation** (Augmented Paper Principle 2) and **integrate seamlessly without disruption** (Principle 4).

Interestingly, they welcomed retrospective augmentation—such as applying tools to already-written material—but rejected real-time feedback like automated spellcheck:

> "What matters is that I can follow my thoughts, not that it's spelled correctly."

This perspective reflects a broader tension between productivity and autonomy in hybrid design, underscoring the importance of optional, non-disruptive augmentation for neurodivergent users.

Embodied Learning and Aphantasia Workarounds

A central theme in Participant 3's reflection was the cognitive role of embodied strategies in overcoming visual limitations associated with aphantasia. Unable to mentally visualise geometric forms, they instead relied heavily on hand gestures and spatial enactments to "approximate" three-dimensional constructs. This embodied approach to spatial understanding aligns with the principle to **activate the body and senses** (Augmented Paper Principle 6) and with embodied cognition research made by Abrahamson et al. [1].

During their bachelor's project, which involved high-dimensional topology, they repeatedly used gesture, pencil-movement in midair, and physical enactments to reason about abstract forms. They also speculated that augmented paper systems enabling 3D interaction or projection might provide valuable support:

> "If I could draw a box and it turned into a manipulable 3D model... that would help."

Such speculative needs illustrate new directions for inclusive augmentation that not only support but extend embodied scaffolds for spatially disabled learners.

Digital Reading and Spatial Anchoring

While Participant 3 experienced spatial disorientation during both analogue and digital reading, they reported that annotation—especially highlighting and marginal note-taking—was a key strategy for maintaining place and focus. Digital readers with integrated markup features (e.g., PDF viewers) were used regularly:

> "Highlighting and notes are things I do often."

However, they also noted that physical page-flipping and manual layouting provided a better sense of document structure.

This duality supports the principle of **restoring spatial sensemaking** (Digital Reading Principle 3) and **supporting diverse**

reading goals and attentional states (Principle 1). In particular, their reflections emphasise that for neurodivergent readers with aphantasia, visual orientation is not just about navigation but about spatial cognition as such.

Collaborative Practices and Privacy of Understanding

Unlike Participant 1 or 2, Participant 3 expressed a marked preference for individual cognitive work. While they had previously participated in study groups, they actively avoided shared digital documents during collaborative reading or exam prep:

> "My notes are tailored to how I understand things... I'd be concerned about whether others could follow or if they would be good enough."

They described their annotation and documentation style as highly idiosyncratic—optimised for personal recall rather than public clarity. This stance reiterates the need for hybrid systems that maintain **modular cognitive space** (cf. Augmented Paper Principle 7) and support private, self-referential workflows even when collaboration is desirable.

Augmentation Preferences and Theoretical Fit

Participant 3 was open to experimenting with augmented paper tools under specific constraints: minimal visual separation from handwriting, high configurability, and non-intrusive design. They expressed particular interest in systems that could support 3D modeling in mathematical contexts, provided that these remained optional and reactive. Their remarks thus affirm nearly all seven augmented paper design principles—especially those centered on control, coherence, and bodily engagement.

Summary: Control, Abstraction, and Cognitive Anchoring

Participant 3's interview surfaces three central themes:

- **Manual abstraction as cognitive filter:** The analogue-to-digital transition serves as a phase of deliberate conceptual distillation.
- **Control as cognitive affordance:** Autonomy over when and how digital systems appear is a precondition for effective augmentation.
- **Embodiment as workaround:** Bodily strategies compensate for aphantasia and support spatial reasoning in abstract domains.

Together, these practices strongly reinforce the principles outlined in Chapter 2 and validate the analytical integration of embodied cognition, hybrid tool continuity, and neurodivergent-centered design. Participant 3's experience foregrounds a critical design imperative: support structured independence, without sacrificing tactile fluency or embodied depth.

Summary: Deliberate Control, Embodied Workarounds, and Emerging Contradictions

While the interview with Participant 3 generally reinforced themes from the diary study—especially regarding modular workflows and analogue-to-digital transitions—it also surfaced critical reinterpretations and nuanced tensions that problematise earlier assumptions in the proposed design principles from Chapter 2.

First, Participant 3 powerfully affirmed the need for **user-governed digital augmentation** (Augmented Paper Principle 2), but the interview introduced a sharper temporal distinction than previously recognised. While the diary suggested seamless analogue-to-digital pipelines, the interview clarified a strong preference for *retrospective*, user-initiated augmentation only. Participant 3 explicitly rejected any form of real-time correction, especially spellchecking or pop-up prompts: "Unless I've asked for it, it should stay out of the way." This positions Principle 2 as a precondition for cognitive trust, not a mere convenience—and simultaneously challenges Digital Reading Principle 4's endorsement of real-time interactive feedback. For this participant, productivity-enhancing suggestions were not neutral but cognitively intrusive.

Second, their strong reliance on embodied gestures to compensate for aphantasia complicates Augmented Paper Principle 6 (Activate the body and senses). While Participant 3 affirmed the value of embodiment, they redefined it not as a general benefit but as a *compensatory workaround* for specific neurocognitive constraints. This distinction matters: embodiment was not inherently desirable, but necessary in the absence of visual mental imagery. They speculated about future augmented paper systems that might translate 2D sketches into interactive 3D models—a concept that goes beyond current design principles, suggesting an expansion toward dynamic spatial scaffolding in math-based learning environments.

Third, although Participant 3 expressed high appreciation for analogue reading and annotation practices—affirming Digital

Reading Principles 1 and 3—the interview highlighted persistent spatial disorientation even in physical formats. While annotation helped ground their reading, it did not fully resolve orientation challenges: "Whether digital or paper, I still lose my place sometimes." This complicates the assumption that analogue formats inherently resolve spatial tracking issues and points to the need for hybrid designs that combine tactile interaction with digital navigational supports.

Fourth, and most unexpectedly, the interview revealed an ambivalence toward collaborative environments that had not emerged in the diary. While the participant had occasionally engaged in study groups, they resisted shared digital workspaces due to concerns about idiosyncratic understanding and quality of expression. "I'd be concerned about whether others could follow," they noted—indicating that Digital Reading Principle 6 (Facilitate co-located collaborative reading) must account for learners who require private cognitive space even within collaborative settings. This reveals a broader tension between shared augmentation and solitary cognition.

Finally, Participant 3 offered novel insight into how abstraction itself becomes a cognitive burden. In high-dimensional mathematics, they described a recurring breakdown when abstract concepts had to be rendered into formally rigorous structures. This points to a potential refinement of Cognitive Load Theory integration: in their case, intrinsic and germane loads overlapped problematically, particularly when stringency followed abstraction. Analogue scaffolding mitigated this, but only partially—a finding that underscores the need for design interventions that can bridge the gap between intuitive ideation and formal expression.

In summary, the interview repositions Participant 3 not just as a structured and modular learner, but as one who demands full **temporal control**, uses **embodiment tactically**, and experiences hybrid transitions as both enabling and cognitively taxing. Their case highlights key design tensions: augmentation must not only defer to the user—it must remain emotionally and cognitively silent unless summoned. This reinforces a critical design imperative for hybrid systems targeting neurodivergent learners: support structured independence, honor embodied workarounds, and preserve quiet modularity.

4.2.4 *Cross-Participant Comparison: Semi-Structured Interviews*

This section synthesises the key findings from the individual interviews, focusing on both common patterns and diverging strategies across the three participants. While each participant demonstrated a distinct cognitive ecology, several shared values, tensions, and reinterpretations emerged—particularly in relation to the design principles proposed in Chapter 2. The following subsections unpack these cross-cutting themes, emphasising where design assumptions were affirmed, redefined, or contested.

Shared Emphasis on Analogue Tools for Conceptual Grounding

All three participants articulated a clear cognitive preference for analogue tools in the early phases of comprehension and ideation. Paper was described as a "thinking surface" (Participant 2), a memory anchor (Participant 3), and a flexible layout space (Participant 1). These insights affirm **Augmented Paper Principle 1 (Preserve and amplify the role of paper)**. However, tensions arose around what it means to "preserve." Participant 2 rejected passive scanning workflows, and Participant 3 framed paper not as a bridge to digital output but as a discrete conceptual stage.
Emerging Tension: Principle 1 must move beyond preservation toward **active transformation**—supporting workflows where analogue inputs are not merely retained but restructured into new cognitive artifacts. This reframing impacts Principle 3 (Adapt to rhythms), which must acknowledge transitions as epistemic events rather than technical handovers.

Universal Desire for User-Controlled Augmentation

All participants expressed strong preferences for **opt-in digital augmentation** (Principle 2), emphasising that augmentation should never occur unsolicited. Participant 3's remark—"Unless I've asked for it, it should stay out of the way"—was echoed in Participant 1's rejection of autocorrect and Participant 2's skepticism toward overly intelligent interfaces.
Reinterpretation: Principle 2 is not just about user control but about **cognitive sovereignty** and emotional safety. Violating control—even with well-intentioned feedback—was perceived as an affront to autonomy, especially among participants with sensory or attentional sensitivities.
Contradiction: These responses challenge **Digital Reading Principle 4 (Enable active processing and feedback)**. For these

participants, real-time feedback (e.g., spellcheck, tracking aids) was cognitively intrusive, not beneficial. Feedback must be user-triggered, retrospective, and context-aware—not automated or behaviourally inferred.

Embodied Learning: Shared but Divergently Motivated

Embodiment was a consistent but variably motivated strategy. Participant 1 used whole-body movement to regulate arousal. Participant 2 practiced spatial rehearsal via blackboard gesturing. Participant 3 used microgestures to compensate for aphantasia, indicating that **embodiment can be either a strength or a workaround**.
Reinterpretation: While **Augmented Paper Principle 6 (Harness bodily engagement)** was affirmed, its rationale must be recontextualised. For some users, embodiment is sensorimotor enrichment; for others, it is compensatory. Tools must be **modular and non-normative** in how they activate bodily engagement.
Implication: This complicates Principle 6's original framing as universally beneficial. Embodied design must distinguish between voluntary kinesthetic scaffolding and forced sensory augmentation—Participant 1, for example, rejected haptics as intrusive "noise."

Shared Frustration with Digital Reading Interfaces

Each participant described spatial disorientation when reading digitally—ranging from mild (Participant 1) to deeply frustrating (Participant 2). While annotation and layouting helped, they did not fully resolve the issue. Kindle-style progress bars, page anchors, and viewport constraints were proposed as alternatives.
Affirmation and Tension: These accounts affirm **Digital Reading Principle 3 (Restore spatial sense-making)** but introduce a caveat: even analogue formats did not fully resolve disorientation (Participant 3), suggesting that the issue is not medium-specific but a deeper neurocognitive need for **anchored navigation**.
Contradiction: Participants largely resisted Principle 4's implementation when it took the form of proactive, system-driven interaction (e.g., eye-tracking, pop-ups). Instead, they desired **low-interference support**—such as visual markers or manual placeholders—which do not presume behavioural prediction.

Ecological Divergence in Rhythmicity, Fragmentation, and Modularity

While all participants maintained hybrid ecologies, their modulation strategies diverged sharply:

- Participant 1 relied on **rhythmic, embodied routines** and a trusted peer.
- Participant 2 described **emotionally driven fragmentation**, toggling tools based on state.
- Participant 3 enforced **modular segmentation**, keeping cognitive domains spatially and procedurally discrete.

Reinterpretation: This challenges static interpretations of **Augmented Paper Principle 3 (Adapt to rhythms)**. Rhythms may be affective (P2), social (P1), or spatial (P3), requiring systems to differentiate between **routine dependence** and **adaptive unpredictability**. There is no universal "rhythm"—only idiosyncratic modulation profiles.
Design Implication: Hybrid tools should scaffold multiple modulation logics—routine anchors, emotional diagnostics, or procedural segmentation—rather than assuming a single pacing schema.

Divergent Attitudes Toward Collaboration and Publicness

Collaboration was not universally desirable. Participant 1 welcomed co-authoring when access and boundaries were transparent. Participant 2 remained mostly independent but leveraged emotional co-regulation via chat. Participant 3 actively avoided shared documents, citing quality anxiety and idiosyncratic expression.
Tension: These findings complicate **Digital Reading Principle 6 (Facilitate co-located collaborative reading)** and refine **Augmented Paper Principle 7 (Design for modularity, stability, and simplicity)**. While collaboration is essential in some contexts, participants revealed distinct discomforts around shared digital environments.
Reinterpretation: Principle 6 must shift from co-location to **configurable co-presence**—enabling collaboration that respects individual pacing, privacy, and annotation styles. Principle 7 must expand beyond technical modularity to include **social modularity**, accommodating learners who require collaboration without shared screens or synchronised inputs.
Design Implication: Systems should support flexible, asynchronous collaboration features such as optional shared canvases, toggled visibility of notes, and user-controlled engagement levels.

Cognitive safety depends on being able to opt in and out of social exposure.

Varied Interpretations of Augmented Paper Potential

All three participants supported the idea of augmented paper—but under different conditions:

- Participant 1 demanded emotional neutrality and undo capability.
- Participant 2 welcomed structured translation (e.g., handwritten to code) but feared feature overload.
- Participant 3 imagined 3D visualisations to support aphantasia but rejected real-time interventions.

Contradiction and Redirection: These stances challenge assumptions in **Principle 4 (Integrate seamlessly)**. Seamlessness cannot be defined purely in terms of interface fluidity; it must also be measured against **emotional opacity, reversibility, and user tempo**.
Conclusion: Augmented paper systems must not only minimise friction but avoid perceptual overreach. They must be tools of invitation, not imposition—quietly aligned with users' epistemic thresholds.

Taken together, the cross-participant analysis highlights both strong affirmations and pointed rejections of the original design principles introduced in Chapter 2. While participants validated several core ideas—such as the cognitive value of analogue media, the need for spatially coherent reading environments, and the importance of embodied learning—these affirmations were frequently accompanied by clarifying constraints, reinterpretations, or tensions. Certain principles were upheld only under strict conditions (e.g., delayed augmentation, modular collaboration), while others—such as real-time feedback or default system intelligence—were consistently resisted. These divergences reveal not only the cognitive specificity of neurodivergent learners but also structural gaps in how the literature conceptualises user agency, emotional safety, and epistemic sovereignty.

In response, the following section presents a revised and expanded set of design principles. These principles synthesise existing research with the lived cognitive strategies observed across diary and interview data. Rather than discarding the

literature-based framework, this synthesis builds upon it—retaining what proved robust, refining where participant experiences suggested nuance, and introducing new principles where empirical friction revealed missing dimensions. The resulting framework serves as a bridge between theory and practice, foregrounding neurodivergent needs while maintaining continuity with established design scholarship.

4.3 REVISED DESIGN PRINCIPLES FOR HYBRID LEARNING ENVIRONMENTS

This section presents a refined and expanded set of design principles that synthesise insights from the literature with empirical findings from three neurodivergent participants. Each principle includes a rationale for its revision or expansion and is grounded in participant feedback from diary studies and semi-structured interviews.

4.3.1 *Digital Reading Design Principles*

1. **Support diverse reading goals and attentional states**
   Enable transitions between deep focus, overview, and affect-driven modes through adaptable layouts, progress anchors, and minimal visual clutter.
   Empirical Basis: Participant 2's affect-modulated tool use and all participants' shifting attentional rhythms necessitate flexible interfaces that respond to varying engagement states.

2. **Centre accessibility, personalisation, and emotional legibility**
   Systems must accommodate sensory profiles, dyslexia-friendly layouts, and affective safety. Interfaces should feel emotionally neutral and cognitively predictable.
   Empirical Basis: Participant 1's reliance on emotionally simple guides and rejection of intrusive systems demonstrates the need for emotionally legible onboarding and interface behaviours.

3. **Restore and stabilise spatial sense-making**
   Reintroduce stable layouts, visible location cues, and low-interference spatial supports across digital formats.
   Empirical Basis: All participants experienced spatial disorientation in digital reading. Participant 3 noted disorientation even with paper, emphasising a broader need for spatial anchoring tools.

4. **Support user-triggered processing—not automated feedback**
   Emphasise optional annotation tools, manual summarisation, and delayed feedback that users can summon as needed. Avoid unsolicited corrections or predictive feedback.
   Empirical Basis: All participants rejected real-time behaviour tracking or automated corrections. Participant 3 requested retrospective, user-initiated augmentation only.

5. **Activate the body through opt-in sensorimotor scaffolds**
   Support embodied strategies such as gestural rehearsal, stylus interaction, and spatial layouting, but avoid enforced digital embodiment like haptics unless explicitly desired.
   Empirical Basis: Embodied learning was present across all cases, but motivations differed. Participant 1 explicitly rejected digital haptics; Participant 3 used embodiment tactically due to aphantasia.

6. **Facilitate configurable co-presence, not forced collaboration**
   Enable asynchronous, semi-private collaboration (e.g., shared canvases, comment toggling, access control) rather than assuming synchronous co-reading.
   Empirical Basis: Participant 3 avoided shared digital work due to cognitive anxiety; Participant 1 accepted collaboration only with clear boundaries and transparent access.

4.3.2 *Augmented Paper Design Principles*

1. **Preserve and actively transform paper workflows**
   Augment paper not only by retaining its materiality but by enabling meaningful transitions—rewriting, restructuring, and concept distillation.
   Empirical Basis: Participants 2 and 3 found passive digitisation (e.g., scanning) cognitively ineffective; only active transformation supported learning.

2. **Let users initiate and time augmentation**
   Maintain user governance over when and how digital layers appear. Avoid unsolicited interventions or automatic enhancements.
   Empirical Basis: All participants insisted on cognitive sovereignty and rejected real-time, non-consensual augmentation.

3. **Adapt augmentation to user modulation profiles**
   Support diverse patterns of rhythm and pacing—routine-based, affect-driven, or spatially segmented—through adap-

tive input/output channels.
Empirical Basis: Participants displayed distinct ecological rhythms: P1 followed embodied routines; P2 was mood-driven; P3 operated through modular segmentation.

4. **Design invisibly until summoned**
Augmentation must feel cognitively silent and emotionally neutral until explicitly activated.
Empirical Basis: Participants preferred passive systems. Participant 3 emphasised invisibility as essential for cognitive trust; P1 tolerated OneNote due to its passive augmentation style.

5. **Design for emotional simplicity, not just technical robustness**
Ensure affective onboarding, error forgiveness, and emotionally legible system behaviour.
Empirical Basis: Participant 1 required emotionally simple guides; Participant 2 feared overwhelming systems—even minimal ones—unless emotionally intuitive.

6. **Harness embodied learning as latent user expertise**
Scaffold user-defined embodied strategies (e.g., gesture, pacing, layout), rather than imposing embodied features.
Empirical Basis: Participants used embodiment intuitively. P2 rehearsed at blackboards without consciously identifying it as strategy; P3 used gesture to compensate for aphantasia.

7. **Maintain both technical and social modularity**
Design stable, interoperable systems that preserve private space while enabling controlled collaboration.
Empirical Basis: Participant 3 rejected shared environments due to self-expression anxiety. All participants required stable ecosystems with opt-in social features.

4.3.3 *New Principles Emerging from Empirical Friction*

8. **Design for abandonment thresholds**
Tools must fail gracefully, provide quick exits, and allow recovery without cognitive penalty.
Empirical Basis: Participant 1 ceased note-taking entirely when OneNote malfunctioned; systems must respect neurodivergent overload boundaries.

9. **Enable epistemic transformations, not just format shifts**
Facilitate knowledge restructuring during transitions be-

tween analogue and digital, rather than merely replicating content.
Empirical Basis: Participants 2 and 3 described analogue-to-digital transitions as sites of conceptual clarity—not reproduction.

10. **Protect cognitive and emotional sovereignty**
Avoid predictive systems, behavioural nudges, or automatic adaptations. Prioritise user privacy, idiosyncratic logic, and individual pacing.
Empirical Basis: All participants valued autonomy and resisted systems that assumed intentions or imposed workflow interpretations.

The preceding analysis has illuminated the complex, embodied, and cognitively diverse practices that characterise hybrid learning among the three neurodivergent participants. Through detailed diary and interview data, a refined set of design principles has emerged—ones that both align with and diverge from the theoretical principles originally outlined in Chapter 2. While many empirical insights affirm the foundational value of literature-based principles—such as the need for spatial coherence, embodied engagement, and user control—others introduce critical tensions that challenge their underlying assumptions. Participants consistently emphasised the importance of affective safety, friction thresholds, retrospective augmentation, and epistemic autonomy—factors only partially accounted for in the existing research.

This tension invites a broader discussion: to what extent should the original, literature-grounded principles be retained, refined, or replaced in light of these lived experiences? The following chapter undertakes this evaluative task. It systematically contrasts each principle derived from prior work with the empirically grounded revisions surfaced through participant practice. In doing so, the discussion will explore whether a hybrid framework—one that integrates theoretical foresight with real-world neurodivergent needs—can offer a more inclusive and practically viable design model. The chapter ultimately aims to converge on a final set of design principles that are both theoretically robust and experientially grounded.

Design Principle	Participant 1	Participant 2	Participant 3
Digital Reading Principle 1: Support diverse reading goals and attentional states	Affirmed via embodied pacing, but variable focus was not addressed directly.	Affirmed but noted state-dependent variability; mood tools must adapt in real time.	Affirmed; used modular tool use for attentional fit.
Digital Reading Principle 3: Restore spatial sense-making	Affirmed tactile layouting; digital formats disrupted spatial sense.	Affirmed in principle, but suffered digital disorientation; preferred low-interference aids.	Affirmed but complicated; disorientation persisted even in analogue.
Digital Reading Principle 4: Offer interactive feedback on reading behaviour	Rejected as intrusive (e.g., readingspeed calculator); seen as emotionally dysregulating.	Rejected real-time feedback; preferred minimal, opt-in spatial tools.	Rejected real-time prompts; insisted on retrospective, user-initiated feedback only.
Digital Reading Principle 6: Facilitate co-located collaborative reading	Open to collaboration, but only under transparent, low-pressure conditions.	Preferred individual expression over shared screens; flexible collaboration needed.	Rejected shared documents; required private space even in study groups.
Augmented Paper Principle 1: Preserve and strengthen the role of paper	Affirmed; analogue tools central to rhythm and pacing.	Affirmed for memory/ideation; noted that scanning alone offers no benefit.	Affirmed, but required active rewriting for memory encoding.
Augmented Paper Principle 2: Let users control the digital layer	Affirmed with stricter control needs; unsolicited augmentation rejected.	Affirmed; required nonintrusive control over when/what is augmented.	Affirmed with temporal caveat—control must be retrospective, never live.
Augmented Paper Principle 4: Integrate digital feedback without disrupting	Partially affirmed; seamlessness required strict user consent.	Preferred dedicated tools over general-purpose devices (e.g., iPads).	Affirmed only if augmentation remains invisible until summoned.
Augmented Paper Principle 6: Activate the body and senses	Rejected digital embodiment (e.g., haptics); preferred physical gesture in the physical 3D space.	Affirmed embodied strategies (e.g., blackboard rehearsal); not initially self-identified as such.	Affirmed tactically; embodiment used to compensate for aphantasia.
Augmented Paper Principle 7: Design with simplicity and coherence across tools	Affirmed: Future systems were required to fit into already existing ecology,	Affirmed: To avoid cluttering, new systems needed to fit into existing technologies.	Affirmed: Future tools would need to be integrated into existing tool ecology.

5 DISCUSSION

This chapter critically reflects on the revised design principles developed in Chapter 4.3, situating them in relation to the original theoretical framework presented in Chapter 2.7. While the literature on digital reading, augmented paper, and hybrid learning environments offers a compelling normative agenda—one focused on personalisation, embodiment, and multimodal coherence—the empirical data gathered in this study reveals that such principles are not always sufficient when confronted with the lived experiences of neurodivergent learners.

Across the diary and interview data, participants both affirmed and problematised the ideals found in the existing research. Some principles, such as restoring spatial structure or preserving the role of paper, were strongly validated. Others, like real-time feedback and seamless integration, were challenged for being emotionally intrusive, cognitively disruptive, or insufficiently responsive to breakdown and overload. These empirical-theoretical tensions necessitate a more nuanced framework: one that foregrounds the practical, emotional, and cognitive strategies employed by learners to maintain autonomy, rhythm, and understanding in hybrid academic spaces.

To structure this synthesis, the revised principles have been grouped into five thematic clusters that reflect the central design challenges emerging from this study:

1. **Cognitive and Affective Adaptability:** Supporting fluctuating energy, mood, and focus.
2. **Cognitive Sovereignty and User Control:** Ensuring users retain agency over when and how digital systems act.
3. **Embodied and Situated Learning:** Recognising sensorimotor engagement as a latent but critical learning modality.
4. **Spatial Anchoring and Workflow Continuity:** Maintaining spatial logic across hybrid modalities and transitions.
5. **Configurable Social Presence and Privacy:** Allowing collaboration without compromising individual pace or expression.

An optional sixth theme—**Friction Sensitivity and Graceful Design**—further underscores how emotional and cognitive safety play a central role in technology adoption, especially among users with low tolerance for failure, noise, or overload.

This chapter proceeds by examining each thematic cluster in detail, contrasting the empirical findings with the original literature-based principles. Where tensions exist, it explores whether those tensions warrant refinement, recontextualisation, or replacement of earlier frameworks. The goal is not to discard prior research but to deepen it—to move from generalised design recommendations toward a neurodivergent-informed perspective grounded in lived complexity. Through this, the chapter aims to establish a final set of design principles that are not only theoretically sound but practically resonant with the realities of hybrid learning among cognitively diverse students.

5.1 COGNITIVE AND AFFECTIVE ADAPTABILITY

Among the most consistent cross-cutting themes in this study was the participants' need to modulate their study practices based on fluctuating cognitive and emotional states. Unlike conventional design assumptions that treat user behaviour as relatively stable, the diary and interview data revealed that energy levels, mood, and attentional capacity often varied not just day-to-day, but within short study sessions. These intra-individual shifts shaped how, when, and whether participants engaged with digital or analogue tools. Systems that failed to accommodate this variability were quickly sidelined in favor of more forgiving or neutral alternatives.

Participants each navigated this adaptability in idiosyncratic ways. Participant 1 relied on embodied routines to maintain consistency, often integrating analogue tools to structure time and attention. Participant 2 toggled tools based on mood, avoiding cognitively demanding platforms during periods of low affective energy. Participant 3 structured their work environment modularly, isolating tools to specific cognitive domains and switching between them to manage focus fatigue. These strategies were not merely preferences—they were adaptive responses to complex internal rhythms.

The literature does support the notion of adaptive interfaces and context-aware personalisation. Research on attentional pacing, affective computing, and cognitive load-sensitive design

offers compelling frameworks for aligning system behaviour with user readiness. This is reflected in the original design principles, particularly in Digital Reading Principle 1 (Support diverse reading goals and attentional states) and Augmented Paper Principle 3 (Adapt to user needs and rhythms). However, most such work assumes a relatively narrow form of adaptability—typically adjusting layouts, contrast, or task length.

In contrast, participants in this study required more holistic modulation: the ability to shift modality (from screen to paper), social engagement (from collaboration to solitude), and even the ontology of tasks (from linear planning to associative sketching). This was echoed in the empirically revised principles, where rhythm-based switching and idiosyncratic pacing emerged as central concerns. These findings suggest that existing design strategies may underestimate the complexity and volatility of neurodivergent learners' modulation profiles.

To reflect this complexity, a new principle is introduced that goes beyond interface-level customisation. It calls for systems that recognise *modulation profiles*: individualised patterns of cognitive and emotional pacing that shape how tools are adopted, avoided, or repurposed. Rather than assuming consistency, designers should build for rhythmic discontinuity. Tools must allow users to downshift or reorient without penalty, offering stable waypoints and flexible modes that match the learner's momentary needs.

Final Principle: *Design for cognitive and affective adaptability. Support user-specific modulation profiles by enabling rhythm-based tool switching, energy-aware interaction pacing, and affect-sensitive interface flexibility.*

5.2 COGNITIVE SOVEREIGNTY AND USER CONTROL

One of the most decisive insights emerging from this study is the reframing of control not merely as a technical preference, but as a cognitive and emotional necessity. While literature in educational technology often celebrates real-time feedback, predictive annotation, and automatic scaffolding as beneficial features, all three participants consistently rejected such forms of unsolicited assistance. Rather than experiencing these functions as supportive, participants described them as intrusive, emotionally taxing, and disruptive to their autonomy. This divergence signals a more foundational design requirement: what

this thesis terms *cognitive sovereignty*—the learner's right to decide when, how, and whether digital systems intervene in their workflow.

Analogue tools were consistently described as cognitively "safe" because of their inherent passivity and predictability. In contrast, digital systems that initiated interaction or assumed user intent without permission were perceived as risky. This sheds light on a deeper analogue-digital dynamic: the mediums are not only distinct in form but also in their perceived relational stance toward the user. Analogue tools recede into the background unless summoned; digital tools, unless carefully restrained, risk asserting unwanted agency. Thus, the analogue-digital divide becomes a proxy for perceived friction and epistemic control.

This finding reframes Digital Reading Principles 2 and 4 (Center accessibility and personalization, and Enable active processing and feedback) and Augmented Paper Principle 2 (Let the user control digital augmentation), shifting from an emphasis on proactive support to an insistence on retrospective, user-invited engagement. Although metacognitive scaffolding remains valuable in both literature and practice, its implementation must prioritise reversibility, timing, and user consent. Where literature tends to assume that assistance is helpful by default, participants highlighted how even well-intentioned support can erode trust when misaligned with their pacing or affective readiness. This tension underscores a recurring theme: design must respect not only cognitive load but also the emotional boundaries that define safe learning environments for neurodivergent users.

Yet this rejection of real-time augmentation need not imply a wholesale dismissal of responsive systems. A more nuanced interpretation suggests that the issue lies not in the timing of support per se, but in how that support is introduced, perceived, and experienced. If a system could be designed to appear cognitively passive—offering help only when patterns of need are deeply attuned to the user's rhythms, and doing so in a way that feels like intuitive extension rather than external imposition—it may be possible to reconcile real-time functionality with cognitive sovereignty. In this sense, the ideal is not the elimination of automation, but the development of "invisible responsiveness": a design ethos where system behaviour is so well-aligned with user logic and emotional pacing that its in-

terventions feel user-authored rather than system-imposed. For neurodivergent learners, this would require not only adaptive modeling but also affectively legible interfaces, opt-out safeguards, and ongoing calibration to evolving study states. In short, responsive tools may be viable—but only if responsiveness is reframed as relational attunement, not algorithmic anticipation.

Final Principle: *Support cognitive sovereignty through retrospective, user-invited augmentation. Responsive systems must appear passive, emotionally legible, and fully reversible, ensuring that control remains with the learner/user at all times.*

5.3 EMBODIED AND SITUATED LEARNING

Embodiment, another pillar of learning design, was affirmed by participants but with important qualifications that challenge literature norms. Rather than serving as a universally enriching modality, bodily engagement was deployed tactically. For Participant 1, movement helped manage arousal; Participant 2 used gestural rehearsal to reinforce memory; and Participant 3 used micro-gestures as a workaround for aphantasia. The body, then, was not merely a channel of enrichment but often a form of regulation or compensation.

While the literature—drawing from sources such as Abrahamson and Flood—often idealises embodied interaction through tools like haptics or gesture tracking, participants critiqued these features when they felt forced or artificial. Haptics, for example, were described by Participant 1 as "noise," while Participant 3 emphasised the necessity for bodily strategies to emerge from individual need rather than be pre-imposed by system design. This underscores a key distinction: embodiment must be *attuned*, not imposed. The revised design principles reflect this nuance by reframing bodily interaction as a form of latent user expertise—something to be scaffolded, not simulated.

In this light, Digital Reading Principle 5 and Augmented Paper Principle 6 (Activate the body and senses, and Harness bodily engagement in learning) require recontextualisation. While both frameworks acknowledge the value of embodied learning, they differ in their assumptions about how such engagement should be activated. Participants preferred analogue acts—writing, spatial layouting, gesturing—as self-directed and meaningful, in contrast to digital systems that attempted to replicate these

effects without preserving the user's intentionality. This highlights a broader issue in hybrid learning design: embodiment is not merely a mode of input, but a reflection of cognitive strategy, emotional pacing, and personal meaning-making.

Final Principle: *Scaffold embodied learning as user-defined and opt-in. Avoid enforced digital embodiment; instead, support gesture, layout, and movement as latent cognitive strategies emerging from user practice.*

5.4 SPATIAL ANCHORING AND WORKFLOW CONTINUITY

Spatial disorientation emerged as a significant challenge, but in ways that exceeded the scope of existing literature. While prior research primarily addresses spatial mapping in digital contexts—citing scrolling, layout shifts, and fragmented navigation as issues—participants in this study also reported spatial confusion in analogue settings when structure was absent. Participant 3, for example, noted difficulties even with paper documents that lacked consistent layouting. This indicates that spatial sense-making is not medium-specific but a cross-modal necessity.

Crucially, participants described transitions between analogue and digital environments not as technical handovers but as *epistemic events*—moments in which their understanding could either crystallise or fracture. These transitions became sites of cognitive transformation or breakdown depending on whether visual anchors and workflow coherence were preserved. As such, the revised design principles expand the concept of spatial design beyond individual interfaces to encompass entire learning ecologies.

Digital Reading Principle 3 (Restore spatial sense-making), originally focused on restoring spatial mapping in digital reading, is thus insufficient. It must be rearticulated as a principle of *workflow continuity*, emphasising the need for cognitive bridges that maintain orientation across modalities. Visual cues, predictable structure, and anchoring mechanisms are not just usability features—they are cognitive scaffolds that enable meaning to travel across space, tool, and time. This refinement affirms the literature's recognition of spatial navigation, while extending its scope to better match the hybrid and transitional realities of neurodivergent learners.

Final Principle: *Design spatial supports that ensure workflow continuity across analogue and digital tools. Treat transitions as epistemic events requiring visual anchors and structural predictability to preserve cognitive flow.*

5.5 CONFIGURABLE SOCIAL PRESENCE AND PRIVACY

Collaboration, though often valorised in the design of educational technologies, was perceived with ambivalence or outright resistance by participants in this study. While literature frequently supports features like shared annotation, synchronous co-reading, and collaborative whiteboards, these were seen by participants as threats to epistemic safety rather than enhancements. Participant 3, for instance, explicitly rejected all forms of shared digital documents due to anxiety over expression and loss of control. Participant 1 accepted collaboration only when clear boundaries and access transparency were in place.

However, this does not invalidate the literature's emphasis on the importance of collaborative learning. Numerous studies have demonstrated that co-reading, shared annotation, and dialogic meaning-making can enhance comprehension, foster accountability, and build deeper engagement—particularly when learners are scaffolded in their roles and responsibilities. From this perspective, the problem may lie less in the premise of collaboration than in how it is implemented. If collaboration is presented as mandatory, synchronous, or unfiltered, it risks undermining autonomy. But when designed with opt-in structures, clear social cues, and flexible timing, collaborative tools can foster both social learning and emotional safety.

The revised principles thus introduce the concept of *social modularity*, not as a replacement but as a reframing of Digital Reading Principle 6 (Facilitate co-located collaborative reading) and Augmented Paper Principle 7 (Maintain coherence across tools). Rather than discarding collaborative features, designers are encouraged to create layered participation structures—supporting both independent work and optional co-presence. Participants themselves did not reject the value of communication or shared effort entirely; rather, they emphasised that these processes must be user-curated, emotionally safe, and reversible.

Final Principle: *Support configurable social presence through social modularity. Enable user-defined collaboration with opt-in controls,*

asynchronous modes, and emotional safety mechanisms. Balance the benefits of co-presence with the necessity of epistemic privacy.

5.6 FRICTION SENSITIVITY AND GRACEFUL DESIGN

Participants in this study consistently emphasised that their decisions to adopt or abandon tools were shaped not only by technical functionality, but by emotional readiness and onboarding experiences. Contrary to assumptions that neurodivergent users reflexively avoid unfamiliar tools, the data revealed a more nuanced process: abandonment often followed the slow accumulation of friction — glitches, lag, unexpected behaviours — compounded by unmet expectations and unclear guidance. Participant 1, for instance, needed what they described as "kindergarten - level" walkthroughs to feel safe exploring new systems. In the absence of such onboarding, even minor interface confusion could trigger permanent disengagement.

This dynamic points to a deeper form of sensitivity—one rooted not in technical limitation but in *affective thresholds*: emotional limits beyond which a system becomes intolerable. Participants did not necessarily require perfect reliability; rather, they needed emotionally legible systems that failed in ways they could understand and repair. In this light, successful onboarding emerged not as a luxury but as a foundational mechanism for building emotional trust and mitigating the volatility of early tool experiences. Where onboarding was emotionally attuned, participants were more likely to persevere through early friction and develop a sustainable relationship with the tool.

The literature on usability and hybrid interface design tends to emphasise robustness, simplicity, and continuity of experience. However, these frameworks rarely account for the emotional toll of malfunction or the need for systems to "fail gracefully"—to make breakdowns legible, reversible, and minimally disruptive to cognitive flow. In neurodivergent contexts, graceful failure is not a bonus feature but a core precondition for continued use. Revised design principles (8–10) respond to this gap by reframing friction sensitivity as a function of both affective safety and cognitive trust.

Analogue tools were praised for their predictability and tactile consistency, but not idealised. Participants acknowledged the limits of paper-based workflows, particularly around archiving and revision. Yet analogue tools rarely introduced hidden

complexity or unanticipated behaviour. They failed in familiar, repairable ways. Digital systems, in contrast, carried the risk of inscrutable behaviour—an update, a popup, an error message—which could exceed a user's emotional threshold even when technically minor. Critically, such volatility was tolerated only when trust had already been established—often through emotionally legible onboarding that framed the system as safe, navigable, and recoverable.

These findings suggest that friction sensitivity is not just a usability issue—it is a design imperative that centers emotional safety and cognitive pacing. This reframes Augmented Paper Principle 5 (Ensure simplicity and robustness), expanding it to address not only technical robustness but also *emotional predictability*, *fail-safe mechanisms*, and *pacing coherence*. Participant 1's experience makes clear that without onboarding designed for emotional safety, even simple tools risk being permanently rejected. Likewise, Participant 3 described abandoning otherwise promising digital features simply because they introduced more confusion than clarity. What these moments reveal is that trust in hybrid tools is slow to build and easily lost—and once broken, rarely restored.

In this context, the literature's emphasis on simplicity and robustness must evolve to prioritise *graceful failure*—the ability of systems to recover from errors without escalating user distress or cognitive overload. Emotional legibility, quick recovery paths, and reversible actions are not extras; they are core design functions in neurodivergent ecologies. This includes not only preventing breakdowns but responding to them in ways that respect the user's emotional and epistemic boundaries.

Taken together, these insights challenge any model of hybrid learning that treats system reliability as purely computational. Instead, reliability must be understood as both affective and procedural. To be usable, a system must be emotionally coherent. To be trusted, it must fail legibly. And to be sustainable, it must scaffold confidence from the first interaction onward. These are not optional features—they are preconditions for adoption in ecologies where emotional volatility and cognitive pacing are central to tool negotiation.

Final Principle: *Design for friction sensitivity and graceful failure. Prioritise emotional predictability, error forgiveness, and affective onboarding as core elements of system stability. Stability must be under-*

stood not only as technical reliability, but as emotional coherence that scaffolds trust from the first interaction.

5.7 GENERAL DISCUSSION

The thematic synthesis of literature-based and empirically grounded design principles surfaces key tensions—and opportunities—for reimagining hybrid learning systems. While existing scholarship offers a solid conceptual foundation in areas like spatial cognition, embodiment, and accessibility, the lived realities of the participants in this study highlight gaps in how these theories translate into practice.

5.7.1 *From Design Intentions to Lived Realities*

A central finding across themes is that features intended to support learning—such as predictive feedback, system-driven augmentation, and enforced collaboration—often disrupted rather than assisted participants. These elements, commonly endorsed in design literature, were perceived as intrusive or misaligned with individual needs, particularly around pacing, privacy, and sensory thresholds. This underscores a persistent disconnect between *intended affordances* and *experienced effects*, especially for neurodivergent learners navigating hybrid environments.

The revised principles introduced here reflect more than technical adjustments; they mark a reorientation toward affectively aware, socially respectful, and epistemically humble design. Such systems must respect diverse rhythms, non-linear workflows, and intentional disengagement—treating these not as breakdowns, but as valid strategies.

It is worth noting, however, that these findings are derived from a small but deeply contextualised sample. While less generalisable than large-scale studies, their specificity offers critical insights into a group often overlooked by broader research. Where conventional design frameworks rely on scale, these principles gain strength from relevance. They are not universal solutions but precise responses to specific cognitive and emotional landscapes.

Rather than dismissing literature-derived principles, the goal is to recalibrate their application. General frameworks offer reach; lived experience offers depth. In contexts of neurodiversity, speci-

ficity may prove not only more ethical but more effective as a design basis.

5.7.2 *Hybrid Systems as Sites of Negotiation*

Hybrid learning environments are often framed in the literature as seamless integrations of analogue and digital modalities—a kind of linear progression from paper to screen, from gesture to tap, from static text to interactive affordance. This rhetoric of fluidity suggests that technological transitions should ideally disappear, replaced by uninterrupted workflows that privilege efficiency and interface harmony. Yet, the lived experiences of participants in this study tell a more complex story—one in which the act of transitioning is not a problem to be solved but a strategy to be preserved.

Participants did not adhere to a singular modality or a stable configuration of tools. Instead, their academic routines were shaped by continual shifts—between paper, tablet, blackboard, and screen—each selected not for its technological affordance alone, but for its cognitive, emotional, and sensory fit within a given moment. These transitions were often motivated by internal states: the need to decompress after digital overstimulation, to regain focus through tactile engagement, or to externalise abstract reasoning through spatial gesture. Far from signaling breakdown or inefficiency, these shifts were part of a practiced repertoire of self-regulation. They allowed participants to modulate cognitive load, re-establish pacing, or shift epistemic stance.

Such findings challenge dominant assumptions in design discourse that frame friction as failure and consistency as virtue. From a conventional standpoint, frequent media-switching might be read as a breakdown in system integration—an inefficiency to be minimised. But this interpretation overlooks the adaptive intelligence embedded in these transitions. Rather than disruptions, they operate as micro-negotiations between the learner and their material environment—deliberate reconfigurations of sensory and cognitive engagement that support continuity on the learner's own terms.

This reveals a deeper design tension: whether hybrid systems should prioritise coherence and continuity, or instead be built to support modulation, rupture, and reorientation. Both paradigms have merit, but they serve different user needs. For some learn-

ers, uninterrupted flow may indeed support deep work. For others—particularly neurodivergent learners—flexibility, reversibility, and moment-to-moment responsiveness may prove more valuable than seamless integration.

Accordingly, hybrid systems should not be conceptualised as pipelines for transitioning from analogue to digital, nor as linear progressions toward ever-greater interactivity. They are better understood as ecologies of practice—spatial, cognitive, and emotional environments in which the learner orchestrates their own mode-switching logic. Within such ecologies, friction need not be eliminated but should instead be made legible and navigable: a medium through which learners can regain agency, not lose it. Design, in this view, becomes less about removing obstacles and more about enabling negotiation—supporting transitions not as breakdowns but as part of the user's own cognitive choreography.

5.7.3 *Toward a Neurodivergent-Centered Design Paradigm*

The study's contribution extends beyond revised design guidelines—it advocates for a fundamental shift in how we conceptualise learning technologies. Rather than optimising for seamlessness, automation, or universal standards, a neurodivergent-centered paradigm emphasises emotional regulation, user autonomy, and modular adaptability.

Skeptics might argue that principles derived from three individuals risk lacking general validity. Traditional design theory often prizes generalisability, favoring findings from large, diverse datasets. From this standpoint, the principles here might appear too context-bound.

Yet this critique obscures a key asymmetry: mainstream design research rarely centers neurodivergent users. Its generality often stems from excluding those whose cognitive needs deviate from the norm. In contrast, this study offers rare insight into precisely those experiences, providing granular understanding of adaptive strategies that standard models overlook.

Moreover, neurodivergence challenges prevailing metrics of design success. Criteria like efficiency, sustained engagement, or seamless interaction may be misaligned—or even harmful—when applied uncritically. As shown here, behaviours such as disengaging, reverting to analogue tools, or rejecting augmentation

can be intelligent, context-sensitive responses. Recognising and supporting such strategies requires rethinking what counts as "good design."

This does not entail discarding broader design principles but re-examining their relevance and limits. For marginalised groups, specificity can illuminate where generality fails. Lived experience becomes not just illustrative, but foundational for developing new theoretical commitments.

Ultimately, the aim is not to universalise these findings but to legitimise them as a valid design basis. A neurodivergent-centered approach reframes friction, modulation, and emotional safety not as anomalies but as essential design considerations. These users do not simply need adapted tools—they reveal how existing paradigms fall short and point toward more inclusive, generative directions.

5.8 SYNTHESIS OF THEMATIC REVISIONS AND FINAL DESIGN COMMITMENTS

The preceding sections examined six core thematic clusters that emerged from the intersection of theory and empirical data. Each cluster reflects not only a design challenge but also a diagnostic lens through which the limits of existing frameworks become visible. These revisions do not discard the original literature-based principles; rather, they refine and re-situate them in light of neurodivergent learners' situated practices. The shift is not away from personalisation, embodiment, or multimodality, but toward a more contingent and relational interpretation of those values—grounded in real-world negotiation, breakdown, and adaptation.

Across participants, a common pattern emerged: tools were not judged solely on their features but on how well they aligned with users' internal rhythms, emotional thresholds, and spatial-cognitive strategies. Systems were embraced when they allowed learners to set their own terms—when modulation was not penalised, when friction was legible, and when support respected boundaries. Conversely, tools were abandoned when they acted uninvited, collapsed workflows, or disrupted embodied control. This study thus affirms a basic tenet: design is not only a technical intervention but a relational act. For neurodivergent learners, its success depends on how gracefully it supports autonomy, pacing, and epistemic safety.

In response, the final design principles consolidate the thematic revisions into a cohesive yet flexible framework. These principles are not directives for universal application but scaffolds for context-sensitive design. They are meant to guide the development of hybrid learning tools that are responsive not only to tasks, but to learners' fluctuating needs, material preferences, and emotional trajectories.

1. **Cognitive and Affective Adaptability:** *Design for user-specific modulation profiles. Enable rhythm-based tool switching, energy-aware pacing, and affect-sensitive flexibility.*

2. **Cognitive Sovereignty and User Control:** *Support retrospective, user-invited augmentation. Systems must remain emotionally legible, reversible, and subordinate to learner control.*

3. **Embodied and Situated Learning:** *Scaffold bodily engagement as opt-in and user-defined. Avoid imposed digital embodiment; support analogue gestures, layout, and movement as latent cognitive strategies.*

4. **Spatial Anchoring and Workflow Continuity:** *Preserve orientation across modalities. Design transitions as epistemic events, ensuring stable visual anchors and consistent structure.*

5. **Configurable Social Presence and Privacy:** *Implement social modularity. Provide opt-in collaboration, asynchronous options, and mechanisms that maintain epistemic safety.*

6. **Friction Sensitivity and Graceful Design:** *Design for emotional predictability and fail-safe interaction. Prioritise trust-building onboarding and graceful recovery from error or confusion.*

Together, these principles propose a recalibrated vision for hybrid learning technologies: one that privileges lived complexity over theoretical neatness, and that measures success not in terms of optimisation, but in the degree to which systems respect and respond to difference. In doing so, they chart a path toward a more inclusive, humane, and neurodivergent-informed design paradigm.

As this thesis moves into its final chapter, the challenge is no longer to reconcile theory and practice, but to articulate a coherent framework that grows from their productive tensions. The revised design principles presented here are not universal prescriptions, but context-responsive tools—rooted in the specific

needs, strategies, and vulnerabilities of neurodivergent learners. In foregrounding emotional safety, rhythmic modulation, and epistemic agency, they propose a new baseline for inclusive design: one where difference is not accommodated after the fact, but centered from the start.

6 CONCLUSION

6.1 TOWARD A NEURODIVERGENT-CENTERED FRAMEWORK FOR HYBRID LEARNING DESIGN

This thesis has explored the intersection between theoretical design principles for hybrid learning and the lived experiences of neurodivergent students. Through diary studies, interviews, and grounded thematic analysis, it has become clear that while existing frameworks promote personalisation, embodiment, and digital-analogue integration, they often fall short when confronted with the volatility, sensitivity, and strategy-driven adaptations of neurodivergent learners.

Rather than dismissing established design knowledge, this study has sought to extend it—recasting general principles in light of specific needs. The Discussion Chapter 5 articulated six revised design principles, each emerging from a cluster of empirical challenges: fluctuating cognitive states, threats to autonomy, tactically deployed embodiment, spatial disorientation, social overstimulation, and affective fragility in moments of tool breakdown. Each principle stands as a corrective—not to the intent of prior frameworks, but to their scope and assumptions.

What now emerges is a neurodivergent-centered framework for hybrid learning design: a model that does not assume seamlessness, automation, or consistency as design virtues, but instead builds around variability, negotiation, and emotional coherence. The framework is not a static checklist, but a set of interrelated commitments that reposition the learner not as a passive recipient of support, but as an active orchestrator of their learning ecology.

6.1.1 *Core Commitments of the Framework*

The framework rests on five interdependent commitments, each reflecting both critical literature gaps and user-reported tensions. To better illustrate these commitments, each theme is now accompanied by a hypothetical design scenario and contextualised with relevant literature references.

1. **Rhythmic Modulation over Seamlessness:** Design should allow users to shift modes, tools, and roles in response to momentary cognitive and affective needs. Modulation is not a sign of failure—it is a core learning strategy.
   Design Scenario: Drawing on Bødker and Klokmose's concept of artifact ecologies as layered, purpose-specific configurations of tools [5], a hybrid learning dashboard supports user-defined "cognitive modes"—such as *focus*, *review*, and *reset*. Each mode dynamically reconfigures interface parameters (e.g., text density, notification suppression, annotation affordances) to align with the learner's current cognitive and emotional state. Mode-switching is learner-initiated through a physical token, stylus gesture, or low-friction self-assessment slider. This enables seamless transitions without context loss, in line with Flood's findings on media-switching as an embodied regulation strategy rather than disruption [12]. Rather than penalising rhythm changes, the system scaffolds them—treating cognitive modulation as a primary design value.

2. **Opt-In Augmentation over Predictive Assistance:** Digital systems should never assume permission. Instead of proactive feedback, design must support user-invited augmentation—ensuring that help is emotionally legible, temporally reversible, and epistemically humble.
   Design Scenario: Responding to Singer and Alexander's findings that unsolicited feedback during digital reading disrupts comprehension and increases cognitive load [32], an annotation interface is built around a "quiet dock" system. Augmentation tools—such as automated summarisation, keyword definitions, or contextual timelines—are present but visually dormant until manually dragged into the active workspace. This ensures the system remains passive until explicitly invited. Echoing Le Cunff's critique of affective computing systems that overestimate user readiness and reduce autonomy [21], the platform avoids predictive activation entirely. Instead, it adapts to user routines only in terms of interface placement—not timing or content delivery. All augmentation features include a clearly visible "retract" function, and every interaction is stored in a reversible state log, ensuring temporal reversibility and epistemic humility.

3. **Embodied Scaffolding over Simulation:** Support embodied practices without simulating them artificially. Gesture, layout, and analogue routines are not enhancements to be

digitised, but native strategies to be scaffolded within hybrid systems.
Design Scenario: Building on Abrahamson's framework of embodied design as a means of conceptual grounding through bodily engagement [1], this learning environment integrates analogue-first practices such as paper sketching and tangible card sorting with digital augmentation that preserves embodied logic. Using passive NFC tags and overhead cameras—similar to the approach in Mackay et al.'s augmented notebooks [24]—the system does not simulate the gestures but captures their spatial configuration as a replayable "workspace snapshot history." This allows learners to review and reconstruct their embodied reasoning without flattening actions into symbolic digital equivalents, thus retaining the cognitive richness of analogue workflows.

4. **Cross-Modal Anchoring over Interface Optimisation:** Transitions between digital and analogue must be treated as cognitive events. Spatial logic, consistent structure, and anchoring cues should be preserved across modalities to protect workflow integrity.
Design Scenario: Extending Mangen et al.'s insights into spatial disorientation in digital environments [25] and Delgado et al.'s findings on modality-linked memory gaps in neurodivergent readers [11], this hybrid annotation system treats analogue-to-digital transitions as epistemic reframing points. Learners manually "migrate" handwritten ideas into a digital canvas, but rather than reproducing layout, the system initiates restructuring prompts (e.g., "cluster related terms," "rephrase for clarity," "draw a next-step diagram"). Scannable visual anchors in paper margins do not mirror content but open interactive, context-sensitive elaboration zones. In doing so, the system turns transition into transformation—deepening conceptual understanding while preserving spatial orientation through intentional anchoring and structural continuity.

5. **Emotional Legibility over Robustness Alone:** Tools must not only work—they must fail well. Design must centre onboarding, recovery paths, and emotional predictability as foundational elements of usability.
Design Scenario: Informed by Wollscheid et al.'s emphasis on user trust and affective safety during digital learning [37], and Prieto et al.'s work on the importance of transparent feedback in collaborative environments [30], this

> platform introduces a "soft start" onboarding flow. Features are revealed progressively, based on user curiosity rather than enforced tutorials. When errors occur, messages use affectively neutral or validating language (e.g., "Let's rewind and try again" instead of system alerts). Every feature includes a "calm interface" toggle that mutes visual transitions, suppresses feedback loops, and maintains UI stasis for moments when sensory regulation or recovery is needed. This fosters emotional predictability and trust—crucial for users with heightened sensitivity to breakdown, ambiguity, or overstimulation.

Together, these commitments consolidate the user-informed logic of the framework, transforming empirical divergences into actionable design priorities.

6.1.2 *Designing for Epistemic Safety*

Underlying all five commitments is a unifying concern: epistemic safety. For neurodivergent learners, this entails not just access to information, but the ability to engage with it on their own terms—without being overwhelmed, misread, or prematurely corrected. Epistemic safety is maintained when users feel trusted, respected, and in control of their pacing, attention, and expressive modalities.

In this light, the framework offers more than design suggestions. It proposes a shift in what counts as good learning technology: not the optimisation of flow or retention, but the cultivation of environments where learners can meaningfully regulate how they learn, express, and interact.

6.1.3 *Implications and Future Work*

While grounded in a small empirical base, this framework contributes to a growing body of research advocating for inclusive, situated design and is a valuable contribution to neurodivergent HCI. It challenges the universality of dominant metrics—such as efficiency, automation, and seamless integration—and instead proposes a paradigm built around difference, flexibility, and emotional attunement.

Future work could test and refine this framework in co-design settings, broader participant cohorts, or longitudinal studies. Equally, it may serve as a critical lens through which to audit

existing learning platforms for their inclusivity and epistemic sensitivity.

6.1.4 *Final Reflection*

Designing for neurodivergent learners does not require entirely new systems—it requires rethinking how existing systems operate by default. Many common design assumptions, such as the value of immediate feedback, seamless transitions, or proactive support, may benefit some users but can create barriers for others. As this study has shown, these features can become sources of friction or disengagement when they do not align with a learner's pacing, sensory thresholds, or cognitive strategies.

The design principles developed here suggest a shift in emphasis: from optimising for general efficiency to supporting individual variability. This is not about discarding established design goals, but about adjusting them to better account for diverse learning needs. Starting from neurodivergent experience surfaces design insights that are broadly valuable, particularly in contexts where flexibility, trust, and user control matter.

Rather than treating these considerations as edge cases or accommodations, they should inform the foundation of hybrid learning design. Doing so can lead to systems that are not only more accessible but also more adaptable, responsive, and respectful of learners' autonomy.

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7 APPENDIX A: DIARY GUIDE

HEJ, OG TAK FORDI DU VIL HJÆLPE MED AT SKRIVE DAGBOG TIL MIT SPECIALE

På grund af sygdom kommer denne besked lidt senere end planlagt, men jeg håber stadig, at du er villig til at deltage i dette eksperiment.

HVAD SKRIVER JEG OM?

Jeg skriver speciale om værktøjer, der befinder sig i gråzonen mellem det digitale og det analoge. Det digitale dækker over ting som computer, mobil, webservices, iPad, PDF'er osv., mens det analoge omfatter blyant, papir, tavle og kridt (måske tusch), fysiske bøger osv. Mit fokus er på de værktøjer, der ligger imellem disse to kategorier, eller de, der anvendes i overgangen mellem dem (det skal du ikke tænke for meget over – det er mit job at finde ud af, hvad de præcist er).

HVORFOR GØR JEG DET?

Jeg blev opmærksom på dette emne, da jeg i et specialeforberedende projekt sidste semester fandt et interessant sammenspil mellem det analoge og det digitale blandt de studerende, jeg talte med. Det vil jeg gerne undersøge nærmere for at se, om jeg kan bidrage til at optimere eller klargøre nogle af de processer, der opstår i dette samspil.

HVAD SKAL DU GØRE?

Du skal skrive dagbog. Hver dag – eller når du bruger et værktøj – skal du tage din dagbog frem (den kan være fysisk eller digital, det bestemmer du selv) og besvare følgende spørgsmål:

1. **Hvilket værktøj har du brugt?** (Se listen ovenfor for eksempler på værktøjer)
2. **Hvordan har du brugt det, og hvad har du brugt det til?** (Vær gerne så detaljeret som muligt)

3. **I hvilken sammenhæng har du brugt værktøjet?** (Fx til studier, hjemmebrug osv.)
4. **Hvor har du brugt værktøjet?** (Fx på universitetet, derhjemme osv.)

Dette hjælper mig med at kortlægge din *artifact ecology* – altså et netværk af de værktøjer, du bruger i din studietid – som jeg derefter kan analysere.

HVOR LÆNGE SKAL DU GØRE DETTE?

I 14 dage (Hvis du ikke ønsker at skrive dagbog i weekenderne, er det helt okay).

AFSLUTTENDE ORD

Endnu en gang tak, fordi du vil hjælpe mig! Jeg ser frem til at modtage din dagbog og komme i gang med at analysere dataene til mit speciale.

8 APPENDIX B: DIARY ANALYSIS

PARTICIPANT 1

Open Coding

- Color-coded note techniques using both physical and digital materials
- Use of audiobooks and sitcoms for auditory regulation
- Collaboration with study partner for assignments and comprehension
- Rhythmic structuring through poledance, self-imposed deadlines, and calendar routines
- Strategic recovery (breaks, support from network)

Open Coding of Participant 1

CODE: MULTIMODAL NOTETAKING STRATEGIES

- **Tactile Structuring:** Use of physical textbook + color-coded highlighters (green, yellow, red, blue) with distinct semantic roles.
- **Digital-Auditory Input:** OneNote used with stylus on foldable tablet-PC for drawing molecular structures.
- **Selective Attention Strategy:** Alternates between writing, listening, and understanding due to difficulty maintaining all three simultaneously.

Emerging Dimensions: Modality (analog/digital), Task Type (chemistry/math), Focus Distribution (visual/manual/auditory).

CODE: COGNITIVE REGULATION AND LOAD MANAGEMENT

- **End-of-Day Shutdown:** Self-prescribed recovery time after back-to-back lectures (24/02).
- **White Noise Tuning:** Uses familiar audiobooks and sitcoms to create concentration-conducive environments, especially when medication fades.

- **Environmental Sensitivity:** Notes reduced focus in silence and overstimulation in unfamiliar noise.
- **Medication Calibration:** Reports differing environmental needs depending on whether ADHD medication is active.

Emerging Dimensions: Dependency (medication, setting), Load Type (intrinsic/extraneous), Coping Mode (passive/active).

CODE: PEER ANCHORING AND DISTRIBUTED COGNITION

- **Collaboration With Study Partner:** Used for idea formulation, error correction, and shared LaTeX editing.
- **Synchronous Writing Sessions:** "Writing together/Solving academic tasks together" as a form of emotional and cognitive support (e.g., 27/02–03/03).
- **Accountability Structures:** Agreement with study partner to "hold each other accountable" during high-stress periods.

Emerging Dimensions: Interaction Format (co-presence, digital), Function (motivation, correction, emotional buffering).

CODE: TEMPORAL STRUCTURING AND INTERNALIZED DEADLINES

- **Poledance as Temporal Anchor:** Used as motivational constraint to increase work efficiency (25/02).
- **Self-assigned Milestones:** Uses "self-imposed deadlines" to structure unaided study sessions.
- **Preemptive Overviews:** Mid-semester planning used to regain overview and prioritize catch-up (06/03).

Emerging Dimensions: Source of Constraint (external/internal), Motivation Type (deadline-driven, fear of falling behind).

CODE: EMOTIONAL BURNOUT AND SOMATIC STRESS RESPONSE

- **Stress-Linked Symptoms:** Reports facial rash during peak workload (03/03–05/03).
- **Recognition of Cognitive Limits:** Declares explicit rest day to avoid collapse despite course backlog (05/03).
- **Emotional Cost of Missteps:** Demotivation and frustration after 7 hours on an incorrect problem approach (03/03).

Emerging Dimensions: Stress Manifestation (physical/affective), Timing (acute vs. cumulative), Relief Strategies (rest, baking, support network).

CODE: SOCIAL AND FAMILIAL SUPPORT STRUCTURES

- **Partner Support:** Emotional and practical help, e.g., food prep during submission stress (28/02).
- **Parental Scaffolding:** Mother provides strategic warning about cognitive cycles and neurodivergent traits (06/03).

Emerging Dimensions: Relation Type (peer, familial), Function (logistical, emotional, anticipatory regulation).

CODE: ACADEMIC PRIORITIZATION AND TACTICAL SKIPPING

- **Intentional Skipping:** Skips Theoretical Exercises to focus on more urgent Modeling revision (26/02).
- **Forecasting Focus Shifts:** Predicts upcoming shift in effort due to incoming major assignment (Pointless Topology).

Emerging Dimensions: Skipping Rationale (emotional energy, workload forecasting), Trade-off Awareness (explicit/implicit).

CODE: TOOL-SPECIFIC AFFORDANCES

- **LaTeX as Finalization Tool:** Reserved for high-stakes writing tasks; used after conceptual work is complete.
- **LøbeSeddel for Real-Time Retrieval:** Designed to cue note quality improvement during lecture decoding (25/02).
- **Calendar Use for Pacing:** Used to scaffold transitions between tasks and commitments.

Emerging Dimensions: Task Phase (planning, execution, submission), Modality (visual, spatial, temporal).

Axial Coding

Axial Coding of Participant 1

CODE: MULTIMODAL NOTETAKING STRATEGIES

- **Conditions:** High information density in back-to-back lectures; modality-specific demands (visualization in chemistry, precision in math).
- **Contextual Factors:** Need to remain cognitively engaged throughout long teaching days; variation in subject type and tool compatibility.

- **Strategies:** Alternates between OneNote for chemistry (tablet + stylus for diagrams) and physical textbooks with highlighter annotations for math. Uses color coding with specific semantic rules to deepen processing.
- **Consequences:** Supports real-time comprehension but also reveals cognitive strain—must switch between listening and writing, leading to selective attention trade-offs.

CODE: COGNITIVE REGULATION AND LOAD MANAGEMENT

- **Conditions:** Extended cognitive exertion (e.g., 8–17 lecture days), fading medication effects in the evening.
- **Contextual Factors:** Sensory sensitivity; task complexity; time of day; medication state.
- **Strategies:** Deploys white noise (e.g., familiar audiobooks or sitcoms) to reduce auditory overload; schedules passive media for decompression; avoids silence or unpredictable stimuli.
- **Consequences:** Enables sustained attention during late-day work sessions, but also reveals adaptive dependency on external modulation. Environmental alignment becomes a prerequisite for productivity.

CODE: PEER ANCHORING AND DISTRIBUTED COGNITION

- **Conditions:** High-stakes assignments, complex conceptual formulation, emotional overload.
- **Contextual Factors:** Collaborative compatibility with study partner; shared subject knowledge; looming deadlines.
- **Strategies:** Uses co-writing, real-time sparring, and shared LaTeX sessions to externalize and stabilize complex reasoning. Also leverages peer check-ins for emotional regulation and accountability.
- **Consequences:** Increases conceptual clarity and motivation; prevents cognitive isolation; may offload language formulation to others, compensating for self-identified articulation difficulties.

CODE: TEMPORAL STRUCTURING AND INTERNALIZED DEADLINES

- **Conditions:** Limited executive functioning windows; variable motivation across the day.

- **Contextual Factors:** Fixed activities (e.g., poledance, meetings); absence of external deadlines.
- **Strategies:** Constructs self-imposed time constraints (e.g., "self-imposed deadlines"); uses physical activities like poledance to generate natural endpoints and drive task completion.
- **Consequences:** Converts temporal pressure into momentum. Enhances efficiency but may increase stress if goals aren't met. Adds structure to unbounded self-study time.

CODE: EMOTIONAL BURNOUT AND SOMATIC STRESS RESPONSE

- **Conditions:** Accumulated workload (Pointless Topology + backlog); social and academic responsibilities.
- **Contextual Factors:** Overlapping deadlines; difficulty articulating problems; past academic trauma (fear of failure, ADHD-related stigma).
- **Strategies:** Pushes through long workdays (e.g., 7–8 hour sessions), then recognizes signs of breakdown (e.g., skin rash) and institutes rest days or passive recovery (baking).
- **Consequences:** Demonstrates self-awareness of cognitive limits, yet pattern indicates reactive rather than proactive stress management. Also shows successful emotional reset when rest is prioritized.

CODE: SOCIAL AND FAMILIAL SUPPORT STRUCTURES

- **Conditions:** Periods of stress, overwhelm, and cumulative academic pressure.
- **Contextual Factors:** Ongoing support from romantic partner and mother; shared household tasks; long-standing familial understanding of neurodivergence.
- **Strategies:** Accepts emotional and logistical help (e.g., food prep during deadline crunch); consults mother on cognitive-emotional patterns during semester transitions.
- **Consequences:** Emotional buffering prevents collapse; aids in re-establishing realistic expectations; long-term impact includes improved self-awareness and reduced shame around pacing needs.

CODE: ACADEMIC PRIORITIZATION AND TACTICAL SKIPPING

- **Conditions:** Conflicting deadlines; perception of higher value in one task over another.
- **Contextual Factors:** Personal energy levels; prior understanding of subjects; social expectations (e.g., tutorforening meetings).
- **Strategies:** Skips or defers lower-priority work (e.g., Pointless Topology Theoretical Exercises) in favor of emergent needs (e.g., Modellering catch-up); applies strategic triage.
- **Consequences:** Prevents overextension but leads to cumulative task burden; requires catch-up planning later in semester; signals strong metacognitive awareness of limits.

CODE: TOOL-SPECIFIC AFFORDANCES

- **Conditions:** Different phases of academic workflow (notetaking, review, writing).
- **Contextual Factors:** Subject-specific needs (e.g., math vs. chemistry), device affordances (stylus, LaTeX compatibility), spatial preferences (e.g., kontor vs. home).
- **Strategies:** Uses LøbeSeddel to preemptively mark key concepts for later elaboration; LaTeX only during finalization; color annotation to scaffold memory; calendar to integrate social, academic, and physiological rhythms.
- **Consequences:** Creates efficient, layered workflow with clear task separation. However, reliance on particular tools may increase friction if tools malfunction or are unavailable.

Selective Coding

Rhythm and self-regulation as coping strategy in hybrid study practices

Selective Coding: Participant 1

Core Category: *Situated Cognitive Self-Regulation in Hybrid Learning Environments*

This core category captures Participant 1's continuous and adaptive effort to regulate cognitive load, energy, and performance across dynamic learning contexts. Participant 1's diary

reveals a layered strategy for managing multimodal input, executive function, and emotional sustainability while negotiating the demands of complex academic work. Participant 1's hybrid learning ecology is shaped by deliberate modulation of tools, environments, routines, and support systems.

CONDITIONS

- **High cognitive demand and content abstraction:** Courses like Pointless Topology and Modellering require abstract reasoning and sustained effort, creating intensive *intrinsic cognitive load*.
- **Executive function fluctuations:** Medication windows, time of day, and accumulated fatigue shape Participant 1's cognitive availability and determine which tools or strategies are viable at a given time.
- **Environmental and sensory sensitivity:** Noise levels, social surroundings, and background stimuli impact Participant 1's ability to concentrate and require active environmental calibration.

CONTEXTUAL INFLUENCES

- **Structured-unstructured hybrid rhythms:** Participant 1 thrives in externally structured contexts (e.g., lectures, meetings) but must scaffold structure during open-ended study periods.
- **Tool affordances and constraints:** Participant 1 selectively uses tools like OneNote (for drawing), physical books (for annotation), and LaTeX (for formal output) to align with specific cognitive tasks.
- **Social scaffolding:** Study partner collaboration and familial/partner support provide emotional anchoring and co-regulation of workload, motivation, and perspective.

STRATEGIES (ACTIONS/INTERACTIONS)

- **Multimodal notetaking and annotation:** Uses differentiated tools (OneNote, physical book, color-coded highlighters) to break down complexity and support memory retention—facilitates *germane cognitive load*.
- **Temporal structuring through embodied anchors:** Aligns productivity around known endpoints (e.g., poledance), treating them as implicit deadlines that enhance motivation and focus.

- **Environmental self-modulation:** Uses background sitcoms or audiobooks to generate white noise and regulate attention, especially when medication efficacy wanes—manages *extraneous cognitive load*.
- **Emotional offloading via rest and reframing:** Chooses strategic disengagement (e.g., full rest day, baking) when burnout manifests somatically—interrupts escalating cognitive overload.
- **Task triage and academic prioritization:** Skips low-yield sessions or defers tasks in favor of urgent conceptual recovery, guided by metacognitive assessment of her own bandwidth.

OUTCOMES

- **Adaptive continuity in learning:** Participant 1 demonstrates resilient recovery from overload by toggling between active work, sensory regulation, and decompression—enables ongoing engagement despite setbacks.
- **Metacognitive clarity and pacing realism:** Participant 1 articulates patterns of stress and decline over the semester, leveraging past experience to mitigate spirals of shame or collapse.
- **Context-specific fluency across modalities:** Participant 1's ability to shift between digital and analog tools, solo and collaborative tasks, and structured vs. unstructured time indicates advanced ecological flexibility.
- **Emergent learning ecology shaped by cognitive economy:** Rather than optimizing for maximum efficiency, Participant 1 builds a system that sustains them across time, emotionally and intellectually, by actively managing all three types of cognitive load.

COGNITIVE LOAD THEORY INTEGRATION Cognitive Load Theory provides a scaffold for understanding Participant 1's learning practices:

- **Intrinsic Load:** Manifest in their engagement with abstract, logic-heavy content like Topology and recursive computation—managed via multimodal breakdowns and external support.
- **Extraneous Load:** Reduced via customized environments (e.g., sitcoms as white noise), sensory tools, and consistent

routines. Increased when system failures (e.g., misunderstanding a task) occur.

- **Germane Load:** Deliberately cultivated through color-coded annotation systems, structured co-writing with peers, and reflective study planning that links present tasks to exam preparation.

This integrated coding structure shows how Participant 1's tool-mediated study practices form a responsive ecology of support—one that dynamically aligns internal cognitive states with external academic demands, allowing them to stay afloat in a cognitively complex learning environment.

Cognitive Load Theory

- **Intrinsic load:** High in abstract subjects, managed through externalization
- **Extraneous load:** Minimized through predictable environments and familiar noise
- **Germane load:** Supported by color-coding, collaboration, and structuring

Intrinsic Cognitive Load

Intrinsic load stems from the complexity of the material or task itself—especially those with high element interactivity or abstract reasoning demands.

- **Pointless Topology:** Repeated mentions of conceptual difficulty and exhaustion from working with abstract mathematical content (e.g., "en hel dag med svær eksamens aflevering").
- **Modellering (chemistry course):** Participant 1 expresses concern about falling behind and the overwhelming nature of catching up ("det virker overvældende at skulle begynde på").
- **Back-to-back lectures (e.g., Mandag 24/02):** Struggling to both listen, understand, and take notes during complex teaching segments suggests high intrinsic demand.
- **Simultaneous modality use (writing, listening, understanding):** Especially during Theoretical Exercises, where Participant 1 must extract keywords in real time and apply them immediately.

- **Reengaging with previously missed material:** Particularly on 06/03, where Participant tries to catch up mid-semester and feels cognitive overload from accumulated gaps.

Extraneous Cognitive Load

Extraneous load arises from the way content or tasks are presented, including environmental distractions, interface issues, or disorganized workflows.

- **Multitasking across tools and modalities:** Switching between OneNote, LaTeX, physical books, etc., with limited integration (e.g., notes on paper vs. digital write-ups) adds interface friction.
- **Environmental noise and sensory modulation:** Needs familiar white noise (e.g., sitcoms, audiobook they has heard before) to reduce cognitive overload from uncontrolled auditory input. This is a compensatory strategy against distracting environments.
- **Unpredictable tool transitions:** Difficulty navigating spontaneous shifts between teaching formats (e.g., having to process Theoretical Exercises right after dense lectures) adds to non-essential cognitive strain.
- **Poor timing and feedback alignment:** Not having Theoretical Exercises prep info until Monday evening, despite early Tuesday sessions, forces reactive rather than proactive engagement.
- **Stress from workload mismanagement and late insight:** The "wall of frustration" that arises when Participant 1 suddenly sees the scale of what they have missed in a subject.

Germane Cognitive Load

Germane load is the productive mental effort used to build and refine understanding or structure knowledge meaningfully.

- **Color-coded annotation system:** Especially in math, Participant 1 uses yellow, red, and green to differentiate types of information—this reflects deliberate schema-building.
- **Use of the "LøbeSeddel" during Theoretical Exercises:** Writing down keywords in advance and highlighting them later helps Participant 1 manage complexity and focus on conceptual integration.

- **Poledance as embodied cognitive structuring:** Acts as a time boundary to increase efficiency and focus—this supports mental resource allocation for deeper processing.
- **Collaboration with study partner:** This aids in comprehension, formulation, and debugging—particularly helpful when Participant 1 struggles with articulation.
- **Schema recovery and planning routines:** Especially during catch-up phases, Participant 1 breaks material into manageable chunks, identifies priority topics, and seeks social support. These are metacognitive strategies that channel cognitive effort into sense-making rather than overwhelm.

PARTICIPANT 2

Open Coding

- Frequent multitasking and dispersed tool use
- Use of pen and paper for pseudocode and comprehension
- Digital/analog combinations in work routines
- Use of music, podcasts, and breathing exercises for emotional regulation
- Social communication as both support and distraction

Open Coding of Participant 2

CODE: FRAGMENTED DIGITAL ECOLOGIES

- **Multiplatform Troubleshooting:** Switches between VS Code, Github, PDF viewers, Brightspace, and Stack Overflow to solve assignments.
- **Tool Complementarity:** Combines digital reference materials and editing environments for code debugging and coursework.
- **Navigation Overhead:** Reports delays and confusion due to spread of information across unintegrated tools.

Emerging Dimensions: Tool Fragmentation, Navigation Cost, Task Continuity.

CODE: AUDITORY SCAFFOLDING

- **Podcasts as Rhythmic Anchors:** Uses Spotify and YouTube during commuting, study, and decompression to modulate pacing.
- **Mood Regulation via Audio:** Selects familiar or repetitive soundtracks to maintain concentration and emotional stability.
- **Environment Adaptation:** Uses audio to reduce silence-induced distraction during solitary work sessions.

Emerging Dimensions: Modality (auditory), Regulation Target (focus/mood), Routine Embeddedness.

CODE: COGNITIVE OFFLOADING

- **Passive Recovery Tools:** Engages with YouTube, Netflix, Disney+, and games like Minesweeper during stress or low-energy periods.
- **Affective Disengagement:** Screens used to distance from feelings of pressure or overload.
- **Productive Avoidance:** Sometimes uses media or puzzles (e.g., Sudoku) as a low-cognitive-load bridge toward later re-engagement.

Emerging Dimensions: Load Type (affective/cognitive), Activity Engagement Level (passive/interactive), Functional Outcome (delay vs. regulation).

CODE: ANALOG-FIRST STRUCTURING

- **Sketching Pseudocode:** Draws recursive logic and problem breakdowns on paper before transferring to digital formats.
- **Visual Planning for Writing Tasks:** Uses hand-drawn dramaturgical diagrams to scaffold written assignments.
- **Analog–Digital Transitioning:** Paper precedes formalization in Word, Overleaf, or code editors.

Emerging Dimensions: Modality Sequencing, Externalization Function, Planning Depth.

CODE: MOOD-CONTINGENT RESOURCE MANAGEMENT

- **Tool Use Based on Energy Level:** Chooses between code, e-books, videos, or puzzles depending on mental state.
- **Adaptive Engagement:** Switches between work and rest tools to match current affective load.
- **Temporal Rhythm Awareness:** Tracks motivational fluctuations across the week to plan accordingly.

Emerging Dimensions: Energy-State Coupling, Self-Regulatory Flexibility, Temporal Self-Awareness.

CODE: INSTITUTIONAL PLATFORM DEPENDENCY

- **Brightspace as Reference Anchor:** Constantly checks for updates, task clarifications, and course overviews.

- **Github as Production Environment:** Uses version control and online repo as cognitive scaffolding.
- **Workflow Structuring via Platforms:** Dependencies on institutional interfaces shape pacing and visibility of tasks.

Emerging Dimensions: Platform Centrality, Dependence on Digital Infrastructure, Structural Anchoring.

CODE: DELIBERATE SENSORY DOWNREGULATION

- **Tactile and Visual Tools for Recovery:** Engages in puzzles, simple games, or light fiction as sensory decompression.
- **Strategic Screen Time:** Uses screen-based downtime to compartmentalize academic and personal energy.
- **Rest-Work Boundary Enforcement:** Selective use of low-demand tasks to maintain distinction between study and rest.

Emerging Dimensions: Sensory Regulation, Cognitive Reset, Contextual Withdrawal.

CODE: LIGHTWEIGHT SOCIAL CO-REGULATION

- **Memes and Messaging:** Uses Instagram and Messenger to share humorous content and sustain affective connection.
- **Digital Peer Anchoring:** Maintains light social engagement to buffer emotional isolation during study.
- **Informal Accountability:** Social media acts as passive structure for staying on track via shared reference.

Emerging Dimensions: Relational Anchors, Emotional Buffering, Informal Peer Structures.

Axial Coding

Axial Coding: Participant 2

In this phase, categories developed during open coding were relationally elaborated by linking contextual triggers, strategic responses, and resultant outcomes. These axial models provide a foundation for identifying deeper patterns of cognitive regulation, tool coordination, and self-management.

CATEGORY: FRAGMENTED DIGITAL ECOLOGIES

- **Conditions:** High technical complexity of programming tasks (e.g., recursion, re-handins); need to reference multiple types of content (slides, repositories, documentation).
- **Context:** Assignment-driven days; distributed across university and home settings.
- **Strategies:** Switching between Github, Brightspace, PDF viewers, Stack Overflow, and VS Code to piece together solutions.
- **Consequences:** Increased extraneous cognitive load; interruptions in cognitive flow; reliance on physical sketching or auditory scaffolds to restore coherence.

CATEGORY: AUDITORY SCAFFOLDING

- **Conditions:** Need to maintain mental pacing during transitions (e.g., cycling, returning home); energy dips; difficulty focusing in silence.
- **Context:** Commuting, self-study, late-evening sessions.
- **Strategies:** Recurrent use of music, podcasts, and familiar audio stimuli to bracket learning time or decompress.
- **Consequences:** Improved task initiation and sustained focus; more consistent affect regulation across physical locations.

CATEGORY: COGNITIVE OFFLOADING

- **Conditions:** Emotional overwhelm, workload pressure, low energy days (e.g., 12/03 and 17/03); unresolved tasks piling up.
- **Context:** At-home study environments with minimal accountability.
- **Strategies:** Use of passive or semi-passive media (Netflix, Disney+, YouTube), puzzle-solving, and e-books to disengage or regain calm.
- **Consequences:** Temporary mental relief; however, potentially delays re-engagement and can reinforce avoidant patterns.

CATEGORY: ANALOG-FIRST STRUCTURING

- **Conditions:** Conceptually difficult problems (e.g., Theoretical Exercises coding, dramaturgical analysis); digital interfaces feel inadequate for sensemaking.
- **Context:** Formal education settings (classroom, lecture); study prep sessions.
- **Strategies:** Diagramming, sketching pseudocode, highlighting texts prior to digitization.
- **Consequences:** Reduces intrinsic load by decomposing tasks; supports better digital transitions later (e.g., Word, Overleaf).

CATEGORY: MOOD-CONTINGENT RESOURCE MANAGEMENT

- **Conditions:** Energy fluctuations; mood instability; sense of overload or underload.
- **Context:** Mid- to late-week drop in motivation or post-deadline crash.
- **Strategies:** Choosing study tools and media types that match current cognitive or affective state (e.g., using e-book or YouTube when depleted).
- **Consequences:** Maintains partial engagement on low-energy days; fosters self-awareness but may prolong cycles of inefficiency.

CATEGORY: INSTITUTIONAL PLATFORM DEPENDENCY

- **Conditions:** Course logistics tied to digital platforms (e.g., deadlines, materials, announcements).
- **Context:** Core academic workflow; especially programming and dramaturgy courses.
- **Strategies:** Regular checking of Brightspace and Github; use of these platforms as mental "anchors" for task management.
- **Consequences:** Enables orientation and progress tracking; however, risks of overload when platform content is poorly organized.

CATEGORY: DELIBERATE SENSORY DOWNREGULATION

- **Conditions:** Sensory overstimulation; mental exhaustion; difficulty winding down after work sessions.
- **Context:** Late evenings, weekends, or post-task decompression.
- **Strategies:** Use of tactile or low-demand visual tasks (puzzles, Minesweeper), light fiction, or ambient media.
- **Consequences:** Supports recovery and rebalancing; maintains a boundary between academic and personal time.

CATEGORY: LIGHTWEIGHT SOCIAL CO-REGULATION

- **Conditions:** Emotional need for affirmation, shared experience, or levity during solo study.
- **Context:** Between-task moments; breaks; mobile-first environments.
- **Strategies:** Sharing memes, chatting on Messenger, scrolling Instagram.
- **Consequences:** Strengthens peer connection; offers low-cost relational support; may delay re-engagement.

Selective Coding

Multimodal coping and fragmented tool use in a complex academic environment

Selective Coding: Participant 2

The final phase of the grounded theory process integrated all prior findings by identifying a *core category* that accounted for the largest range of variation across contexts, conditions, strategies, and consequences. Through comparison of axial coding categories, theoretical memos, and diary data, the following central theme emerged:

CORE CATEGORY: SITUATED MODULATION OF COGNITIVE LOAD

Participant 2's diary data consistently points to a central strategy of modulating cognitive engagement based on contextual, emotional, and sensory conditions. Rather than adopting a single study mode, they fluidly adapt their environment, media, and modality choices in response to changing internal and external demands. This modulation is not arbitrary but structured

around perceived cognitive load, motivational rhythms, and the affordances of specific tools.

Integration of Major Categories

Each axial category was positioned in relation to the core as either a *condition*, *strategy*, *contextual influence*, or *outcome* within the broader pattern of situated modulation:

- **Conditions:**
  - Task complexity (e.g., recursion, dramaturgical analysis)
  - Emotional state (e.g., stress, low energy, overload)
  - Temporal structure (e.g., deadlines, recovery days)
- **Strategies:**
  - Analog-first structuring (e.g., pseudocode sketches)
  - Media-switching for affective regulation (e.g., YouTube, Spotify)
  - Tactile and visual downregulation (e.g., puzzles, e-books)
  - Lightweight social interaction (e.g., Messenger, Instagram)
- **Contextual Influences:**
  - Fragmented platform architecture (Brightspace, Github, Stack Overflow)
  - Hybrid academic environments (university, classroom + home-based learning)
  - Institutional dependence on digital systems for task orchestration
- **Outcomes:**
  - Temporary relief from extraneous load (via offloading or rest)
  - Adaptive maintenance of productivity during energy dips
  - Risk of engagement fragmentation and inefficient task transitions
  - Emergent self-awareness and metacognitive skill in navigating complexity

Theoretical Integration: Cognitive Load Theory (CLT)

Participant 2's study practices illustrate a highly dynamic interaction between intrinsic, extraneous, and germane cognitive load. Their modulation strategies map closely onto CLT's load types:

- **Intrinsic Load:**
  - High intrinsic load in programming tasks, Theoretical Exercises, and dramaturgical synthesis (e.g., N-Queens problem, manuscript analysis)
- **Extraneous Load:**
  - Extraneous load introduced by fragmented digital ecosystems, sensory distraction, and emotional dysregulation
  - Actively managed through background audio, rest days, or simplified media
- **Germane Load:**
  - Optimized through analog note-making, repeated tool use (e.g., VS Code), and metacognitive regulation via mood-contingent choices

Theoretical Proposition

For neurodivergent learners operating in complex hybrid learning environments, effective navigation is driven not by static tool mastery but by situated modulation of cognitive load—balancing affective states, task demands, and media affordances in real time to preserve cognitive continuity.

Cognitive Load Theory

- **Intrinsic load:** High in recursion and coding, mitigated by analog externalization
- **Extraneous load:** Increased due to frequent tool-switching and overlap
- **Germane load:** Partially supported through physical reflection and visual aids

Intrinsic Cognitive Load

Intrinsic load relates to the complexity of the learning material and its element interactivity.

- **Programming tasks and recursion concepts:** Use of VS Code, pseudocode, and problem-solving in Theoretical Exercises and handins/re-handins involves abstract, multi-step reasoning and imposes high intrinsic demand.
- **Simultaneous media navigation:** Reading slides in PDF while solving programming assignments places demand on working memory across modalities.
- **Dramaturgical analysis and N-Queens problem (10/03):** Involves synthesis of theoretical and computational concepts, increasing conceptual complexity.
- **Multi-genre academic engagement:** Working on both revue-song text in Overleaf, song searching in Google, and technical assignments in VS Code requires high cognitive flexibility and task switching.
- **Catch-up sessions on Brightspace after absence (17/03):** Engaging with missed content and preparing for re-handins increases intrinsic load due to cumulative content density.

Extraneous Cognitive Load

Extraneous load is caused by the way information is presented or the surrounding environment—not the material itself.

- **Platform fragmentation:** Switching between Brightspace, Github, PDF viewers, VS Code, Stack Overflow, etc., adds navigation overhead and disrupts cognitive flow.
- **Distractions and procrastination:** Facebook, Instagram, Google Solitaire, Minesweeper, and Post.au (12/03) are explicitly noted as distractions that break task engagement.
- **Mentimeter and projector transitions (10/03):** While potentially engaging, abrupt mode-switching in classroom tech may demand non-productive focus shifts.
- **Emotional and sensory dysregulation:** On days marked by stress or low energy (e.g., 12/03 and 17/03), extraneous stimuli compound cognitive effort, forcing reliance on passive media or breathing exercises to regain composure.
- **Self-reported low energy states:** "Lav energi og behov for ro" (17/03) hinders focus, increasing load not from material but from environmental and internal states.

Germane Cognitive Load

Germane load reflects productive effort spent on constructing and automating schemas (deep learning).

- **Paper-based pseudocode and illustration (06/03):** Used to visualize and plan code, aiding in conceptual clarity and mental model building.
- **Integration of leisure and learning tools:** E-book reading for rest and sudoku for pattern recognition enhance schema fluency by alternating cognitive effort and recovery.
- **Reflection through multimodal creation:** Writing revuesong texts in Overleaf engages metacognitive processes like reformulation and synthesis.
- **Walk and talk pedagogy (10/03):** Encourages embodied cognition and verbal processing, contributing to meaningful engagement.
- **Strategic media use for self-regulation:** Repeated Spotify playlists or familiar YouTube series serve to facilitate focus and decompression.
- **Consistent tool repurposing:** VS Code is used across diverse contexts—from Theoretical Exercises preparation to re-handins —reflecting internalized workflows that reduce future cognitive load.

PARTICIPANT 3

Open Coding

- Problem-solving through pen and chalk as primary media
- Digital supplements via Texmaker, Wikipedia, Brightspace, PDFs
- Structured time management using Google Calendar
- Focus regulation through YouTube and physical proofreading
- Sensory shielding for chores (headphones, gloves)

Open Coding: Participant 3

The following codes were developed through iterative open coding of Participant 3's diary entries, following the analytic strategy described in the methods chapter. Individual actions, interactions, and reflective choices were conceptualized and clustered into thematic categories that will later inform axial coding.

CODE: ANALOG-FIRST REASONING

- **Problem Solving on Paper/Chalkboard:** Consistently starts with pen and paper or chalkboard when tackling complex math tasks (e.g., Morse Theory, Theoretical Exercises).
- **Physical Proofreading:** Uses highlighters on printed documents to identify and correct errors during revision.
- **Analog-Digital Bridging:** Takes photos of chalk work for later transcription and uses it as a scaffold in LaTeX.

Emerging Dimensions: Modality Sequence (analog → digital), Cognitive Externalization, Error Awareness.

CODE: HYBRID DOCUMENTATION ECOLOGY

- **Texmaker for Formalization:** Used for final versions of assignments; signals transition from ideation to presentation.
- **Tool Specialization:** Employs Notepad and Obsidian to organize notes by semantic structure or theme.

- **Workflow Continuity:** Moves smoothly from analog ideation to digital synthesis across multiple platforms.

Emerging Dimensions: Workflow Modularity, Semantic Layering, Tool-Task Specificity.

CODE: FOCUS MODULATION STRATEGIES

- **Auditory Rhythm with YouTube:** Uses ambient or familiar videos to sustain attention during solo sessions.
- **Time Structuring with Calendar:** Relies on Google Calendar for planning, pacing, and commitment tracking.
- **Sensory Shielding:** Wears headphones and gloves to reduce sensory distractions during chores or transitions.

Emerging Dimensions: Arousal Regulation, Temporal Anchoring, Sensory Filtering.

CODE: TASK-CONTEXT ALIGNMENT

- **Space-Specific Tasking:** Reserves academic work for university settings and avoids deep study tasks at home.
- **Modality-Matched Tool Use:** Uses analog tools for ideation and digital tools for production and delivery.
- **Intentional Simplicity in Digital Tools:** Chooses low-friction environments to minimize unnecessary interactions.

Emerging Dimensions: Spatial Segmentation, Cognitive Environment Matching, Friction Reduction.

CODE: ACADEMIC PHASE AWARENESS

- **Time-Point Segmentation:** Acknowledges major academic transitions (e.g., post-17/03) and reorganizes work accordingly.
- **Boundaries on Documentation:** Explicitly suspends diary entries during personal or unstructured time to protect cognitive resources.
- **Strategic Reflection on Shifts:** Uses diary and planning tools to reflect on mental load and scheduling balance.

Emerging Dimensions: Temporal Meta-awareness, Cognitive Boundary-Making, Reflective Pacing.

Axial Coding

Axial Coding: Participant 3

The axial coding phase elaborates on the structural logic of Participant 3's recurring practices by relating open codes to each other through causal conditions, contextual influences, actions/interactions, and consequences. This enables a shift from descriptive categories to preliminary theoretical propositions.

Category: Analog-first Reasoning

- **Causal Conditions:** High intrinsic complexity of math problems (e.g., Morse Theory, Theoretical Exercises) necessitates clarity and spatial structuring before formalization.
- **Contextual Factors:** Access to chalkboard environments at university; academic culture emphasizing hand-written derivations.
- **Strategies:** Begins problem-solving with pen and paper or chalk; uses diagrams, handwritten notes, and iterative visual sketches to externalize reasoning.
- **Consequences:** Reduced working memory load during abstraction; preserves conceptual continuity during LaTeX transcription; enables physical engagement with abstract problems.

Category: Hybrid Documentation Ecology

- **Causal Conditions:** Institutional requirements for digital submission; personal preference for analog structuring.
- **Contextual Factors:** Availability of modular tools (Texmaker, Obsidian); predictable workflows that support analog-to-digital transitions.
- **Strategies:** Captures analog drafts with phone camera; reformulates written content in LaTeX; uses Notepad and Obsidian to structure and index conceptual material.
- **Consequences:** Increased fluency in tool-switching; consistent documentation of thought processes; creates persistent, legible, and assessable deliverables.

Category: Focus Modulation Strategies

- **Causal Conditions:** Long, uninterrupted study periods; susceptibility to attentional fatigue.
- **Contextual Factors:** Solo work settings; flexible task timing; background noise sensitivity.
- **Strategies:** Plays familiar YouTube videos to maintain arousal levels; wears headphones and gloves during chores; uses Google Calendar for task pacing.
- **Consequences:** Extends cognitive endurance; maintains productivity in repetitive or solitary work; supports rhythm and momentum during large projects.

Category: Task-Context Alignment

- **Causal Conditions:** Preference for environmental consistency; awareness of sensory-cognitive coupling.
- **Contextual Factors:** University spaces afford focus and social legitimacy; home is reserved for rest or light tasks.
- **Strategies:** Restricts heavy cognitive tasks to the university; separates ideation (analog) from production (digital); avoids multitasking environments.
- **Consequences:** Maintains clarity of focus; reduces transition costs between tools and mental states; reinforces task identity through spatial separation.

Category: Academic Phase Awareness

- **Causal Conditions:** Shift in course calendar (post-project submission and new course start).
- **Contextual Factors:** Periodic academic overload; strong personal boundaries around untracked leisure time.
- **Strategies:** Uses calendar to navigate workload spikes; consciously suspends data collection during periods of rest; marks time retrospectively through diary.
- **Consequences:** Preserves reflective capacity without burnout; maintains autonomy in documentation; supports temporal organization of academic experience.

Selective Coding

Strategically balanced hybrid work in cognitively demanding contexts

Selective Coding: Participant 3

Core Category: Structured Modularity as Cognitive Strategy

The analysis culminated in the identification of a core category: **structured modularity as cognitive strategy**. This concept captures how Participant 3 navigates hybrid academic environments by segmenting tasks, spaces, and tools into clearly defined modules that correspond to specific phases of learning. This modularity is not merely a preference but an active cognitive strategy for reducing extraneous load and enhancing germane processing in high-intrinsic-load contexts. The remaining categories function as interrelated components that support or result from this central structuring mechanism.

Conditions:

- **Intrinsic complexity of subject matter:** Morse Theory, abstract algebra, and Theoretical Exercises assignments present high element interactivity, requiring clarity of representation and phased processing.
- **Temporal fluctuations in workload:** Shifts between project delivery and new course preparation necessitate reorganization of resources and routines.
- **Sensory sensitivity and environmental constraints:** Participant 3 must manage cognitive input not only from tasks but also from spatial and sensory contexts.

Strategies:

- **Analog-first reasoning:** Complex tasks are initially worked out on paper or chalkboard, which externalizes cognition and scaffolds conceptual integration (supports reduction of intrinsic load).
- **Hybrid documentation workflow:** Solutions are later formalized in LaTeX via Texmaker, with references managed via Obsidian or Notepad, preserving semantic continuity.
- **Focus modulation techniques:** Tools such as YouTube, headphones, gloves, and Google Calendar are used to self-regulate energy, reduce distraction, and support temporal pacing (addresses extraneous load).
- **Task-context alignment:** Academic labor is spatially anchored to the university setting; home is reserved for untracked, restorative activities—strengthening role clarity across environments.

Contextual Influences:

- **Institutional structure:** University expectations around digital submissions shape the necessity for hybrid workflows.
- **Availability of modular tools:** The participant selectively adopts tools that afford clean transitions between stages of a task (e.g., analog sketching → LaTeX documentation).
- **Social norms and support cultures:** The absence of social distraction in the data suggests a solitary but self-regulated academic identity.

Outcomes:

- **Cognitive economy:** By structuring tasks modularly, Participant 3 reduces transition costs and working memory overload across modalities.
- **Schema reinforcement:** Repetition of analog-to-digital transitions, coupled with structured note tools (Obsidian), supports germane load by fostering meaningful abstraction and integration.
- **Temporal sustainability:** Participant 3's routines preserve mental energy over long cycles, enabling deep work without burnout, even in intensive academic phases.

Integration with Cognitive Load Theory

Structured modularity appears to function as an implicit design principle for managing the three types of cognitive load:

- **Intrinsic Load:** Mitigated by decomposing complex tasks into analog sketches and iterative reasoning stages.
- **Extraneous Load:** Minimized through consistent tool choices, environment separation, and sensory regulation.
- **Germane Load:** Amplified through schema-building routines such as analog-to-digital transcription, printed proofreading, and semantic note organization.

This pattern suggests that the participant's hybrid ecology is not simply adaptive, but reflectively optimized to maximize deep learning under neurodivergent conditions.

Cognitive Load Theory

- **Intrinsic load:** High in Morse Theory, reduced through analog externalization
- **Extraneous load:** Minimized through planning and calm
- **Germane load:** Supported through reflection, auditory rhythm, and proofreading

Intrinsic Cognitive Load

Arises from the inherent complexity of the material and the cognitive effort required to understand it.

- **Work in Morse Theory:** This highly abstract mathematical domain naturally involves heavy intrinsic load due to its conceptual difficulty and symbolic complexity.
- **Multi-phase problem solving:** Participant 3 consistently starts with analog work (paper, chalkboard) before translating into LaTeX, requiring mental transfer across modalities and representations.
- **Theoretical Exercises prep and exam work (e.g., 17/03):** Solving exercises that require conceptual understanding and procedural fluency increases element interactivity.
- **Lectures and new course start:** Managing new content streams during course transitions (especially post-17/03) imposes additional cognitive complexity, particularly when understanding builds cumulatively.

Extraneous Cognitive Load

Generated by the learning environment or task design—elements not essential to the material itself.

- **Fragmented toolchain:** Switching between chalkboard, paper, LaTeX, PDFs, Obsidian, and camera-based documentation (e.g., photos of chalk work) can cause friction, especially without automation or integration between steps.
- **Digital reference navigation:** Use of Wikipedia, Brightspace, and PDF readers to access theoretical material could impose navigational overhead, especially if content is not well-aligned or indexed.

- **Manual transfer of solutions:** Transferring complex math from chalkboard/paper into LaTeX manually (Texmaker) without WYSIWYG feedback may introduce unnecessary memory strain and error-checking loops.
- **Environmental stimuli:** Use of YouTube as a focus aid suggests a need for external modulation—if suboptimally matched, it may also distract, especially during longer work sessions.
- **Sensory regulation(Gloves and headphones):** Helpful for reducing extraneous sensory load, but also indicative of environments that are otherwise overstimulating.

Germane Cognitive Load

Represents effort that contributes productively to schema construction and long-term learning.

- **Chalkboard use for conceptual articulation:** Participant 3 externalizes thinking by sketching out solutions on a physical surface, supporting structured reasoning and schema development.
- **Use of Obsidian and Notepad:** Indicates intentional organization and capture of personal insights, promoting deeper processing and knowledge retention.
- **Iterative analog-to-digital workflow:** Beginning on paper and moving to LaTeX demonstrates a meaningful, phased structuring of tasks that supports internalization.
- **Google Calendar for time management:** Facilitates metacognitive control of cognitive resources by aligning effort with task goals.
- **Feedback and self-correction through printout review (13/03):** Highlighter-based correction supports error detection and reflective learning.
- **Honest documentation and self-awareness:** Participant 3's deliberate boundary between structured study and unstructured rest reflects deep insight into cognitive sustainability.